

GRAS Determination
for the Use of Kluvy® Protein
in Selected Conventional Foods

Prepared for:
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FOREWORD

Shanghai Changing Biotechnology Co., Ltd. (referred to as “Changing Bio”) is a company based in Shanghai, China, and was established in 2017. The company specializes in researching, developing, and manufacturing microbial and synthetic proteins. The subject of this notice is Kluvy® Protein, a yeast protein product produced by the fermentation of *Kluyveromyces marxianus*.

ACC’s evaluation focused primarily on composite safety information, which includes scientific procedures corroborated by historical use. The review covered the safety and toxicity of the *Kluyveromyces marxianus*, its history of use, Kluvy® Protein compositional details, specifications, the method of preparation of the substance, and allergenicity evaluation. A thorough search of the scientific and regulatory literature was conducted up to September 2024, with a focus on adverse reports and studies that support safety conclusions. Pertinent references from this review are listed in 7. The integration of safety and toxicity studies with dietary exposure information forms the scientific basis for the GRAS conclusion.

At Changing Bio’s request, the Anchor Center for Certification (referred to as “ACC”) convened an Expert Panel to complete an independent safety evaluation of the yeast protein. In addition, Changing Bio has appointed ACC as the Agent for the submission of this GRAS notification to the FDA.

1 SIGNED STATEMENTS AND CERTIFICATION

1.1 Submission of GRAS Notice

Shanghai Changing Biotechnology Co., Ltd. is hereby submitting a GRAS notice in accordance with Subpart E of 21 CFR Part 170.

1.2 Name And Address of The Sponsor

Shanghai Changing Biotechnology Co., Ltd.
3rd Floor, Building 4, No.789/799/855 Zi Ping Road,
Pudong District, Shanghai, China 201318

As the Responsible Party, Changing Bio accepts the responsibility for the GRAS conclusion that has been made for Kluvy® Protein as described in the subject safety evaluation; consequently, the Kluvy® Protein having an acceptable composition that meets the conditions described herein, is not subject to the premarket approval requirements for food ingredients.

1.3 Common or Usual Name

The common name of the substance to be used on food labels is yeast protein and Changing Bio plans to market this product under the trade name: Kluvy® Protein.

1.4 Trade Secret or Confidential Information

This notification does not contain any trade secret or confidential information.

1.5 Intended Use

Changing Bio's Kluvy® Protein is intended for use as a non-animal source protein in a variety of food applications as detailed in **Table 10** at levels determined by this safety review and current Good Manufacturing Practices (cGMP). This substance is not intended for use in products regulated by the U.S. Department of Agriculture. Furthermore, Kluvy® Protein is not intended for use in infant formula.

1.6 Basis of Exclusion from the Requirement for Premarket Approval Pursuant to Subpart E of 21 CFR 170.30¹

Changing Bio has determined that Kluvy® Protein, which meets the specifications described herein, is GRAS in accordance with Section 201(s) of the Federal Food, Drug, and Cosmetic Act (FD&C Act). The GRAS determination is based on scientific procedures as described in the following sections. The

¹ See 81 FR 54960, 17 August 2016. Accessible at: <https://www.govinfo.gov/content/pkg/FR-2016-08-17/pdf/2016-19164.pdf>
(Accessed October 3, 2024)

evaluation accurately reflects the intended conditions of use for the designated ingredient.

1.7 Premarket Approval

The notified Kluvy® Protein is not subject to the premarket approval requirements of the FD&C Act based on our conclusion that the yeast protein is GRAS under the conditions of intended use.

1.8 Availability of Information

The data and information that serve as the basis for this GRAS determination will be available for review and copying at reasonable times at the office of Anchor Center for Certification*:

Contact:

Anchor Center for Certification

No 1295 Chuanqiao Road, Building# , Suite 302,

Pudong, Shanghai, China 201206

Email: Stella.si@acclab.cn; Phone: +1-925-204-9901

or be sent to FDA upon request

*ACC is the authorized representative of Shanghai Changing Biotechnology Co., Ltd. for GRAS Notification for the Use of Kluvy® Protein in Selected Conventional Food.

1.9 Freedom of Information Act (FOIA)

Parts 2 through 7 of this notification do not contain data or information that is exempt from disclosure under the FOIA.

1.10 Information Included in the GRAS Notification

To the best of our knowledge, the information contained in this GRAS notification is complete, representative, and balanced. It contains both favorable and unfavorable information, known to the notifier and ACC, and pertinent to the evaluation of the safety and GRAS status of Kluvy® Protein.

Signed:



Yu Xu

Director of Science Center

Shanghai Changing Biotechnology Co., Ltd.

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Date: 28/10/2024

Signed:



Agent for Shanghai Changing Biotechnology Co., Ltd.

Stella Si

Date: 28/10/2024

Managing Director

Anchor Center for Certification (ACC)

No. 1295 Chuan Qiao Road, Building 2, Suite 302,
Pudong, Shanghai, China 201206

2 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT

This section reviews and cites publicly available information, as well as data provided by ACC and Changing Bio, pertaining to the composition, manufacturing process, and specifications of Kluvy® Protein that is the subject of this GRAS determination. This information characterizes the notified substance intended for use as an ingredient in food products.

2.1 Identity

The subject of this GRAS determination is Kluvy® Protein, a yeast protein derived from the CJ3113 strain of *Kluyveromyces marxianus* (referred to as “*K. marxianus*”). The strain was isolated from a special type of Mongolian cheese, locally called “milk tofu” and is used to produce yeast protein via aerobic fermentation in a defined medium. Following fermentation, the entire biomass of the CJ3113 strain is collected, autoclaved, and dried into powder, which results in a product containing $\geq 55\%$ w/w yeast protein, as well as natural fermentative by-products and yeast cell debris. Such a product is named Kluvy® Protein and it is intended for use as a substitute for animal-source protein in conventional foods. Kluvy® Protein is predominantly composed of protein ($>55\%$), along with dietary fiber ($\sim 30\%$), moisture ($<7\%$), and fat ($\sim 5\%$).

2.2 Manufacturing Process

2.2.1 Strain Used as Production Strain

K. marxianus, a nonpathogenic and nontoxigenic yeast, is well known due to its frequent connection with traditional dairy products like fermented milk, kefir, yogurt, cheese, and so on (Coloretti et al., 2017). It is a dairy yeast that has also been previously described by synonyms in scientific literature, notably *Kluyveromyces fragilis* and *Saccharomyces keyfr* (Karim et al., 2020, Morrissey et al., 2014). In general, *K. marxianus* is considered a safe yeast used in the food and biotechnology sectors (Varela et al., 2017, Wu et al., 2024). Because of its history of association with food, *K. marxianus*, like *Kluyveromyces lactis* (referred to as “*K. lactis*”), has been designated European Food Safety Authority (EFSA) Qualified Presumption of Safety (QPS) status (EFSA BIOHAZ, 2024, Varela et al., 2017, Wu et al., 2024). In general, yeast, including *K. marxianus*, are not known to produce toxic metabolites (Alasmar et al., 2020).

The subject of this GRAS Determination is manufactured from a non-genetically modified *K. marxianus* CJ3113 strain. This strain was isolated from self-made edible “milk tofu” collected from herdsmen in Mongolia, China, by culturing it in *Kluyveromyces* Differential Medium (KDM), containing yeast extract, peptone, malt extract, dextrose, agar, 5-bromo-4-chloro-3-indolyl-b-D-galactopyranoside, isopropyl-beta-D-thiogalactoside (IPTG), and chloramphenicol. This medium is a screening and identification medium that can be used to selectively screen wild-type yeast (*K. marxianus* and *K. lactis*) with β -galactosidase activity. Single, blue-colored colonies growing on the medium are selected as candidate strains. The ideal strain was selected based on rapid growth rate,

high biomass production, high protein content, balanced amino acid composition, and suitability for large-scale fermentation production and named CJ3113. The CJ3113 strain has been deposited into the China Center for Type Culture Collection (CCTCC) with deposition number M 20211265.

To further characterize the CJ3113 strain, polyphase identification analysis was conducted. When grown on malt extract agar (MEA) plate at 28°C for 72 hrs, the streak culture was light cream colored. The colonies' surface was flat with a sticky texture and irregular margins. Under a microscope, CJ3113 budding cells were ellipsoidal, kidney to cylindrical shaped, with dimensions of 3.5 – 11.5µm × 1.8 – 8.5µm. The physical observation of the CJ3113 strain meets the expectation of the morphology of *K. marxianus* species.

The physiological and biochemical characterization of the CJ3113 strain was done via a carbon source assimilation study. The outcome of the study is tabulated in **Table 1**. The assimilation results of the CJ3113 strain show similar characteristics to those expected of a *K. marxianus* species (Lappe-Oliveras et al., 2023).

Table 1. Carbon Source Assimilation of CJ3113 Strain

Carbon Source Assimilation							
D-glucose	+	Glycerol	+	Calcium 2-keto-gluconate	-	L-arabinose	-
D-xylose	-	Adonitol	-	Xylitol	-	D-galactose	+
Inositol	-	D-sorbitol	-	Methyl-αD-glucopyranoside	-	N-acetyl-glucosamine	-
D-cellobiose	-	D-lactose (bovine origin)	+	D-maltose	-	D-saccharose (sucrose)	-
D-trehalose	-	D-melezitose	-	D-raffinose	-		

Note: “+”: positive, “-”: negative

Furthermore, ITS sequencing of CJ3113 was conducted and the sequence alignment was performed by comparing it with publicly available sequences in the NCBI database. The sequence alignment shows that CJ3113 is closely related to *K. marxianus* CBS 712^T (AY046214), with 99.8% similarity. A phylogenetic tree was drawn based on ITS rDNA sequence by the neighbor-joining method in the MEGA program version 5.0 (**Figure 1**).

The outcome of this polyphase identification analysis indicates that the CJ3113 strain, naturally isolated strain from a food source, is a *Kluyveromyces marxianus* species and is closely related to *K. marxianus* CBS 712^T. Its taxonomic classification is described in **Table 2**.

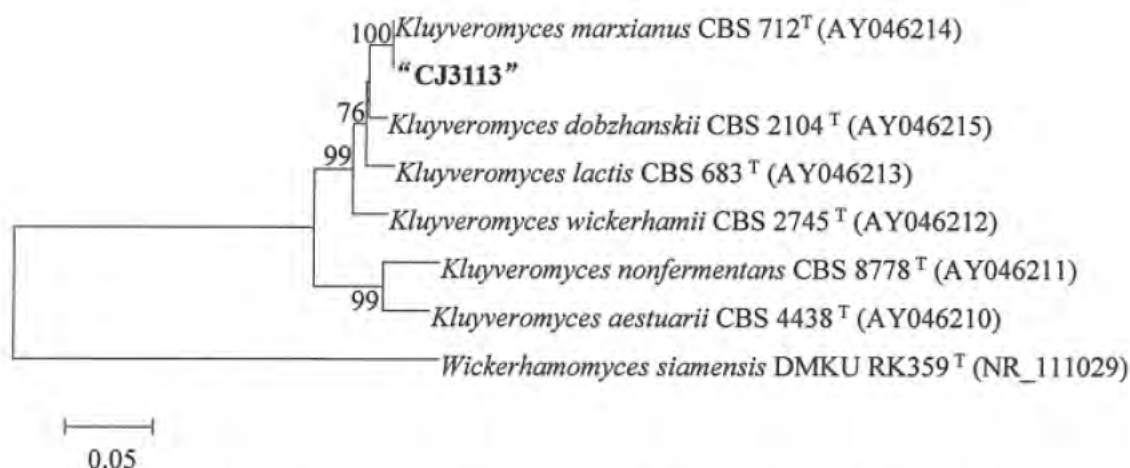


Figure 1. ITS rDNA phylogenetic tree of CJ3113 with related species

Bootstrap values (>70%) after 1000 replicates were shown at each branch point. The letter “T” indicates the type strain.

Table 2. Taxonomic Classification of *K. marxianus* CJ3113

Class	Scientific Classification
Kingdom	Fungi
Phylum	<i>Ascomycota</i>
Class	<i>Saccharomycetes</i>
Order	<i>Saccharomycetales</i>
Family	<i>Saccharomycetaceae</i>
Genus	<i>Kluyveromyces</i>
Species	<i>Kluyveromyces marxianus</i>
Strain	CJ3113

Available at: <https://www.ncbi.nlm.nih.gov/datasets/taxonomy/4911/> (Accessed on August 20, 2024)

2.2.2 Quality Control

Kluby® Protein is manufactured in compliance with cGMP and the principles of Hazard Analysis Critical Control Point (HACCP). Changing Bio’s prerequisite programs are developed according to the cGMP components listed in US FDA 21 CFR Part 117 Subpart B. The Good Manufacturing Practice (GMP) programs establish the conditions and practices followed by Changing Bio to ensure the production of Kluby® Protein is under strict sanitary conditions.

Additionally, the food safety plan of Changing Bio’s Kluby® Protein is developed in compliance with the requirements defined in the US FDA Food Safety Modernization Act (FSMA). Potential hazards arising from raw materials, processing aids, and each manufacturing step are reviewed and evaluated.

Allergen management and contamination control procedures are implemented and executed throughout procurement, storage, production, packaging, and product management. All raw materials used in the fermentation are neither major allergens nor derived from major allergens. All equipment is carefully designed, constructed, operated, cleaned, and maintained to prevent contamination. Raw

materials used are sterilized at the start of manufacturing to avoid microbial contamination. The manufacturing process is conducted under strict and vigorous sanitary practices to ensure the highest quality and safety of the finished product. All personnel are also well-trained to comply with the established manufacturing procedures and to uphold the highest standards of cleanliness.

The purity of the production strain is ensured by performing a strain confirmation test as per the established standard operating procedures (SOPs). The production strain identity is verified throughout the manufacturing process, including during seed bank, flask inoculation, and tank fermentation, by evaluating the morphological characteristics, microscopic characteristics, and rDNA sequencing of the production yeast.

All raw materials, food additives, processing aids, and food contact surfaces used meet US food-grade standards. SOPs and work instructions are followed, batch records are maintained, numerous in-process controls are performed, and the final Kluvy® Protein product is verified through certificates of analysis (COA) and batch release routines.

During the production process, operating parameters such as temperature, fermentation time, pH, and airflow velocity are controlled to ensure product quality. Changing Bio conducts hazard analyses for manufacturing processes, raw materials, products, packaging materials, equipment, and food contact surfaces. Based on Changing Bio's HACCP plan, the autoclaving step and metal detection after packaging have been identified as critical control points (CCPs). During autoclaving, the production organism CJ3113 is heat-killed, ensuring that the finished product does not contain live yeast. The absence of live production yeast is a quality attribute included in the product specification and is tested via yeast analysis using FDA's Bacteriological Analytical Manual (BAM) Chapter 18. This method has been verified to be suitable to effectively detect the presence of live production yeast (see Appendix 1). The second critical control point is metal detection, which is used to detect metal impurities in the product. The critical control limits for metal detectors are iron ≤ 1.5 mm, non-iron ≤ 2.0 mm, and stainless steel ≤ 2.0 mm. Final products containing metal contaminants greater than critical control limits will be detected by the metal detector.

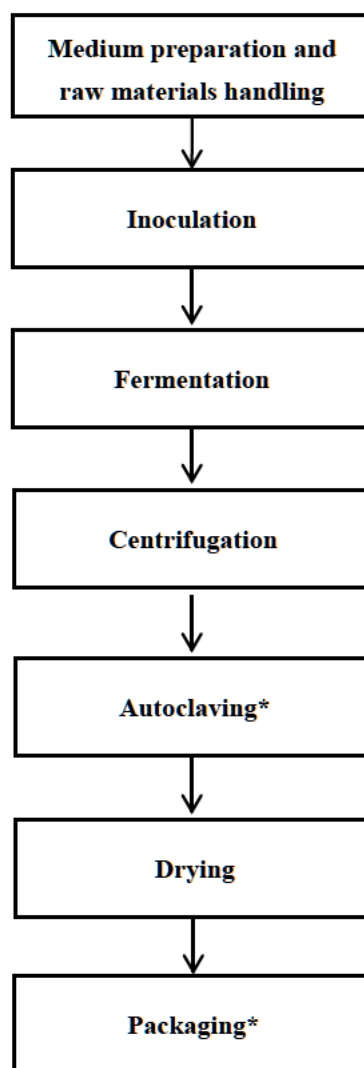
Furthermore, production water, sourced from the city water supply, is tested annually by a third-party laboratory to ensure compliance with the China National drinking water standards. Internal monitoring of production water is conducted monthly, including visual inspection, odor and taste evaluation, pH measurement, total plate counts, and total coliform testing.

Solid production waste is collected and treated before being disposed of by qualified third-party contractors. Production wastewater is collected and sent to a sewage treatment facility.

The quality system established at Changing Bio is certified by globally recognized food safety verification programs, such as the Food Safety System Certification (FSSC) 22000 and the British Retail Consortium Global Standard Food Safety. The certificates are attached in Appendix 2.

2.2.3 Method of Manufacture

The subject of this GRAS determination is produced via fermentation using a non-genetically modified *K. marxianus* strain. The key manufacturing steps of Kluvy® Protein include medium preparation and raw materials handling, inoculation, fermentation, centrifugation, autoclaving, drying, and packaging. Detailed descriptions of each manufacturing step are provided in the following section. The production process of Kluvy® Protein is illustrated in **Figure 2**.



*Represent critical control points

Figure 2 Manufacturing process of Kluvy® Protein

Medium Preparation and Raw Materials Handling

The raw materials used in the fermentation of *K. marxianus* are ingredients commonly used in the food fermentation and enzyme production industries (**Table 3**). All the raw materials are food grade. None of the materials used in the production of Kluvy® Protein are derived from major allergens. The fermentation medium is prepared in accordance with cGMP (21 CFR Part 117). The media used for

inoculation and fermentation are the same. The required raw materials for medium preparations are placed into respective stainless-steel fermentation tanks, sterilized at 115-121°C for 20-30 mins, and cooled before use. This ensures the absence of contamination of other microorganisms.

Table 3. List of Raw Materials Used for Medium Preparation

Materials Name	Function	Regulatory Status
Glucose	Carbon source	21 CFR 184.1865
Ammonia	pH adjustment	21 CFR 184.1139
Sodium hydroxide	pH adjustment	21 CFR 184.1763
Magnesium Sulfate Heptahydrate	Fermentation medium ingredient	21 CFR 184.1443
Ammonium phosphate, monobasic	Fermentation medium ingredient	21 CFR 184.1141a
Citric acid monohydrate	Fermentation medium ingredient	21 CFR 184.1033

Inoculation

The working seed of CJ3113 is transferred to a shake flask containing 400 mL sterilized medium and incubated at 30±0.5°C, 200 rpm for 10 – 16 hrs, producing the primary seed liquid. This primary seed liquid is then inoculated to a 50 L seed fermenter tank (containing 25 L sterilized medium) with a 10% inoculation and is incubated at 30±0.5°C for 8 – 12 hrs to obtain the secondary seed liquid. The secondary seed liquid is subsequently transferred into a 500 L seed fermenter tank (containing 250 L sterilized medium) at 10% inoculation and incubated at 30±0.5°C for 8 – 12 hrs until sufficient biomass is obtained. The entire cultivation of seed liquid is done in enclosed and sterile environments to avoid cross-contamination.

Fermentation

The seed liquid from the 500 L tank is then transferred to a 5000 L fermentation tank containing sterilized medium (8 – 10% inoculation). The main fermentation parameters are set at 30±5°C, 65 rpm stirring speed, pH 5.5±0.5, air flow rate of 200 m³/h, and dissolved oxygen level at 30%. The fermentation process is conducted in a closed vessel and a contained and sterile environment for 18 to 36 hrs.

Centrifugation

The fermentation broth containing the cell of CJ3113 is then harvested by centrifugation (separation factor: 12,000 – 15,000; rotational speed: 5000 – 6000 rpm; processing capacity: 2000 – 3000 L/h) to remove the fermentation liquid supernatant and obtain the cell precipitate.

Autoclaving

The cell precipitate is then resuspended in water to get a cell suspension with a solid content range of 12% to 15%. The suspension is then transferred to a clean 5000 L fermentation tank and autoclaved

with high-temperature steam at 105 – 125°C, with a pressure of 0.1 – 0.15 MPa, 65 rpm stirring for 15 – 30 mins. This process effectively heat-kills the production yeast. The efficiency of autoclaving step was verified by evaluating the presence of total colony count, yeast count, and mold count when the in-process product was heat-treated at 105 or 125°C for 15 or 30 mins. The outcome of the study is summarized in **Table 4** below. The results show that autoclaving at 105 – 125°C for 15 – 30 mins effectively heat-killed the production yeast.

Table 4. Autoclave Efficiency Study Outcome of Samples from Three Non-Consecutive Batches

Sample	Autoclave temperature (°C)	Autoclave time (min)	Total colony count (CFU/g)	Yeast count (CFU/g)	Mold count (CFU/g)
Sterile saline solution	105	15	<10	<10	<10
Sterile saline solution	125	15	<10	<10	<10
Sterile saline solution	125	30	<10	<10	<10
Batch No. 20240308	105	15	<10	<10	<10
Batch No. 20240413	125	15	<10	<10	<10
Batch No. 20240518	125	30	<10	<10	<10

Drying

The autoclaved cell suspension is then spray dried with an inlet temperature of 180°C±5 °C and an outlet temperature of 100°C±5 °C to remove moisture from the final product. The finished product is off-white to light brown powder with moisture content of ≤7.0%.

Packaging and Storage

The final Kluvy® Protein product is packed in a hot-sealed polyethylene (PE) bag, which is then hot-sealed with an outer aluminum foil bag along with a desiccant. The packaging system protects the product from light, moisture, and oxygen. The packaged product is run through a metal detector before it is stored under ambient conditions prior to delivery. Each production lot is sampled and tested as per product specifications. A COA is generated for every production lot and evaluated by quality control before release.

Kluvy® Protein manufacturing involves a non-genetically modified, food-derived *K. marxianus* strain free of antibiotic resistance markers and any other transgenes. Furthermore, the entire manufacturing process, including the fermentation and purification steps, complies with cGMP and HACCP. Changing Bio also conducts hazard analysis for the manufacturing process, final product, raw materials, packaging materials, and equipment. Moreover, the absence of live production yeast is ensured by

testing for the presence of yeast as per compendial method (FDA's BAM Chapter 18) with a specification limit of <10 CFU/g. These precautions are taken to ensure that the Kluvy® Protein product is of the highest quality and safe for human consumption.

2.3 Product Specification

Kluvy® Protein consists primarily of protein (over 55%), along with dietary fiber (around 30%), moisture (less than 7%), and fat (approximately 5%). To ensure product quality, Changing Bio conducts analytical tests on every batch of Kluvy® Protein. The product specifications for Kluvy® Protein are established, as detailed in **Table 5**. Each lot must meet the acceptance criteria before release. The parameters tested for every batch of the final product include appearance, protein content, ash, moisture, pH, heavy metals, and microbial limits. All analytical methods have been validated for the intended use.

Table 5. Changing Bio's Product Specification for Kluvy® Protein.

Test Item	Acceptance Criteria	Reference Method
Appearance	Off white to light brown powder	Visual
Moisture (%)	≤7.0	AOAC 930.04
Protein (%)	≥55	AOAC 979.09
Ash (%)	<3	AOAC 930.05
Lead (ppm)	≤0.1	J. AOAC vol. 90 (2007)
Arsenic (ppm)	≤0.1	J. AOAC vol. 90 (2007)
Mercury (ppm)	≤0.1	J. AOAC vol. 90 (2007)
Cadmium (ppm)	≤0.1	J. AOAC vol. 90 (2007)
Total plate Count (CFU/g)	<10000	AOAC 966.23
Yeast (CFU/g)	<10	FDA BAM Chapter 18
Mold (CFU/g)	<100	FDA BAM Chapter 18
<i>Escherichia coli</i> (CFU/g)	<10	AOAC 991.14
<i>Enterobacteriaceae</i>	Negative in 25 g	AOAC 2003.01
<i>Salmonella</i>	Negative in 25 g	AOAC 2003.09
<i>Staphylococcus aureus</i> (CFU/g)	<10	AOAC 2003.08

Five non-consecutive batches of Kluvy® Protein were manufactured and tested. The results, as shown in **Table 6**, demonstrates that all five batches meet the specifications. The respective COAs are attached in Appendix 3. The collection of these reports demonstrated that the substance is well-characterized and meets the established specification limits.

Table 6. Analytical Results of Five Batches of Kluvy® Protein.

Test Items	Specification	Test Results				
		Batch #20221011	Batch #20221212	Batch #20230618	Batch #20230919	Batch #20240308
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	3.85	4.07	3.66	4.19	3.79
Protein (%)	≥55	59.7	62.8	60.5	62.1	62.9
Ash (%)	<3	2.5	2.8	2.7	2.3	2.5
Lead (ppm)	≤0.1	0.011	0.011	0.012	0.012	0.011
Arsenic (ppm)	≤0.1	ND	ND	ND	ND	ND
Mercury (ppm)	≤0.1	ND	ND	ND	ND	ND
Cadmium (ppm)	≤0.1	ND	ND	ND	ND	ND
Total plate count (CFU/g)	<10000	58	15	67	37	66
Yeast (CFU/g)	<10	ND	ND	ND	ND	ND
Mold (CFU/g)	<100	ND	ND	ND	ND	ND
<i>E. coli</i> (CFU/g)	<10	ND	ND	ND	ND	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	Negative	Negative	Negative	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	Negative	Negative	Negative	Negative
<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	ND	ND	ND	ND

Note: ND = not detected within the limit of detection (LOD), <10 CFU/g for microbiological tests, 0.01 ppm for arsenic; 0.01 ppm for cadmium; 0.01 ppm for lead; 0.01 ppm for mercury.

2.4 Other Quality Attributes

2.4.1 Amino Acids

Kluy® Protein is a complete protein, consisting of all twenty naturally occurring amino acids. This includes nine essential amino acids - those that the human body cannot produce and must be supplemented by food: isoleucine, leucine, histidine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. Based on the analyses of three non-consecutive batches of Kluy® Protein, the average amino acid content is tabulated in **Table 7**, wherein the essential amino acids make up about 27% of the total dry weight, the branched-chain amino acids (leucine, isoleucine, and valine) make up 11 – 12% of the total dry weight, and the total amino acids make up 55 – 60% of the total dry weight.

The essential and non-essential amino acids combined comprise all the organic nitrogen content in the protein. In addition, Kluvy® Protein compares favorably to other protein replacement products used as meat and dairy analogs.

Table 7. Amino Acid Composition of Kluvy® Protein and Comparison to Alternative Protein Replacement Products

Test item (% of total, as dry weight basis)	Kluyv® Protein	GRN 000904 Mycoprotein ^a	Tofu ^b	Pea protein ^c
Alanine	3.80 ± 0.06	2.99	2.56	1.14
Arginine	3.41 ± 0.04	2.96	4.54	2.02
Aspartic acid	5.51 ± 1.04	4.67	6.75	2.35
Glutamic acid	7.01 ± 0.09	5.51	10.90	3.51
Glycine	2.81 ± 0.12	2.30	2.43	0.87
Histidine	1.56 ± 0.04	1.17	1.43	0.51
Isoleucine	3.20 ± 0.05	2.32	2.82	0.92
Leucine	5.02 ± 0.09	3.83	4.61	1.53
Lysine	5.06 ± 0.15	4.04	2.93	1.50
Phenylalanine	3.19 ± 0.05	2.13	2.77	0.95
Proline	2.39 ± 0.06	2.22	3.59	0.82
Serine	3.14 ± 0.05	2.36	3.36	0.86
Threonine	3.20 ± 0.05	2.57	2.60	0.96
Tyrosine	2.30 ± 0.04	1.64	2.32	0.54
Valine	3.84 ± 0.09	5.12	2.89	1.11
Methionine	1.12 ± 0.02	0.88	0.70	0.39
Cysteine & Cystine	0.49 ± 0.01	0.40*	0.19*	0.15*
Tryptophan	0.94 ± 0.02	0.85	0.78	0.18

^{a b c} Values obtained from GRAS Notice GRN 000904, page 16, Table 6; *As cystine only

2.4.2 Nutritional Profile of Kluvy® Protein

Changing Bio has also conducted nutritional profile tests for Kluvy® Protein. The typical nutritional profile results of Kluvy® Protein is shown in **Table 8**.

Table 8. Nutritional Profile of Five Lots of Kluvy® Protein.

Parameters	Test method	Test Result				
		Batch #20221011	Batch #20221212	Batch #20230618	Batch #20230919	Batch #20240308
Moisture (%)	AOAC 930.04	3.85	4.07	3.66	4.19	3.79
Protein (%)	AOAC 979.09	59.7	62.8	60.5	62.1	62.9
Sugar (Glucose + Fructose + Maltose + Lactose + Sucrose)	AOAC 982.14	ND	ND	ND	ND	ND
Dietary fiber (g/100 g)	AOAC 991.43	29.1	29.0	28.7	28.9	29.6
Insoluble dietary fiber (g/100 g)	AOAC 991.43	19.6	19.5	18.9	19.3	19.5

Parameters	Test method	Test Result				
		Batch #20221011	Batch #20221212	Batch #20230618	Batch #20230919	Batch #20240308
Soluble dietary fiber (g/100 g)	AOAC 991.43	9.23	9.10	9.51	9.33	9.65
Total carbohydrate (g/100 g)	AOAC 982.14:1983	30.2	29.8	29.3	30.0	31.5
Calories (KJ/100 g)	REGULATION (EU) No 1169/2011	1438	1471	1501	1429	1456
Total fat (g/100 g)	AOAC 996.06	5.36	4.21	4.12	5.41	4.25
Saturated fat	AOAC 996.06	1.34	1.08	0.98	1.3	1.12
Monounsaturated	AOAC 996.06	1.79	1.55	1.47	1.93	1.52
Polyunsaturated	AOAC 996.06	2.17	1.51	1.6	2.11	1.55
Trans fat	AOAC 996.06	0.04	0.04	0.04	0.04	0.03
Cholesterol (mg/100g)	AOAC 976.26	ND	ND	ND	ND	ND
Sodium (mg/kg)	AOAC 996.06	64.63	65.29	73.09	68.78	65.12
Ash (%)	AOAC 930.05	2.5	2.8	2.7	2.3	2.5
Note: ND: not detected within the LOD, which is <0.4 g/100 g for Sugar (Glucose, Fructose, Maltose, Lactose and Sucrose); <1 mg/100g for Cholesterol						

2.5 Stability

Changing Bio has conducted an accelerated stability study on five non-consecutive batches of Kluvy® Protein. The storage conditions of Kluvy® Protein are at ambient conditions. Hence, the accelerated stability study was conducted at 40±2°C and 75±5% relative humidity (RH) for 6 months. The results of the 6-month accelerated stability test, as shown in **Table 9**, indicate that the product quality has not changed significantly and is stable under the described conditions.

The proposed shelf-life for Kluvy® Protein is 12 months. The currently available stability data supports a shelf-life of 12 months at ambient conditions.

Table 9. Stability Data of Kluvy® Protein At 40±2°C and 75±5%RH

Test items	Specification	Batch No.: 20220201 at 40±2°C and 75±5%RH			
		0 month	1 month	3 months	6 months
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	3.84	3.87	3.90	3.90
Protein (%)	≥55	58.7	58.2	58.1	58.2
Ash (%)	<3	2.61	NT	NT	2.60
Lead (ppm)	≤0.1	0.011	NT	NT	0.011
Arsenic (ppm)	≤0.1	ND	NT	NT	ND
Mercury (ppm)	≤0.1	ND	NT	NT	ND
Cadmium (ppm)	≤0.1	ND	NT	NT	ND
Total plate count (CFU/g)	<10000	36	39	45	105
Yeast (CFU/g)	<10	ND	NT	NT	ND
Mold (CFU/g)	<100	ND	NT	NT	ND
<i>E. coli</i> (CFU/g)	<10	ND	NT	NT	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	NT	NT	ND
Test items	Specification	Batch No.: 20220316 at 40±2°C and 75±5%RH			
		0 month	1 month	3 months	6 months
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	4.05	4.06	4.15	4.15
Protein (%)	≥55	62.3	62.0	61.5	61.1
Ash (%)	<3	2.82	NT	NT	2.77
Lead (ppm)	≤0.1	0.011	NT	NT	0.011

Arsenic (ppm)	≤0.1	ND	NT	NT	ND
Mercury (ppm)	≤0.1	ND	NT	NT	ND
Cadmium (ppm)	≤0.1	ND	NT	NT	ND
Total plate count (CFU/g)	<10000	75	70	78	231
Yeast (CFU/g)	<10	ND	NT	NT	ND
Mold (CFU/g)	<100	ND	NT	NT	ND
<i>E. coli</i> (CFU/g)	<10	ND	NT	NT	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	NT	NT	ND
Test items	Specification	Batch No.: 20220409 at 40±2°C and 75±5%RH			
		0 month	1 month	3 months	6 months
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	3.91	3.99	4.06	4.06
Protein (%)	≥55	60.7	61.7	60.8	60.6
Ash (%)	<3	2.81	NT	NT	2.79
Lead (ppm)	≤0.1	0.012	NT	NT	0.012
Arsenic (ppm)	≤0.1	ND	NT	NT	ND
Mercury (ppm)	≤0.1	ND	NT	NT	ND
Cadmium (ppm)	≤0.1	ND	NT	NT	ND
Total plate count (CFU/g)	<10000	19	35	65	189
Yeast (CFU/g)	<10	ND	NT	NT	ND
Mold (CFU/g)	<100	ND	NT	NT	ND
<i>E. coli</i> (CFU/g)	<10	ND	NT	NT	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	NT	NT	Negative

<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	NT	NT	ND
Test items	Specification	Batch No.: 20220525 at 40±2°C and 75±5%RH			
		0 month	1 month	3 months	6 months
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	4.28	4.35	4.25	4.25
Protein (%)	≥55	61.2	61.5	60.8	61.0
Ash (%)	<3	2.52	NT	NT	2.51
Lead (ppm)	≤0.1	0.012	NT	NT	0.012
Arsenic (ppm)	≤0.1	ND	NT	NT	ND
Mercury (ppm)	≤0.1	ND	NT	NT	ND
Cadmium (ppm)	≤0.1	ND	NT	NT	ND
Total plate count (CFU/g)	<10000	55	66	79	156
Yeast (CFU/g)	<10	ND	NT	NT	ND
Mold (CFU/g)	<100	ND	NT	NT	ND
<i>E. coli</i> (CFU/g)	<10	ND	NT	NT	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	NT	NT	ND
Test items	Specification	Batch No.: 20220618 at 40±2°C and 75±5%RH			
		0 month	1 month	3 months	6 months
Appearance	Off white to light brown powder	Off white powder	Off white powder	Off white powder	Off white powder
Moisture (%)	≤7.0	3.99	4.06	4.16	4.16
Protein (%)	≥55	60.9	60.6	60.5	59.8
Ash (%)	<3	2.77	NT	NT	2.71
Lead (ppm)	≤0.1	0.011	NT	NT	0.011

Arsenic (ppm)	≤0.1	ND	NT	NT	ND
Mercury (ppm)	≤0.1	ND	NT	NT	ND
Cadmium (ppm)	≤0.1	ND	NT	NT	ND
Total plate count (CFU/g)	<10000	88	89	99	325
Yeast (CFU/g)	<10	ND	NT	NT	ND
Mold (CFU/g)	<100	ND	NT	NT	ND
<i>E. coli</i> (CFU/g)	<10	ND	NT	NT	ND
<i>Enterobacteriaceae</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Salmonella</i>	Negative in 25 g	Negative	NT	NT	Negative
<i>Staphylococcus aureus</i> (CFU/g)	<10	ND	NT	NT	ND

Note: ND =not detected within the LOD, which is <10 CFU/g for microbiological tests, 0.01 ppm for arsenic; 0.01 ppm for cadmium; 0.01 ppm for lead; 0.01 ppm for mercury. NT= not tested

3 DIETARY EXPOSURE

3.1 Intended Use

Kluvy® Protein is intended for use in processed food products (excluding infant formulas and fish, meat and poultry products) to increase the level of protein in a product, serving as a substitute for other animal or vegetable protein. Kluvy® Protein is intended for use as a food ingredient and macronutrient in a variety of food categories and the range of intended incorporation rates of Kluvy® Protein in various foods is presented in **Table 10**. The intended use levels are presented below on a dry weight basis, and the anticipated minimum and maximum use levels are provided.

Table 10. Intended Food Use Categories

Food Category	Minimum Use Level (%)	Maximum Use Level (%)
(1) Baked goods and baking mixes, including all ready-to-eat and ready-to-bake products, flours and mixes, requiring preparation before serving.	1	15
(3) Beverages and beverage bases, nonalcoholic, including only special or spiced teas, soft drinks, coffee substitutes, and fruit and vegetable flavored gelatin drinks, drinking water, sport drinks.	1	10
(4) Breakfast cereals, including ready-to-eat and instant and regular hot cereals.	1	15
(8) Condiments and relishes, including plain seasoning sauces and spreads, olives, pickles, and relishes, but not spices or herbs.	1	15
(10) Dairy product analogs, including nondairy milk, frozen or liquid creamers, coffee whiteners, toppings, and other nondairy products.	1	8
(20) Frozen dairy desserts and mixes, including ice cream, ice milks, sherbets, and other frozen dairy desserts and specialties.	1	8
(23) Grain products and pastas, including macaroni and noodle products, rice dishes, and frozen multicourse meals, without meat or vegetables.	1	8
(24) Gravies and sauces, including all meat sauces and gravies, and tomato, milk, buttery, and specialty sauces.	1	10
(31) Milk products, including flavored milks and milk drinks, dry milks, toppings, snack dips, spreads, weight control milk beverages, and other milk origin products.	1	8
(33) Plant protein products, including the National Academy of Sciences/ National Research Council "reconstituted vegetable protein" category, and meat, poultry, and fish substitutes, analogs, and extender products made from plant proteins.	1	20
(35) Processed fruits and fruit juices, including all commercially processed fruits, citrus, berries, and mixtures; salads, juices and juice punches, concentrates, dilutions, "ades", and drink substitutes made therefrom	1	9
(36) Processed vegetables and vegetable juices, including all commercially processed vegetables, vegetable dishes, frozen multicourse vegetable	1	9

Food Category	Minimum Use Level (%)	Maximum Use Level (%)
meals, and vegetable juices and blends.		
(37) Snack foods, including chips, pretzels, and other novelty snacks.	1	15
(38) Soft candy, including candy bars, chocolates, fudge, mints, and other chewy or nougat candies.	1	5
(40) Soups and soup mixes, including commercially prepared meat, fish, poultry, vegetable, and combination soups and soup mixes*.	1	8

3.2 Estimated Daily Intake of Kluvy® Protein

The estimated daily intake (EDI) of Kluvy® Protein was conservatively calculated based on its the maximum intended incorporation rate in representative foods drawn from the intended food categories, in combination with food consumption data provided in the National Health and Nutrition Examination Survey (NHANES) for 2017 – 2018². Data management, statistical analysis, and calculations for the mean and 90th percentile *per capita* and consumer-only intakes, reported per person and per kilogram body weight, were performed using Dazult Ltd.'s DaDiet Software, Version 24.09.

The individual food codes that best reflect the proposed uses were selected for this intake assessment and are listed in Appendix 4.

Per capita and consumer-only estimates of the total daily intake of Kluvy® Protein from the proposed food uses at the maximum use levels listed in **Table 10** were determined on an absolute basis (g/day; **Table 11**) and on a per kilogram body weight basis (g/kg bw/day, **Table 12**).

Table 11. Estimated Daily Intake of Kluvy® Protein (g/day)

Population Group	Age group (Years)	Per Capita Intake (g/day)		Consumer-Only Intake(g/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Children	2-5	21	46	99.0	462	21	46
Children	6-11	25	48	99.0	671	25	49
Female Teenagers	12-19	21	43	98.7	440	21	45
Male Teenagers	12-19	27	59	95.8	422	28	59
Female Adults	20+	21	46	97.9	2104	21	47
Male Adults	20+	28	59	97.4	1903	29	59
Total Population	2+	24	53	97.8	6002	25	53

g = grams; n = sample size

² The NHANES is a program of studies conducted by the National Center for Health Statistics (NCHS) within the Centers for Disease Control and Prevention (CDC) and is designed to assess the health and nutritional status of adults and children in the United States. One component of the NHANES is a 2-day dietary survey of the study participants. The data from this survey is made publicly available and can be used to conduct intake estimates, see

<https://www.cdc.gov/nchs/nhanes/index.htm>.

Table 12. Estimated Daily Intake of Kluvy® Protein (g/kg bw/day)

Population Group	Age group (Years)	Per Capita Intake (g/kg bw/day)		Consumer-Only Intake (g/kg bw/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Children	2-5	1.3	2.7	99.0	454	1.3	2.7
Children	6-11	0.79	1.6	99.0	669	0.80	1.6
Female Teenagers	12-19	0.36	0.67	98.7	433	0.36	0.68
Male Teenagers	12-19	0.42	0.94	95.8	419	0.44	0.95
Female Adults	20+	0.28	0.63	97.9	2086	0.29	0.65
Male Adults	20+	0.32	0.68	97.4	1889	0.33	0.69
Total Population	2+	0.40	0.91	97.8	5950	0.41	0.93

bw = body weight; g= grams; kg = kilogram; n= sample size;

The percentage of users was high among all evaluated age groups in the intake assessment and was >95% of the population groups for products for which Kluvy® Protein would be incorporated under the proposed uses listed in **Table 10**. It should be noted that the estimated daily intakes are considered conservative, as Kluvy® Protein is unlikely to be incorporated into every proposed food product in the marketplace and/or at the maximum proposed use levels.

Among the total population (ages 2 years and older), the mean and 90th percentile consumer-only intakes of Kluvy® Protein were determined to be 25 g/day and 53 g/day, respectively. Male adults (ages 20 years and older) were determined to have the greatest consumer-only mean intakes of Kluvy® Protein, at 29 g/day, while male teenagers (ages 12-19 years) and male adults (ages 20 years and older) were determined to have the greatest consumer-only 90th percentile intakes of 59 g/day. The lowest consumer-only mean intakes were for children (ages 2-5 years), female teenagers (ages 12-19 years), and female adults (ages 20 years and older), at 21 g/day. The lowest consumer-only 90th percentile intakes were for female teenagers (ages 12-19 years), at 45 g/day.

Among the total population (ages 2 years and older) and on a body weight basis, the mean and 90th percentile intakes of Kluvy® Protein were determined to be 0.41 g/kg bw/day and 0.93 g/kg bw/day, respectively. Children (ages 2-5 years) were determined to have the greatest mean and 90th percentile consumer-only intake at 1.3 g/kg bw/day and 2.7 g/kg bw/day, respectively. The lowest mean and 90th percentile consumer-only intakes were for female adults (ages 20 and older), at 0.29 g/kg bw/day and 0.65 g/kg bw/day, respectively.

Table 13. Estimated Daily Intake of Kluvy® Protein as Protein from the Proposed Uses^a (g/day)

Population Group	Age group (Years)	Per Capita Intake (g/day)		Consumer-Only Intake (g/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Children	2-5	13	29	99.0	462	13	29
Children	6-11	16	30	99.0	671	16	31
Female Teenagers	12-19	13	27	98.7	440	13	28

Population Group	Age group (Years)	Per Capita Intake (g/day)		Consumer-Only Intake (g/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Male Teenagers	12-19	17	37	95.8	422	18	37
Female Adults	20+	13	29	97.9	2104	13	30
Male Adults	20+	18	37	97.4	1903	18	37
Total Population	2+	15	33	97.8	6002	16	33

g= grams; n= sample size;

^a Calculated as: (Estimated daily intake of Kluvy® Protein)*(approximate maximum protein content, 63%)

For comparison to dietary reference intakes for protein, the estimated total intake of protein from consumption of Kluvy® Protein from the proposed uses was calculated (see **Table 13**), assuming a maximum protein content of approximately 63% for Kluvy® Protein, based on representative batch results provided in **Table 6**. The conservative consumer-only mean and 90th percentile estimate of the daily consumption of protein from Kluvy® Protein are therefore ~ 16 g/day and 33 g/day, respectively, for the total population (2 years and older). Of the individual consumer-only population groups, male teenagers (ages 12-19 years) and male adults (ages 20 years and older) were determined to have the greatest mean and 90th percentile intakes of protein, at 18 g/day and 37 g/day, respectively. Children (ages 2-5 years), female teenagers (ages 12-19 years), and female adults (ages 20 years and older) were determined to have the lowest mean consumer-only intakes of protein, at 13 g/day, while female teenagers (ages 12-19 years) were determined to have the lowest 90th percentile consumer-only intakes of protein, at 28 g/day.

Based on the information presented herein, the conservative 90th percentile consumer-only estimate of the daily consumption of Kluvy® Protein from the proposed uses is 53 g/day for the total population (2 years and older). Assuming a maximum protein content of ~63% for Kluvy® Protein, the conservative 90th percentile consumer-only estimate of the daily consumption of protein from the proposed uses of Kluvy® Protein is 33 g/day for the total population (ages 2 years and older).

3.3 Background Intake Levels of Protein

Dietary reference intakes (DRIs) for protein have been established by the Food and Nutrition Board of the Institute of Medicine (IOM) of the National Academy of Science (IOM, 2005), as summarized in **Table 14**.

Table 14. DRIs for Protein Established by IOM

Population Group	Age group (Years)	Estimated Average Requirement (EAR) (g/kg bw/day)	Recommended Dietary Allowance (RDA) (g/day)
Children	1-3	0.87	13
Children	4-8	0.76	19
Children	9-13	0.76	34

Population Group	Age group (Years)	Estimated Average Requirement (EAR) (g/kg bw/day)	Recommended Dietary Allowance (RDA) (g/day)
Female Teenagers	14-18	0.71	46
Male Teenagers	14-18	0.73	52
Female Adults	19+	0.66	46
Male Adults	19+	0.66	56

IOM (2005) noted that there are insufficient data to establish a Tolerable Upper Intake Level (UL) for total protein. Further, while the “risk of adverse effects resulting from excess intake of protein from foods appears to be very low”, IOM notes that data on adverse effects associated with high protein intakes are limited, therefore “caution may be warranted.”

FDA has also established a daily reference value (DRV) for protein of 50 g/day based on a 2,000-calorie diet.³

Based on consumption data obtained from the 2007-2008 NHANES survey, the average daily intake of protein by adult men and women is 100 g and 69 g, respectively (Wright and Wang, 2010, as reported by NIH⁴). More recently, Berryman et al. (2018) evaluated protein intake trends from 2001-2014 using NHANES survey data. Protein intake ranged from an average of ~55 g/day in children ages 2-3 years to ~88 g/day in adults ages 19-30 years. Relative protein intakes were reported to be ~ 3.6 g/kg bw/day for children ages 2-3 years and ~1.1 g/kg bw/day for adults ages 71 years and older. It was noted that relative protein intakes were above the estimated average requirement (EAR) in all demographic groups. Berryman et al. (2018) concluded that the majority of US consumers exceed minimum daily recommendations for protein intake.

Based on the EDI for Kluvy® Protein presented herein, the estimated 90th percentile consumer-only intake of protein from all proposed uses for the total population (i.e., 33 g/day protein from Kluvy® Protein for consumers 2 years and older) is lower than or within the range of the recommended and/or background intake levels of protein in the diet. Furthermore, Kluvy® Protein is intended to substitute for other protein sources in the diet, thus the proposed uses of Kluvy® Protein are not expected to result in an overall increase in protein consumption in the diet. Changing Bio concurs with the conclusion that: “...it is reasonable to expect that most of the U.S. population’s intake of protein is expected to remain in the form of unprocessed foods including meat, poultry, fish, and legumes,” as presented on page 24 in Part 3.2.2 of GRN 001125.

³ See:

https://www.accessdata.fda.gov/scripts/InteractiveNutritionFactsLabel/assets/InteractiveNFL_Protein_October2021.pdf
(Accessed September 19, 2024)

⁴ Available online at: <https://ods.od.nih.gov/factsheets/ExerciseAndAthleticPerformance-HealthProfessional/#protein>
(Accessed September 19, 2024)

3.3.1 Dietary Fiber Intake from Kluvy® Protein

In addition to being a dietary source of protein, Kluvy® Protein also contains ~30% dietary fiber, as noted in **Table 8**. For comparison to dietary reference intakes for dietary fiber, the estimated total intake of dietary fiber from consumption of Kluvy® Protein from the proposed uses was calculated (see **Table 15**), assuming a dietary fiber content of approximately 30% for Kluvy® Protein, based on representative batch results provided in **Table 8**. The conservative consumer-only mean and 90th percentile estimate of the daily consumption of dietary fiber from Kluvy® Protein are therefore ~ 8 g/day and 16 g/day, respectively, for the total population (2 years and older).

Table 15. Estimated Daily Intake of Dietary Fiber from the Proposed Uses of Kluvy® Protein^a
(g/day)

Population Group	Age group (Years)	Per Capita Intake (g/day)		Consumer-Only Intake (g/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Children	2-5	6	14	99.0	462	6	14
Children	6-11	8	14	99.0	671	8	15
Female Teenagers	12-19	6	13	98.7	440	6	14
Male Teenagers	12-19	8	18	95.8	422	8	18
Female Adults	20+	6	14	97.9	2104	6	14
Male Adults	20+	8	18	97.4	1903	9	18
Total Population	2+	7	16	97.8	6002	8	16

g= grams; n= sample size;

^a Calculated as: (Estimated daily intake of Kluvy® Protein)*(approximate dietary fiber content, 30%)

DRIs for dietary fiber have been established by the Food and Nutrition Board of the IOM of the National Academy of Science (IOM, 2005), as summarized in **Table 16**.

Table 16. DRIs for Dietary Fiber Established by IOM

Population Group	Age group (Years)	Total Fiber Adequate Intake (AI) (g/day)
Children	1-3	19
Children	4-8	25
Female Children	9-13	26
Male Children	9-13	31
Female Teenagers	14-18	26
Male Teenagers	14-18	38
Female Adults	19+	21-25
Male Adults	19+	30-38

IOM (2005) noted that there are insufficient data to establish a UL for dietary fiber. IOM (2005) also noted that the median intake of dietary fiber ranges from 16.5 to 17.9 g/day for men and 12.1 to 13.8 g/day for women.

FDA has also established a DRV for dietary fiber of 28 g/day.⁵

Based on consumption data obtained from NHANES surveys from 1999-2008, the mean daily intake of dietary fiber for adults (ages 18 years and older) from 1999-2000 was 15.6 g/day, from 2001-2002 was 16.1 g/day, from 2003-2004 was 15.5 g/day, from 2005-2006 was 15.8 g/day, and from 2007-2008 was 15.9 g/day (King et al., 2012). These values are well below the recommended Adequate Intake (AI) for dietary fiber for adults, as listed in **Table 16**. Furthermore, in 2021, the American Society for Nutrition reported that “only 5% of men and 9% of women are getting the recommended daily amount of dietary fiber.”⁶

The estimated 90th percentile consumer-only intake of dietary fiber from all proposed uses of Kluvy® Protein for the total population (i.e., 16 g/day dietary fiber from Kluvy® Protein for consumers 2 years and older) is lower than the IOM DRIs and FDA DRV's for dietary fiber. Therefore, consumption of Kluvy® Protein would not lead to excessive intake of dietary fiber beyond the recommended daily intake.

3.3.2 Sodium Intake from Kluvy® Protein

In addition to being a dietary source of protein, Kluvy® Protein also contains ~70 mg/kg sodium, as noted in **Table 8**. For comparison to dietary reference intakes for sodium, the estimated total intake of sodium from consumption of Kluvy® Protein from the proposed uses was calculated (see **Table 17**), assuming a sodium content of approximately 70 µg/g for Kluvy® Protein, based on representative batch results provided in **Table 8**. The conservative consumer-only mean and 90th percentile estimate of the daily consumption of sodium from Kluvy® Protein are therefore ~1.75 mg/day and 3.71 mg/day, respectively, for the total population (2 years and older).

Table 17. Estimated Daily Intake of Sodium from the Proposed Uses of Kluvy® Protein^a
(mg/day)

Population Group	Age group (Years)	Per Capita Intake (mg/day)		Consumer-Only Intake (mg/day)			
		Mean	90 th Percentile	%	n	Mean	90 th Percentile
Children	2-5	1.47	3.22	99.0	462	1.47	3.22
Children	6-11	1.75	3.36	99.0	671	1.75	3.43
Female Teenagers	12-19	1.47	3.01	98.7	440	1.47	3.15
Male Teenagers	12-19	1.89	4.13	95.8	422	1.96	4.13
Female Adults	20+	1.47	3.22	97.9	2104	1.47	3.29
Male Adults	20+	1.96	4.13	97.4	1903	2.03	4.13
Total Population	2+	1.68	3.71	97.8	6002	1.75	3.71

mg= milligrams; n= sample size;

⁵ See: <https://www.fda.gov/food/nutrition-facts-label/daily-value-nutrition-and-supplement-facts-labels> (Accessed September 30, 2024)

⁶ <https://nutrition.org/most-americans-are-not-getting-enough-fiber-in-our-diets/> (Accessed September 30, 2024)

^a Calculated as: (Estimated daily intake of Kluvy® Protein)*(approximate sodium content, 70 µg/g)

DRIs for sodium have been established by the Food and Nutrition Board of the Health and Medicine Division of the National Academy of Science (NASEM, 2019), as summarized in **Table 18**.

Table 18. DRIs for Sodium

Population Group	Age group (Years)	Sodium AI (mg/day)
Children	1-3	800
Children	4-8	1000
Female Children	9-13	1200
Male Children	9-13	1200
Female Teenagers	14-18	1500
Male Teenagers	14-18	1500
Female Adults	19+	1500
Male Adults	19+	1500

FDA has also established a DRV for sodium of 2300 mg/day.⁷

Based on consumption data obtained from NHANES surveys from 2009-2012, the Center for Disease Control and Prevention (CDC) reported that 89% of adults and more than 90% of children exceeded recommendations for sodium intake.⁸ The mean daily sodium intake for children (ages 2-3 years) was 2,154 mg/day, for children (ages 4-8 years) was 2,754 mg/day, for children (ages 9-13 years) was 3,126 mg/day, for teenagers (ages 14-18 years) was 3,538 mg/day, and for adults (ages 19 years and older) was 3,552 mg/day.

Based on the EDI for Kluvy® Protein presented herein, the estimated 90th percentile consumer-only intake of sodium from all proposed uses for the total population (i.e., 3.71 mg/day sodium from Kluvy® Protein for consumers 2 years and older) is significantly lower than the recommended intake levels and background intake levels of sodium in the diet, and would therefore provide a negligible increase to total sodium consumption in the diet.

⁷ See: <https://www.fda.gov/food/nutrition-facts-label/daily-value-nutrition-and-supplement-facts-labels> (Accessed September 30, 2024)

⁸ <https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6452a1.htm?platform=hootsuite> (Accessed September 30, 2024)
Anchor Center for Certification (ACC)

4 SELF-LIMITING LEVELS OF USE

No self-limiting levels of use have been identified.

5 EXPERIENCE BASED ON COMMON FOOD USE IN FOOD BEFORE 1958

The statutory basis for Changing Bio's conclusion of the GRAS status of Kluvy® Protein in this document is not based on common use in food before 1958. The GRAS conclusion is based on scientific procedures.

6 NARRATIVE

The data and information provided in this GRAS notice support the conclusion that the notified substance is safe under the conditions of its intended use as described herein. The following subsections cover the history of use of the production strain, the safety of the production strain, the safety of the Kluvy® Protein, and allergenicity information. This information is publicly available, and Part 6 does not contain any confidential information. This section establishes that the notified substance is generally recognized as safe based on the available data and the conditions of its intended use.

6.1 History of Use of *K. marxianus*

K. marxianus was first isolated from grapes (Barone et al., 2021). *K. marxianus* is closely related to the better-known *K. lactis* (Lachance, 1998; Llorente et al., 2000). They are morphologically identical with the exception of their spore shape and other characteristics, most notably the ability to ferment lactose. Their DNA homology, however, is only 15–20%, and as a consequence, they are distinct species (Batt, 2014). Both strains belong to the family Saccharomycetaceae and can assimilate lactose, a trait absent in *Saccharomyces cerevisiae*, making these non-conventional yeasts valuable in the food industry. Additionally, due to its ability to assimilate lactose as a carbon source, *K. marxianus* is commonly found in lactose-rich edible dairy products such as fermented milk, yogurt, and cheeses, and is often referred to as dairy yeast. Recent attempts to characterize the microbial content of fermented milk, that is kefir, find that the major yeast species present in the product is *K. marxianus*, accounting for up to 90% of the yeast population (Kim et al., 2019, Youn et al., 2022). According to the Codex Alimentarius (Codex Stan 243-2003), a typical kefir (fermented milk obtained from kefir grains) should contain not less than 10^4 CFU/mL of the total amount of yeast (including *K. marxianus*). Another study has reported that the population of yeast (including *K. marxianus*) and acetic acid bacteria in kefir are around 10^5 to 10^6 CFU/g (Cho et al., 2018).

Moreover, *K. marxianus* is considered an emerging non-conventional food-grade yeast, possessing a unique set of beneficial traits, such as rapid growth, thermotolerance, and a broad substrate spectrum which makes this yeast a particularly attractive host for applications in a variety of food and biotechnology industries (Bilal et al., 2022). The long history of safe association with food products helped this yeast achieve regulatory acceptance for food ingredient production in the US and European Union (EU), respectively (Fonseca et al., 2008, Lane & Morrissey, 2010).

In 1995, Caballero et al. (1995) assessed two strains of *K. marxianus* (NRRL-Y-2415 and NRRL-Y-1109) as baker's yeasts, finding them to have superior proofing activity compared to commercial strains, which led to their successful use in bread making. Subsequently, *K. marxianus* was integrated into the baker's yeast industry, and various studies explored its applications in food baking. In another study, mixed starter cultures of *K. marxianus* and *Lactobacillus bulgaricus* or *Lactobacillus helveticus* were found to improve the quality, shelf life, and consumer preference of sourdough bread (Plessas et al., 2008). Struyf et al. (2017) found fermentation of dough with an inulinase-secreting *K. marxianus* strain allowed the reduction of fructan levels in the final product by more than 90%, while only 56% reduction was achieved when a control *Saccharomyces cerevisiae* strain was used. The use of *K.*

marxianus yeast was found to be more effective than *S. cerevisiae* in reducing the oligo-, di-, and monosaccharides and polyols (FODMAPs) in wheat bread.

The broad and longstanding use of *K. marxianus* in the food industry, paired with regulatory acceptance for safe food ingredient production by the US and EU as safe (Fonseca et al., 2008, Lane & Morrissey, 2010), points to the common food use of *K. marxianus* and thus demonstrates the safe application of *K. marxianus* in the production of food for human consumption.

6.2 Regulatory History of *K. marxianus*

As mentioned, the use of *K. marxianus* to produce food ingredients has been reviewed by the regulatory bodies in the US and EU (Fonseca et al., 2008, Lane & Morrissey, 2010). *K. marxianus* was used to produce a lactase enzyme preparation as described in GRN 000088. The use of this species as the production organism was reviewed by the FDA and GRN 000088 had received “no questions” letter from the FDA on April 3, 2002. Other than in the US, the safety of *K. marxianus* has also been reviewed by EFSA, an EU regulatory body. *K. marxianus* is included in the current QPS list, version v20 (EFSA BIOHAZ, 2024). The QPS assessment process was first outlined by the Scientific Committee in 2007, establishing the initial list of biological agents. This list is reviewed and updated annually by EFSA's Panel on Biological Hazards (BIOHAZ). This assessment covers *K. marxianus*'s use in enzyme production and novel foods, indicating that the organism's safety has already been pre-assessed. Consequently, no additional toxicological data are required to establish its safety in the European Union.

6.3 Safety of Using *K. marxianus*

Kluyv® Protein is produced from a non-genetically modified, food-derived strain of *K. marxianus* CJ3113. This strain is fermented and inactivated, and the resulting biomass is collected. The final product (i.e., yeast protein) is designed to serve as a protein substitute for both animal and plant proteins. As mentioned in **Part 6.2**, the use of this species to produce food ingredients has already been reviewed by the US FDA and EFSA and has no concerns on its safe use. Other supporting evidence of safety is also included in the following section to further demonstrate the safe use of *K. marxianus* and in turn, justify the safety of yeast protein derived from it, i.e., Kluyv® Protein.

A literature search was conducted on September 1, 2024, using PubMed and Google Scholar to identify relevant safety studies. The search terms “*Kluyveromyces marxianus*” or “*K. marxianus*” were used to identify published toxicology and clinical studies. Studies conducted in toxicology-relevant model organisms or those reporting safety parameters such as anthropometric measures, clinical chemistry, hematology, and adverse events in the abstract were considered relevant and were subsequently summarized.

Most of the studies discussed in the following sections examined strains of *K. marxianus* isolated from either food or clinical sources. As a result, the genetic sequences of these strains are not publicly available for direct genetic homology comparison with the CJ3113 strain. Considering that the CJ3113

strain has been concluded to belong to the *K. marxianus* species (**Part 2.2.1**), is food-derived, and has not undergone genetic modification, it is reasonable to assume that it shares similar characteristics with other *K. marxianus* strains. In summary, the safety data reviewed here can be used to support the safe use of *K. marxianus* CJ3113.

6.3.1 Pathogenicity and Toxigenicity

The results of *in vitro* (gelatin hydrolysis, pseudohyphae formation, hemolysis) and *in vivo* (mammalian model, invertebrate model) experiments of *K. marxianus* are discussed in the literature to assess the pathogenicity and toxigenicity of different sources of *K. marxianus*. These comprehensive analyses demonstrate that *K. marxianus* from various sources is considered safe, supporting its use in food and other applications.

6.3.1.1 *In vitro* Studies with *K. marxianus*

6.3.1.1.1 Gelatin Hydrolysis

Gelatinase is an enzyme that degrades a wide range of host substrates, including collagen, hemoglobin, fibrinogen, fibrin, and other peptides that are crucial components of host tissues (Vergis et al., 2002). It also plays a role in the translocation of pathogenic bacteria across intestinal cell layers (Zeng et al., 2005). Therefore, testing for gelatin hydrolysis can indicate the presence of gelatinase and help assess the safety of *K. marxianus*. The results of such tests for *K. marxianus* strains are summarized in **Table 19**, demonstrating that this yeast does not exhibit gelatinase activity, thus supporting its safety.

Table 19. The Gelatin Hydrolysis of *K. marxianus*

Gelatin hydrolysis		
Study strain	Study Details and Results	Reference
<i>K. marxianus</i> strains, Origin: Kefir	Method: - A single colony of the thirteen isolated strain was stab-inoculated into gelatine-containing medium (120 g/L gelatine, 5 g/L peptone, and 3 g/L beef extract, Smith and Goodner 1958) - After 48 h, the cultures were placed into an ice bath for 30 mins and then observed for zones of clearance. <i>Bacillus megaterium</i> was used as a positive control. Result: <i>K. marxianus</i> showed no gelatinase activity, suggesting that their virulence and, accordingly, their potential to cause infections are low.	Lim et al., 2019
<i>K. marxianus</i> A4 and A5 Origin: Korean Kefir.	Method: <i>K. marxianus</i> A4 and A5 were spiked into tubes containing nutrient gelatin - The inoculated and control tubes were incubated at 37 °C for up to 1 week while being examined daily for gelatin liquefaction.	Youn et al., 2023

Gelatin hydrolysis		
Study strain	Study Details and Results	Reference
	The tubes were taken out from the incubator and tilted at 4 °C for 15-30 min. Then, the tubes were observed visually whether the gelatin was hydrolyzed. Result: <i>K. marxianus</i> A4 and A5 could not degrade gelatin.	

The above studies indicate that *K. marxianus* does not exhibit gelatin hydrolysis ability and, to date, Changing Bio has identified no literature reporting gelatinase activity in *K. marxianus*.

6.3.1.1.2 Pseudohyphae Formation

Pseudohyphae formation is a key indicator of pathogenic potential, as these filamentous structures hinder effective phagocytosis by recruiting neutrophils, monocytes, and cytokines to infection sites (Shoham & Levitz, 2005). This formation allows yeast cells to invade host tissues by penetrating the solid matrix of host cells, particularly affecting individuals with weakened immune systems (Lim et al., 2019). For instance, *Candida albicans* enhances its invasiveness and virulence through hyphal formation, which improves tissue penetration and evasion of phagocytosis. Given that *Candida albicans* is a member of the same family as *K. marxianus*, investigating pseudohyphae formation in *K. marxianus* helps assess its potential pathogenicity. The pseudohyphae formation ability of *K. marxianus* is detailed in **Table 20**.

Table 20. The Pseudohyphae Formation of *K. marxianus*

Pseudohyphae formation		
Study strain	Study Details and Results	Reference
<i>K. marxianus</i> strains, Origin: Kefir	Method: A single colony of each yeast strain was smeared on glass slides, covered with 10 µL of 10% KOH solution for 10 min, gently fixed by heating over a flame, and observed under a light microscope. Result: <i>K. marxianus</i> strains showed positive results for pseudohyphae formation, as evidenced by the KOH test.	Lim et al., 2019

Pseudohyphae formation		
Study strain	Study Details and Results	Reference
<i>K. marxianus</i> strains, Origin: 19 strains were obtained from hospitals, and the remaining strains were isolated from natural habitats and foods.	Method: - Synthetic low ammonia dextrose media (SLAD) was used to assay pseudohyphal growth. - Strains were streaked on SLAD plates and colonies features were observed and photographed after 4 days of growth at 30 °C. Result: 89% of the clinical strains were able to exhibit pseudohyphal growth and no significant differences were found between the clinical and non-clinical groups regarding the capability of agar invasion. - The majority of the strains could grow at 42 °C, showed high pseudohyphal growth, and showed invasive growth.	Pérez-Través et al., 2021
<i>K. marxianus</i> A4 and A5 Origin: Korean Kefir.	Method: - A single colony of each yeast strain was smeared on a glass slide and covered with 10 µL of 10% potassium hydroxide for 10 min. - slides were covered with a cover glass and observed under a light microscope. Result: <i>K. marxianus</i> A4 and A5 can form pseudohyphae.	Youn et al., 2023

Studies have shown that *K. marxianus* can form pseudohyphae, a feature associated with pathogenicity. Certain strains of *K. marxianus* have demonstrated increased adhesion capabilities, which may pose a risk to immunosuppressed individuals, potentially akin to the infection mechanisms of *C. albicans*, which also form pseudohyphae to invade host tissues. However, the specific mechanisms by which *K. marxianus* induces toxicity through pseudohyphae formation have not been reported. Moreover, the subject of the current GRAS determination contains only the heat-inactivated form of *K. marxianus*, which would not form pseudohyphae, and therefore the risk of pathogenicity associated with pseudohyphae formation is negligible.

6.3.1.1.3 Hemolysis

Hemolysis, the rupture and dissolution of red blood cells, can be triggered by various physicochemical factors and toxins. It results in decreased red blood cell (RBC) counts, hemoglobin levels, mean platelet volume, and platelet levels, while also affecting plasma biochemical parameters by reducing creatinine levels and increasing aspartate transaminase (AST), alanine aminotransferase (ALT), and total bilirubin levels (Koseoglu et al., 2011). If a microorganism used in fermented foods exhibits hemolytic

activity, it may pose a risk to human health (Liu et al., 2021). The hemolysis of *K. marxianus* from different sources is summarized in Table 21.

Table 21. The Hemolysis of *K. marxianus*

Hemolysis		
Study strain	Study Details and Results	Reference
<i>K. marxianus</i> strains, Origin: Kefir	Method: ♦ <i>K. marxianus</i> strains grown for 24 hrs were streaked onto blood agar (bioMerieux) ♦ Incubated for 48 hrs at the temperature optimal for the growth of each strain, and then observed. ♦ <i>Staphylococcus aureus</i> ATCC 6538 was used as a positive control. Result: None of the <i>K. marxianus</i> strains showed hemolytic activity.	Lim et al., 2019
<i>Kluyveromyces marxianus</i> A4 and A5 Origin: Korean Kefir.	Method: ♦ <i>K. marxianus</i> strains were grown on PDA (Oxoid) at 30 °C for 24 h. ♦ The grown colony was sub-cultured on Columbia agar with 5% sheep blood (bioMerieux) ♦ Incubated at 37 °C for 48 hrs, then observed for the zone of hemolytic reactions. ♦ <i>S. aureus</i> ATCC 25923 was the positive control for hemolysis experiments. Result: <i>K. marxianus</i> A4 and A5 demonstrated no hemolytic activity.	Youn et al., 2023
<i>K. marxianus</i> JY-1 Origin: traditional fermented milk in Xinjiang District	Method: ♦ The <i>K. marxianus</i> JY-1 was strewed on the Columbia agar medium (mixed with 5 % sheep blood) ♦ Incubated at 28 °C for 48 hrs to observe if there was a clear circle around the colony. ♦ <i>Listeria monocytogenes</i> was used as a positive control and <i>Lactobacillus rhamnosus</i> GG was used as a negative control for hemolysis analysis. Result: <i>K. marxianus</i> JY-1 was not hemolytic.	Li et al., 2023

The above studies showed that *K. marxianus* does not exhibit hemolytic properties, and hence safety risk associated with hemolytic properties does not apply to *K. marxianus*.

6.3.1.2 *In vivo* Studies with *K. marxianus*

6.3.1.2.1 *In vivo* Mammalian Model

To evaluate the safety of *K. marxianus*, *in vivo* tests have been conducted in mice, with the results presented in Table 22.

Table 22. *In vivo* Mammalian Model of *K. marxianus*

<i>In vivo</i> of Mammalian Model		
Study Details	Study Methods and Results	Reference
<i>K. marxianus</i> A4 and A5 Origin: Korean Kefir. Study Design: <i>In vivo</i> Study Length: 2 weeks Animals: Specific pathogen-free BALB/c mice (n = 48; 4 weeks old; female) Dose/Concentration: 200 µL of different concentrations Delivery/Vehicle: oral with plastic syringe/saline	Methods: - The mice, weighing approximately 16-19 g, were randomly assigned to eight experimental groups, each containing six mice. - Mice were orally administered 0.2 mL of 0.9% sterilized saline (NC_S; n = 6), <i>Saccharomyces boulardii</i> ATCC MYA-796 (high concentration, S.b_H; low concentration, S.b_L; n = 6 for each), <i>K. marxianus</i> A4 (high concentration, A4_H; low concentration, A4_L; n = 6 for each), or <i>K. marxianus</i> A5 (high concentration, A5_H; low concentration, A5_L; n = 6 for each) for 2 weeks. - The spleen and liver were sampled to measure weight and length. - Complete blood count tests were performed using whole blood samples to screen the signs of various systemic abnormalities. - Whole blood samples (250 µL) collected from each group were used to measure RBC counts, hemoglobin, hematocrit, mean platelet volume, platelet counts, and white blood cell (WBC) counts with differentiation of neutrophils, lymphocytes, monocytes, eosinophils, and basophils. 150 µL of plasma was collected from each mouse, of which a part (50 µL) was used to measure plasma biochemical parameters, and another part (75 µL) was used to evaluate the levels of proinflammatory cytokines. For plasma biochemical parameters, AST, ALT, total bilirubin, blood urea nitrogen (BUN), creatinine, gammaglutamyl transferase (GGT), glucose, total protein, and total cholesterol were measured mainly to detect liver and kidney abnormalities and metabolic disorders. - examine systemic inflammatory responses, the levels of proinflammatory cytokines, such as interleukin (IL)-1β, IL-6, and tumor necrosis factor-alpha (TNF-α) Result: - All mice showed normal appearance and behavior throughout the whole experimental period. No significant differences were observed in initial body weight among the eight groups.	Youn et al., 2023

In vivo of Mammalian Model		
Study Details	Study Methods and Results	Reference
	- No increase of the weight and length of the spleens and livers of mice were found from different treatment groups. - All biochemical values of A4 and A5 groups were comparable to those of the NC group, indicating that the short-term administration of <i>K. marxianus</i> A4 and A5 showed no adverse effects on the liver and kidney functions and did not have a deteriorating effect on the systemic metabolic activity in healthy mice.	

6.3.1.2.2 In vivo Invertebrate Model Using *Galleria mellonella*

Larvae of the greater wax moth, *Galleria mellonella*, serve as an effective *in vivo* model for evaluating the activity and toxicity of antimicrobial agents and for studying immune responses to pathogens. Their results often correlate with those from mammalian models, making them valuable for identifying and assessing novel antimicrobial agents (Piatek et al., 2021). Since fungal burden is a key marker for virulence in *G. mellonella*, the *in vivo* results for *K. marxianus* using this invertebrate model are presented in Table 23.

Table 23. In vivo Invertebrate Model Using *G. mellonella* of *K. marxianus*

In vivo Invertebrate Model using <i>G. mellonella</i>		
Study Details	Study Methods and Results	Reference
Strain: <i>K. marxianus</i> strains Origin: hospitals, natural habitats and foods. Study Design: <i>In vivo</i> Study Length: 72 hours Animals: n = 20 Six instar larvae (weight 0.22 ± 0.03 g) Dose/Concentration: 20 µL of <i>K. marxianus</i> suspension/10 ⁸ , 10 ⁷ , and 10 ⁶ Delivery/Vehicle: Myjector U-100 insulin syringe /PBS Study index: Larvae survival	Methods: - Groups of 20 larvae were each inoculated with 20 µL of either a 5 × 10⁹, 5 × 10⁸, or 5 × 10⁷ yeast suspension (final inoculum concentration of 10⁸, 10⁷, and 10⁶ yeast cells/larval) in phosphate-buffered saline (PBS) buffer. As a control, 20 larvae were injected with PBS alone. - Each larva was inoculated through the last left pro-leg into the haemocoel with a Myjector U-100 insulin syringe. - Larvae were acclimatized at 30 °C for 1 hr before undertaking all experiments and incubated at 30 °C in the dark post-challenge. - Survival was recorded at 24, 48, and 72 hrs. Larvae survival was confirmed visually and by physical stimulation with a sterile pipette tip. Result: <i>K. marxianus</i> showed high survival, 80% for AQ2934 and AQ1482 and 60% for AQ1411, except for AQ2942 with a survival rate of 20%. (AQ2934, AQ1482, AQ2942 from different origin) - For the <i>K. marxianus</i> strains it was observed that an inoculum of 10⁸ cells/larva is the optimal dose for a 72-hr experiment.	Pérez-Través et al., 2021
Dose/Concentration: 20 µL of <i>K. marxianus</i>	Methods: Fungal burden (CFU/ml) in larvae was determined at 24, 48, and 72	

In vivo Invertebrate Model using <i>G. mellonella</i>		
Study Details	Study Methods and Results	Reference
suspension/ 10^8 Study index: Fungal burden (CFU/ml) in larvae	hrs post-challenge after inoculation with a final dose of 1×10^8 <i>K. marxianus</i>. - Three inoculated larvae from each treatment group were homogenized in 5 mL of sterile PBS using a stomacher digester (60s on max). - Ten-fold dilutions of the resulting homogenate were made in PBS and 100 μL from each was plated onto Sabouraud Dextrose Agar (SDA) plates containing chloramphenicol (50 mg/ml). - Plates were incubated at 30 °C for 48 hrs and the CFU/ml was determined for each strain. Result: The four <i>K. marxianus</i> strains were able to maintain high levels of fungal cells in the larvae during 3 days of incubation, but only strain AQ1411 demonstrated some degree of proliferation in the larvae over 72 hrs.	

The virulence of *K. marxianus* in *G. mellonella* correlates well with the physiological and epithelial cell model factors studied, which suggests a potential risk of these strains in the food industry. However, the conclusion on the potential pathogenicity of *K. marxianus* cannot be determined based on the outcome of a single study. Importantly, the subject of the current GRAS determination is Kluby® Protein, which does not contain live *K. marxianus* strain, and hence the safety risk associated with the potential virulence of live *K. marxianus* strain is negligible. Moreover, the commonly used Baker's yeast, *Saccharomyces cerevisiae*, was also reported to be an emerging opportunistic pathogen (Pérez-Torrado & Querol, 2016) and yet remains well accepted in the food industry as safe for use.

6.3.1.3 Cytotoxicity of *K. marxianus*

As epithelia serve as the primary barrier against infection, understanding the interaction between potential pathogens and epithelial cells is crucial. Adhesion to epithelial cells often represents the initial step in the pathogenesis of various species. Additionally, the ability of strains to breach the human epithelial barrier and inflict damage on epithelial cells is a significant factor in determining their infectious potential (Pérez-Través et al., 2021). The cytotoxicity of *K. marxianus* is detailed in **Table 24**.

Table 24. Cytotoxicity Study of *K. marxianus*

Antibiotic sensitivity		
Study strain	Study Details and Results	Reference
<i>K. marxianus</i> strains (AQ1411, AQ1482, AQ2934, AQ2942) Origin: clinical and foods.	Method: The adherence level to human cells was tested as well as the transcytosis ability, the epithelial integrity by measuring the trans-epithelial electrical resistance (TEER) and the cellular damage by measuring the lactate dehydrogenase (LDH) levels, a compound	Pérez-Través et al., 2021

Antibiotic sensitivity		
Study strain	Study Details and Results	Reference
	released by dead cells, using a Caco-2 monolayer. Result: • <i>K. marxianus</i> strains did not affect monolayer integrity during the 48 hrs of the assay. • The monolayers in contact with strains AQ1482, AQ2934, and AQ2942 released higher levels of LDH activity (27.8–29.1%), producing significantly higher cytotoxicity levels which suggests that cytotoxicity can be related to the pathogenicity of this species, although with statistically lower levels than pathogenic <i>C. albicans</i> SC5314 (91.85%). • All the evaluated <i>K. marxianus</i> strains (except for one strain) are positive in most physiological virulent factors, and also in the epithelial cell culture parameters	

The study indicates that certain strains of *K. marxianus* may pose a potential risk to human health, with food-environmental strains potentially serving as sources of infection. Although no confirmed cases of disease caused by *K. marxianus* have been reported, the organism has been isolated from infection sites. However, it is important to note that the current subject of the GRAS determination is Kluvy® Protein in which the production yeast *K. marxianus* has been effectively heat-killed. Therefore, there is no residue of live *K. marxianus* in the final product and hence the safety concerns arising from the potential cytotoxicity of live *K. marxianus* mentioned above are negligible.

Cumulatively, *K. marxianus* generally are relatively safe and harmless for the general population. Nonetheless, similar to other rare-opportunistic pathogens, populations with compromised immunity may unfortunately be at risk of infection with *K. marxianus*. However, the patients generally recovered from infection with treatment using antifungal medications (Reuter et al., 2005, Seth-Smith et al., 2020, Spiliopoulou et al., 2022).

6.3.2 Clinical Cases of *K. marxianus*

In the recent update of the QPS list, the EFSA Panel reviewed six recent scientific articles (Aboueldahab et al., 2023, Afsarian & Sharafi, 2023, Al-Manei et al., 2023, Bektaş et al., 2023, Schwarz et al., 2022, Sig et al., 2023) that contributed relevant clinical information with regard to *K. marxianus*. Three of the studies reported clinical isolates of *K. marxianus* (Aboueldahab et al. 2023; Bektaş et al. 2023; Sig et al. 2023). However, these studies utilized conventional morphological or physiological identification methods and noted predisposing factors. Al-Manei et al. (2023) used matrix-assisted laser desorption ionization-time of flight mass spectrometry (MALDI-TOF MS) species identification and observed *K. marxianus* in the oral microbiome of cancer patients; however, it is uncertain whether the yeasts caused the infection. Afsarian & Sharafi (2023) isolated *K. marxianus* from infected nails of 51 human subjects in Iran, reporting only superficial fungal infections without

additional illness information. These new studies reported that *K. marxianus* may cause opportunistic or superficial infections in rare cases. Despite all these reports, EFSA still included *K. marxianus* in the updated QPS list, thus affirming its safe status for human food use.

The ingredient that is the subject of this dossier is *K. marxianus* protein, with no live *K. marxianus* present. The absence of live *K. marxianus* is assured by autoclave efficiency verification and confirmation of final product specifications. In both cases, the method used is a general method in FDA's BAM Chapter 18, which has been verified to be suitable for detecting and culturing *K. marxianus*. Even when using a general method to test the presence of live yeasts, the results from five non-consecutive batches demonstrate the absence of any live yeast strains (shown in **Table 6**). Moreover, the accelerated study of Kluvy® Protein also showed no presence of live yeast strains up to 6 months at $40\pm 2^{\circ}\text{C}$ and $75\pm 5\%$ RH. These data confirm that there is no residual live *K. marxianus* present in the final Kluvy® Protein product and thus there is no risk of accidental consumption of live *K. marxianus* that may lead to fungal infections.

6.4 Safety of Kluvy® Protein

In **Part 2** of this GRAS notification, Kluvy® Protein is identified as a source of protein, with a protein content of up to 55%. The safe use of this substance is justified based on: 1) Safe use of production strain *K. marxianus*; 2) Safe manufacturing process; 3) Allergenicity evaluation; and 4) Absorption, distribution, metabolism, and excretion profile.

6.4.1 Safe Use of Production Strain

The safety of *K. marxianus* as a production strain is detailed in Part **6.1**, **6.2**, and **6.3**. Moreover, the specific strain used in this dossier, *K. marxianus* CJ3113, has not undergone genetic modification. Therefore, concerns related to the safety of genetic engineering do not apply.

The safety of *K. marxianus* CJ3113 was assessed according to the guidelines developed by Pariza et al. (2015), which are widely accepted by the scientific community and regulatory agencies as criteria for assessing the safety of microbial cultures for consumption by humans and animals. The decision tree for determining the safety of microbial cultures for consumption by humans and animals by Pariza et al. (2015) was applied to evaluate the safety of Changing Bio's production organism *K. marxianus* CJ3113. Utilization of the Pariza et al. (2015) decision tree resulted in the following conclusion regarding *K. marxianus* CJ3113: "The test article is accepted." The decision tree assessment is as follows:

1. Has the strain been characterized for the purpose of assigning an unambiguous genus and species name using currently accepted methodology? (If YES, go to 2. If NO, the strain must be characterized and unambiguously identified before proceeding).

Answer: YES

2. Has the strain genome been sequenced? (If YES, go to 3. If NO, the genome must be sequenced before proceeding to 3.)

Answer: YES.

3. Is the strain genome free of genetic elements encoding virulence factors and/or toxins associated with pathogenicity? (If YES, go to 4. If NO, go to 15.)

Answer: YES

4. Is the strain genome free of functional and transferable antibiotic resistance gene DNA? (If YES, go to 5. If NO, go to 15.)

Answer: YES

5. Does the strain produce antimicrobial substances? (If NO, go to 6. If YES, go to 15.)

Answer: NO

6. Has the strain been genetically modified using rDNA techniques? (If YES, go to 7a or 7b. If NO, go to 8a or 8b.)

Answer: NO

8a. For strains to be used in human food: Was the strain isolated from a food that has a history of safe consumption for which the species, to which the strain belongs, is a substantial and characterizing component (not simply an 'incidental isolate')? (If YES, go to 9a. If NO, go to 13a.)

Answer: YES

9a. For strains to be used in human food: Has the species, to which the strain belongs, undergone a comprehensive peer-reviewed safety evaluation and been affirmed to be safe for food use by an authoritative group of qualified scientific experts? (If YES, go to 10a. If NO, go to 13a.)

Answer: YES. *K. marxianus* is listed in the QPS list by EFSA.

10a. For strains to be used in human food: Do scientific findings published since completion of the comprehensive peer-reviewed safety evaluation cited in question 9a continue to support the conclusion that the species, to which the strain belongs, is safe for use in food? (If YES, go to 11a. If NO, go to 13a.)

Answer: YES

11a. For strains to be used in human food: Will the intended use of the strain expand exposure to the species beyond the group(s) that typically consume the species in “traditional” food(s) in which it is typically found (for example, will a strain that was isolated from a fermented food typically consumed by healthy adults be used in food intended for an 'at risk' group)? (If NO, go to 12a. If YES, go to 13a.)

Answer: NO

12a. For strains to be used in human food: Will the intended use of the strain expand the intake of the species (for example, increasing the number of foods beyond the traditional foods in which the species

typically found, or using the strain as a probiotic rather than as a fermented food starter culture, which may significantly increase the single dose and/or chronic exposure)? (If NO, go to 14a. If YES, go to 13a.)

Answer: NO. The subject of the current GRAS determination is Kluvy® Protein, which is derived from the fermentation of *K. marxianus*. The final product does not contain live *K. marxianus* and therefore does not increase the intake of the live species beyond the traditional foods in which it is typically found.

14a. The strain is deemed to be safe for use in the manufacture of food, probiotics, and dietary supplements for human consumption.

In conclusion, *K. marxianus* CJ3113 used as Changing Bio's production organisms for Kluvy® Protein is considered safe for use.

6.4.2 Safety of Manufacturing Process

Kluyv® Protein is manufactured via fermentation using *K. marxianus*, followed by extractions, heat inactivation, and drying. The entire manufacturing process is conducted in strict compliance with cGMP. Changing Bio's GMP programs ensure that yeast protein is produced under strict sanitary conditions, with thorough evaluation and management of potential hazards associated with raw materials, processing aids, and each manufacturing step. All raw materials used in the fermentation process are neither major allergens nor derived from major allergens and are of food-grade specification. Additionally, all equipment is meticulously designed, constructed, operated, cleaned, and maintained to prevent contamination by undesired microorganisms. The purity of the production yeast is also verified at multiple points of manufacturing, ensuring no cross-contamination with other microorganisms.

The HACCP plan identifies autoclaving as a CCP to ensure that the microbial content of the autoclaved fermentation broth is below acceptable limits and that live *K. marxianus* strains are absent in the final product. This efficiency of the autoclave step has been verified (see discussion in **Section 2.2.3**) and the absence of live *K. marxianus* is confirmed via specifications of yeast count. Before Kluyv® Protein is batch released, it must meet rigorous testing as per the specifications. The specification designed not only characterizes the substance, but the microbiological and heavy metal limits ensure that the final product does not carry any unwanted impurities/contamination, exceeding the regulatory limits for consumers. Therefore, the safety and quality of Changing Bio's Kluyv® Protein are always verified before it reaches the consumers.

6.4.3 Allergenicity Evaluation of Yeast Protein

Evaluating the potential for a new food protein to cause allergic reactions is a crucial part of safety assessment. To examine the allergenic potential posed by Kluyv® Protein, a biomass containing a mixture of proteins, Changing Bio sponsored an analysis of its protein product and requested Dr Richard E. Goodman, Ph.D., to perform an allergenicity evaluation. The evaluation has been concluded,

and a report prepared on September 20, 2024 is attached in Appendix 5.

First, the product's nutritional properties were reviewed and concluded that the results appear reasonable as a protein source. A literature search for *K. marxianus* for allergy or allergenicity found only two references, and they did not show risks, only the benefits of probiotics (Tullio, 2024; Ameli and Ciprandi, 2019). There are no peer-reviewed publications reporting food allergies from this yeast species.

Kluyv® Protein samples were analyzed via LC-MS/MS to characterize the protein composition. The top 100 proteins that were identified are summarized in Appendix 6 (Table 3). According to the proportion of identified proteins, the top 100 yeast proteins with an abundance of more than 0.24% were screened out. The identified protein accounted for 79.21% of the total yeast protein, and the remaining proteins were low-abundance proteins with an abundance of less than 0.24%.

All proteins identified by LC-MS/MS were compared to the major allergenic proteins in the eight priority allergen sources, including peanuts, tree nuts, egg, milk, crustacean shellfish, fish, soybean, and wheat, and did not identify any significant matches in [AllergenOnline](#) databases or other databases. The absence of reported cases of allergy in the literature suggests that the likelihood of food allergy from Kluyv® Protein is low.

6.4.3.1 Adverse Events Related to Allergenicity of Kluyv® Protein

On September 19, 2024, searches in government databases such as MedWatch, FDA's Recalls, Market Withdrawals, & Safety Alerts, and the FDA's Adverse Event Reporting Program (FAERS) using the terms "*Kluyveromyces marxianus*" and "allergic", "*Kluyveromyces marxianus*" and "allergy", "*Saccharomyces marxianus*" and "allergic", "*Saccharomyces marxianus*" and "allergy", "*Candida kefir*" and "allergic", "*Candida kefir*" and "allergy" yielded no report of adverse effect, especially allergenicity. Therefore, allergic diseases or allergenic reactions resulting from *K. marxianus* ingestion, and in turn, Kluyv® Protein ingestion, are highly unlikely.

6.4.4 Absorption, Distribution, Metabolism, and Elimination Profile

No information on the absorption, distribution, metabolism, and elimination (ADME) of Kluyv® Protein (*K. marxianus* protein) was identified in the literature search. However, Kluyv® Protein is majorly composed of >55% protein and approximately 30% dietary fiber. The ADME profile of Kluyv® Protein could be extrapolated by analyzing the ADME of these two major portions.

The majority of dietary proteins are fully degraded and absorbed in the small intestine: after a meal, proteins are denatured by acid and hydrolyzed by gastric pepsin in the stomach, further hydrolyzed by pancreatic proteases, subsequently degraded by small intestinal enterocyte membrane exopeptidases and absorbed across the small intestinal enterocytes into the bloodstream as individual amino acids for use in the body (Dallas et al., 2017). It is expected that Kluyv® Protein would undergo similar ADME pathways as other protein products, providing all sources of amino acids to the body.

On the other hand, fermentation using *K. marxianus* has been reported to generate angiotensin-converting enzyme (ACE) inhibitory peptides and antioxidant peptides in certain fermentation conditions (García-Tejedor et al., 2014, Mirzaei et al., 2018). These peptides are reported to improve the quality of the end products by enriching the nutritive value and enhancing the flavor characteristics.

Current studies on bioactive peptides in fermented foods primarily focus on general screening and identification. After this, the identified peptides are synthetically produced for further characterization of their biological activity. Quantification of bioactive peptides from food sources has not been studied extensively. Since these bioactive peptides naturally occur in foods, people have already been exposed to them, and to date, no adverse effects have been reported from their consumption. In the case of Kluvy® Protein, it is a yeast protein derived from the fermentation of *K. marxianus*. During the manufacturing process, *K. marxianus* is heat inactivated via an autoclaving step (105 – 125°C, 0.1 – 0.15 MPa, for 15 – 30 mins). As a result, the yeast loses all metabolic activity and is unable to produce any proteins or peptides. Although bioactive peptides may be produced during fermentation, heat treatment can cause protein denaturation, leading to a loss of their biological activity. Therefore, the likelihood of bioactive peptides being present in Kluvy® Protein is low.

Most fiber-containing foods include approximately one-third soluble and two-thirds insoluble fiber (Wong & Jenkins, 2007), including Kluvy® Protein. Therefore, it is expected that Kluvy® Protein would undergo a similar ADME pathway as most fiber-containing foods. Dietary fiber can be broadly categorized according to its solubility into two major types of fiber: water-insoluble/less fermented fibers – cellulose, hemicellulose, lignin; and water-soluble/well-fermented fibers – pectin, gums, and mucilages. Both soluble and insoluble fibers are not digested in the human gastrointestinal tract. Soluble fiber absorbs water to become a gelatinous, viscous substance and is fermented by bacteria in the digestive tract. Insoluble fiber has bulking action, is not fermented, and is excreted in feces without any absorption. Given the lack of metabolism and absorption, insoluble fiber is unlikely to cause any adverse effects. On the contrary, soluble fiber can be readily and quickly metabolized by gut bacteria, which could significantly influence the abundance and diversity of the human gut microbiota (McRorie & McKeown, 2016).

6.4.5 Dietary Intake of Kluvy® Protein

Based on dietary intake data, the estimated protein intake for consumers in the 90th percentile from the proposed uses of Kluvy® Protein is conservatively calculated to be 33 g/day for the total population aged 2 years and older. This intake falls below or within the recommended and/or background dietary protein intake (see **Part 3.3**). Additionally, Kluvy® Protein is intended as a protein substitute for other protein sources in the diet and would not contribute to an overall increase in protein consumption in the diet. Therefore, the risk of adverse effects from excessive protein intake due to consumption of Kluvy® Protein is not relevant.

As Kluvy® Protein consists of ~30% dietary fiber, the estimated dietary fiber intake for consumers in the 90th percentile from the proposed uses is conservatively calculated to be 16 g/day for total population aged 2 years and above. As discussed in **Part 3.3.1**, the total fiber adequate intake for

children and adults ranges between 19 – 38 g/day. Therefore, consumption of Kluvy® Protein is not expected to result in dietary fiber intake exceeding the recommended daily amount, making the risk of adverse effects from excessive fiber intake irrelevant.

According to the nutritional composition results in **Table 8**, Kluvy® Protein contains ~70 mg/kg sodium. The estimated mean and 90th percentile sodium intake from Kluvy® Protein was conservatively calculated to be approximately 1.75 mg/day and 3.71 mg/day, respectively, for the total population aged 2 years and older (See **Part 3.3.2**). This estimated sodium intake from Kluvy® Protein is significantly lower than the recommended intake levels and background intake levels of sodium in the diet. As a result, it would contribute only a negligible increase to overall sodium consumption. Therefore, the risk of adverse effects from excessive sodium intake due to consumption of Kluvy® Protein is not applicable.

6.5 Summary Basis for GRAS Determination

The intended use of Kluvy® Protein produced from the fermentation of *K. marxianus* CJ3113 in selected conventional foods is determined to be safe based on the following:

- Kluvy® Protein is derived from *K. marxianus* CJ3113, which was isolated from a food source. *K. marxianus* is a yeast known for its extensive use in fermentation of food and dairy products. Polyphase identification analysis of the CJ3113 strain confirms that it belongs to the *Kluyveromyces marxianus* species and is closely related to *K. marxianus* CBS 712^T.
- Kluvy® Protein is manufactured according to cGMP using processing aids and food contact substances that conform to the conditions of use specified in Title 21 of the United States Code of Federal Regulations and/or Food Chemicals Codex specifications.
- Process procedures and product specifications are in place to control pivotal quality attributes that ensure a consistent, safe, food-grade finished product.
- The absence of live production yeast *K. marxianus* is ensured in all batches via specification monitoring of yeast count, which has been verified to be suitable for intended use.
- Potential contaminants, such as heavy metals and microbes, are either absent (not detected or below method detection limits) or below toxicological and regulatory limits.
- Kluvy® Protein has a similar amino acid composition to other dietary protein sources and therefore is a suitable dietary protein substitute.
- Kluvy® Protein produced from the fermentation of *K. marxianus* CJ3113 is generally recognized as safe to be consumed at the intended use level. The mean and 90th percentile consumer-only intakes of the total population aged 2 and above for Kluvy® Protein were determined to be 25 g/day and 53 g/day.
- The dietary intake of fiber and sodium from consuming Kluvy® Protein falls within the recommended daily levels and is therefore not a safety concern.
- *K. marxianus* has a long history of safe use as a production organism in food manufacturing, with no evidence of pathogenic or toxigenic concerns. Since CJ3113 has been confirmed to belong to the *K. marxianus* species, is derived from a food source, and has not undergone any genetic modification, the safety information associated with *K. marxianus* species is relevant and can be

applied to support the safe use of the CJ3113 strain.

- The evidence that Kluvy® Protein derived from the fermentation of *K. marxianus* CJ3113 does not pose an increased allergenic risk.

Changing Bio has concluded that Kluvy® Protein, a yeast protein derived from *K. marxianus* for its intended use, is consistent with section 201(s) of the FD&C Act. Changing Bio hereby certifies that, to the best of our knowledge, this independent conclusion of GRAS status is a complete, representative, and balanced evaluation of safety that includes unfavorable information, as well as favorable information, known to us and pertinent to the evaluation of the safety and GRAS status of the use of Kluvy® Protein.

Based on the documentation provided in this GRAS Notice, and as discussed above, Changing Bio has concluded that Kluvy® Protein, a yeast protein produced from the fermentation of *K. marxianus* CJ3113, is GRAS based on scientific procedure for use in selected conventional foods.

7 LIST OF SUPPORTING DATA AND INFORMATION IN THE GRAS NOTICE

7.1 GRAS Expert Panel Report

Generally Recognized As Safe (GRAS) Panel Consensus on the Safety and GRAS Status of Kluvy® Protein for Use as an Ingredient in Foods

EXPERT PANEL STATEMENT

An independent panel of experts (GRAS Panel), qualified by scientific training and experience to assess the safety of food and food ingredients, was convened by the Anchor Center for Certification (ACC) on behalf of Shanghai Changing Biotechnology Co., Ltd (Changing Bio). The panel conducted a critical and comprehensive evaluation of the available data and relevant information to determine whether Kluvy® Protein, when used in selected conventional human foods, is safe and Generally Recognized As Safe (GRAS) under the conditions of its intended use. This determination was made in accordance with scientific procedures outlined in the Federal Food, Drug, and Cosmetic Act (FD&C Act), as specified in Title 21 of the Code of Federal Regulations (21 CFR) §170.30(b).

The GRAS Expert Panel convened by teleconference on October 17, 2024 and unanimously concluded that Kluvy® Protein, meeting the specifications detailed in “GRAS Determination for the Use of Kluvy® Protein in Selected Conventional Foods,” when used as an ingredient in conventional foods as a source of protein, and when manufactured under current Good Manufacturing Practices (cGMP), is GRAS based on scientific procedures. A summary of the basis for the GRAS Expert Panel’s conclusion is provided herein.

Summary and Basis for a GRAS Conclusion

Changing Bio’s Kluvy® Protein is produced by fermentation of non-genetically modified *Kluyveromyces marxianus* (*K. marxianus*). The finished product is composed of protein (>55%), along with dietary fiber (~30%), moisture (<7%), and fat (~5%). Kluvy® Protein is intended for use as an ingredient in select conventional foods (i.e., baked goods and baking mixes, beverages, breakfast cereals, condiments and relishes, dairy product analogs, frozen dairy desserts and mixes, grain products and pastas, gravies and sauces, milk products, plant protein products, processed fruit and vegetable juices, snack foods, soft candies, and soups and soup mixes) at use levels ranging from 1% to 20% by weight.

The GRAS Panel independently, collectively, and critically examined Changing Bio’s GRAS dossier for Kluvy® Protein as well as publicly available scientific information and data compiled from the literature and other published sources through October 17, 2024. The data evaluated by the GRAS Panel included information pertaining to the methods of manufacture and product specifications, analytical data, intended use levels in conventional foods, consumption estimates, and comprehensive

literature related to the safety of Kluvy® Protein.

Composition

Kluvy® Protein consists primarily of protein (over 55%), along with dietary fiber (around 30%), moisture (less than 7%), and fat (approximately 5%). Kluvy® Protein is a complete protein, consisting of all twenty naturally occurring amino acids. This includes nine essential amino acids that the human body cannot produce and must be supplemented by food. The amino acid composition of Kluvy® Protein was presented in the dossier and the GRAS Panel concurs that it compares favorably to other protein replacement products used as meat and dairy analogs.

The GRAS Panel has reviewed the analytical data provided in Changing Bio's GRAS dossier for Kluvy® Protein and confirms that it demonstrates the absence of the *K. marxianus* production organism.

Production Strain

Kluvy® Protein is manufactured from the fermentation of non-genetically modified *K. marxianus* CJ3113. *K. marxianus* has a long history of safe use in food production (e.g., cheese and kefir) and is used to produce GRAS food ingredients including the lactase enzyme (per GRN 000088). It is also recognized by European Food Safety Authority (EFSA) to be safe for human food use and has been included in the latest Qualified Presumption of Safety (QPS) list, version v20 (EFSA BIOHAZ, 2024). *K. marxianus* is nonpathogenic and nontoxic and is not known to produce any toxic secondary metabolites (Alasmar et al., 2020).

Manufacturing Process

Kluvy® Protein is manufactured by Changing Bio under cGMP with food-grade raw materials and processing aids. The GRAS Panel notes Changing Bio's statement that none of the raw materials used in the production process are derived from major allergens. As detailed in the GRAS dossier, the production process consistently yields a reproducible product. In addition, Kluvy® Protein is extensively analyzed to ensure the product meets the established specifications, including protein, moisture, ash, heavy metals, and microbial limits.

Microbiological analyses include total plate count, yeast, mold, *Escherichia coli*, *Enterobacteriaceae*, *Salmonella*, and *Staphylococcus aureus*. The GRAS Panel notes the absence of production organism in the finished product was demonstrated by testing three non-consecutive batches using an appropriately verified method.

The GRAS Panel reviewed the manufacturing process and does not have any concerns with the information provided in the GRAS dossier.

Product Specifications

The GRAS Panel notes that the product specifications established by Changing Bio for Kluvy® Protein are sufficient to characterize the ingredient. The GRAS Panel reviewed batch analysis results for five non-consecutive lots of Kluvy® Protein as presented in the GRAS dossier, and notes that the ingredient

is well-characterized and that the analytical data obtained using internationally standardized methods includes protein, moisture, microbiological, and heavy metal content. Stability data indicate that Kluvy® Protein is shelf-stable for up to one year when stored at ambient conditions.

Proposed Uses and Estimated Intake

Estimated intakes of Kluvy® Protein were based on its proposed use in selected conventional foods and the maximum usage levels outlined in the GRAS dossier. The estimated conservative mean daily intake for consumers only and the 90th percentile intake for consumers only are 25 g/day and 53 g/day, respectively, for the total population aged 2 years and older.

To compare with dietary reference intakes for protein, the estimated total protein intake from the proposed uses of Kluvy® Protein was calculated by assuming a maximum protein content of approximately 63%, based on representative batch results provided in the GRAS dossier. The conservative mean and 90th percentile estimate of daily protein consumption from Kluvy® Protein are approximately 16 g/day and 33 g/day, respectively, for the total population aged 2 years and older. The GRAS Panel notes that this intake falls within or below the recommended and typical dietary protein intake levels. Additionally, Kluvy® Protein is intended to substitute other protein sources in the diet, rather than increase overall protein consumption. Therefore, the GRAS Panel concurs with Changing Bio that the risk of adverse effects from excessing protein intake due to Kluvy® Protein consumption is unlikely. The GRAS Panel also concurs that the dietary intake of fiber and sodium from consuming Kluvy® Protein falls within the recommended daily levels and is therefore not a safety concern.

Pivotal Safety Data

The safety of Kluvy® Protein is supported by several factors:

- 1) The production organism, *K. marxianus*, has a long history of safe use in food manufacturing, with no evidence of pathogenicity or toxigenicity. The production organism has been confirmed to belong to the *K. marxianus* species, is derived from a food source, and has not undergone genetic modification; therefore, the GRAS Panel agrees that the safety information associated with *K. marxianus* is relevant and supports the safe use of this organism.
- 2) Kluvy® Protein is produced in accordance with cGMP standards, using food-grade raw materials and processing aids.
- 3) The absence of live production organisms in all batches is ensured through analytical testing and monitoring of yeast counts according to established specifications, which have been confirmed to be appropriate for the intended use.
- 4) Kluvy® Protein has an amino acid profile similar to other dietary protein sources, making it a suitable dietary protein substitute.
- 5) The protein intake from consuming Kluvy® Protein is within or below recommended dietary protein levels and background dietary protein intakes, ensuring no overall increase in protein consumption.
- 6) The GRAS Panel concurs that there is no evidence to suggest that Kluvy® Protein derived from the fermentation of *K. marxianus* poses an increased allergenic risk.

Conclusion

We, the GRAS Panel, unanimously conclude that the proposed use of Changing Bio's Kluvy® Protein as an ingredient in conventional foods and beverages, at levels specified in the GRAS Dossier, up to consumer-only mean and 90th percentile intakes of 25 g/day and 53 g/day, respectively, for the total population (ages 2 years and older), is safe and GRAS based on scientific procedures and supported by a history of safe use. We, the undersigned independent GRAS Panel, have critically evaluated the safety of Changing Bio's Kluvy® Protein, a dietary protein substitute, and unanimously conclude that the scientific data, information, and methods described herein establish that the ingredient, when produced in accordance with cGMP and meeting the specifications outlined in the supporting dossier, meets the standard of reasonable certainty of no harm and is Generally Recognized As Safe (GRAS) under 21 CFR.

In accordance with Title 21 of the Code of Federal Regulations, Part 170, Subpart E on Generally Recognized As Safe (GRAS) Notices, we, the Expert Panel, have evaluated the scientific evidence and agree with Shanghai Changing Biotechnology Co., Ltd.'s conclusion that Kluvy® Protein is GRAS for use as an ingredient in foods and beverages at the levels specified herein. In our opinion, other experts qualified to assess the safety of food ingredients would concur with this conclusion. Therefore, it is excluded from the definition of a food additive and may be used in the U.S. without requiring the promulgation of a food additive regulation by the FDA under 21 CFR.

Margitta Dziwenka, DVM, D.A.B.T.

President

Toxalta Consulting LLC.

Signature:

Date: October 23, 2024

Katrina Emmel, Ph.D.

President KemmelCal Inc.

Signature:

Date: October 28, 2024

Jianjun (Gene) Yang, Ph.D.

President

Brandywine BioSciences, LLC

Signature:

Date: October 23, 2024

Biographical Sketch of the Panelists

Margitta Dziwenka, DVM, D.A.B.T.

Dr. Dziwenka holds a Doctor of Veterinary Medicine degree from the University of Saskatchewan and is a Diplomat with the American Board of Toxicology. She has over 27 years of experience as a practicing veterinarian and laboratory animal research experience. Dr. Dziwenka has worked in research, preclinical regulatory toxicology, safety evaluation of human food and animal feed additives, GRAS dossier, and other regulatory filing dossier preparation for those 27 years. Dr. Dziwenka also has extensive experience as an Expert Panelist. She has experience in the management of preclinical contract research organizations and in the role of Study Director and Study Monitor for GLP-compliant preclinical studies and has assisted in the design of numerous preclinical studies. Dr. Dziwenka is experienced in the writing of position papers, safety opinions for a variety of products, manuscripts for publication, and critical review of study results used to support regulatory submissions and safety conclusions.

Katrina Emmel, Ph.D.

Dr. Katrina Emmel is the founder and president of KemmelCal Inc. With an MS in Biochemistry (WPI) and a Ph.D. in Chemistry (Northern Illinois University), Dr. Emmel started her career as an analytical chemist, performing routine LC-MS analysis in a clinical laboratory, and then developing analytical methods for a dietary supplement and food manufacturer. She was involved in preparing the first Generally Recognized as Safe (GRAS) Notification for Enzyme Modified Steviol Glycosides on behalf of NOW Foods. As an independent consultant since 2013, Dr. Emmel has been involved in a number of GRAS evaluations as a chemist, technical writer, project manager, Expert Panelist, and Chair of the Expert Panel. She has also served as an Expert Witness to support patent litigation activities for food ingredients.

Jianjun (Gene) Yang, Ph.D.

Dr. Yang has led a distinguished career in food ingredients innovation. He obtained his Ph.D. from University of Kentucky, and then completed his NIH fellowship at Cornell University. He had worked for almost 20 years at the DuPont Central Research and Development as Principal Research Scientist, leading the DuPont Biotech Lab in many innovation projects for food ingredients and industrial materials. He had also worked as the leader and advisor in other SynBio and pharmaceutical companies. Dr. Yang has many U.S. and international patents, patent applications, and peer-reviewed publications. His expertise includes genetic engineering, protein engineering, and metabolic engineering.

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7.2 List of Appendices

Appendix 1: Verification Report - Cultivation Report of *Kluyveromyces Marxianus* CJ3113 in Potato Glucose Agar (PDA) Medium

Appendix 2: Manufacturing Facility Certification

Appendix 3: COAs of Kluvy® Protein

Appendix 4: Food Codes Used for EDI Calculation of Kluvy® Protein

Appendix 5: Allergenicity Evaluation of Kluvy® Protein

Appendix 6: Report of Proteins Identification of Kluvy® Protein and Sequence Comparison with Major Allergens Using AllergenOnline

7.3 List of Acronyms

%	Percentage
°C	Degree Celsius
ACE	angiotensin converting enzyme
ADME	Absorption, Distribution, Metabolism, and Elimination
AI	Adequate Intake
ALT	Alanine aminotransferase
AOAC	AOAC INTERNATIONAL
AST	Aspartate transaminase
ATCC	American Type Culture Collection
BAM	Bacteriological Analytical Manual
BIOHAZ	EFSA's Panel on Biological Hazards
BUN	Blood urea nitrogen
bw	body weight
CCP	critical control point
CCTCC	China Center for Type Culture Collection
CDC	Centers for Disease Control and Prevention
CFR	Code of Federal Regulations
CFU	Colony Forming Unit
cGMP	current Good Manufacturing Practices
COA	Certificate of Analysis
DNA	Deoxyribonucleic acid
DRI	Dietary reference intakes
DRV	daily reference value
EDI	Estimated Daily Intake
EAR	Estimated Average Requirement
EFSA	European Food Safety Authority

EU	European Union
FAERS	FDA's Adverse Event Reporting Program
FDA	Food and Drug Administration
FD&C Act	Federal Food, Drug, and Cosmetics Act
FODMAPs	Oligo-, di-, and monosaccharides and polyols
FOIA	Freedom of Information Act
FSMA	Food Safety Modernization Act
FSSC	Food Safety System Certification
g	gram
GGT	Gammaglutamyl transferase
GMP	Good Manufacturing Practice
GRAS	Generally Recognized as Safe
hrs	hour
HACCP	Hazard Analysis Critical Control Point
I	intermediate sensitive
ICP-MS	Inductively coupled plasma mass spectrometry
IOM	Institute of Medicine
IPTG	Isopropyl-beta-D-thiogalactoside
ITS	Internal Transcribed Spacer
KDM	Kluyveromyces Differential Medium
kg	kilogram
KJ	Kilojoules
KOH	Potassium hydroxide
L	Liter
LDH	Lactate dehydrogenase
LOD	Limit of Detection
m ³	cubic meter
MALDI-TOF MS	Matrix assisted laser desorption ionization-time of flight mass spectrometry
MEA	malt extract agar
mg	milligram
mins	minutes
mL	milliliter
mm	millimeter
MW	Molecular weight
n/No.	Number
NCHS	National Center for Health Statistics
ND	Not detected
NHANES	National Nutrition and Health Examination Survey
NS	Not specified
NT	Not tested
PBS	Phosphate-buffered saline
PDA	Potato Dextrose Agar
ppm	parts per million

QPS	Qualified Presumption of Safety
R	resistant
RBC	Red blood cells
RDA	Recommended Dietary Allowance
RH	Relative Humidity
RNA	Ribonucleic acid
rpm	rotation per minute
S	sensitive
SLAD	Synthetic Low Ammonia Dextrose Media
SOP	standard operating procedures
TEER	Trans-epithelial electrical resistance
UL	Tolerable Upper Intake Level
US	United States
USP	United States Pharmacopeia
WBC	White blood cell
wt	weight
μL	microliter

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Appendix 1: Verification Report – Cultivation Report of *Kluyveromyces marxianus* CJ3113 in Potato Glucose Agar (PDA) Medium

**Cultivation Report of *Kluyveromyces marxianus*
CJ3113 in Potato Glucose Agar (PDA) Medium**

Shanghai Changing Biotechnology Co., LTD

Project number: CJSW-SH-WSW-24-0923



Shanghai Changing Biotechnology Co., Ltd. No: CJSW-SH-WSW-24-0923

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1. Experimental background

Potato glucose agar (PDA) medium is one of the commonly used media for yeast and mold counting in food according to FDA BAM Chapter 18. This experiment intends to use potato glucose agar (PDA) from FDA BAM chapter 18 for counting and analyzing the production process of *Kluyveromyces marxianus* CJ3113. Firstly, the cultivable rows of *Kluyveromyces marxianus* CJ3113 in PDA medium need to be tested.

2. Experimental purpose

Detect the growth performance of *Kluyveromyces marxianus* CJ3113 in PDA medium.

3. Experimental plan

3.1 Prepare PDA solid culture medium according to the formula in FDA BAM chapter 18, and pour the solid plate. PDA formula: Potatoes: 200 g, need to be peeled and cut into thin slices or small pieces. Glucose: 20 g, providing carbon source and energy. Agar: 15-20 g, used as a coagulant to solidify the culture medium. Water: 1000 mL, used for boiling potatoes and dissolving other ingredients. The production method includes boiling potatoes until they are soft and not rotten, filtering the filtrate, adding glucose and agar, heating to dissolve the agar, and then packaging and sterilizing. This type of gene cultivation is rich in nutrients and suitable for the growth of various microorganisms, especially fungi such as yeast and mold. Therefore, it is widely used in laboratory and industrial microbial cultivation.

3.2 Apply or streak *Kluyveromyces marxianus* CJ3113 onto a PDA solid plate.

3.3 Cultivate the coated or streaked PDA plate in the incubator for 5 days and observe the growth of *Kluyveromyces marxianus* CJ3113

4. Experimental results

Kluyveromyces marxianus CJ3113 can be cultured normally in PDA medium, with a milky white colony morphology that can form a single colony, appearing smooth, raised, moist, and sticky. With the extension of cultivation time, the colony surface becomes dry and shows gyrus like folds, which is consistent with the general characteristics of yeast on PDA medium. As shown in the following figure, it indicates that the *Kluyveromyces marxianus* strain can grow normally in PDA medium.



Shanghai Changing Biotechnology Co., Ltd. No: CJSW-SH-WSW-24-0923



Figure 1: Growth of *Kluyveromyces marxianus* CJ3113 in PDA solid medium.

5. Experimental conclusion

Kluyveromyces marxianus CJ3113 can grow normally in PDA medium, that is, potato glucose agar (PDA) in FDA BAM chapter 18 can be used to culture *Kluyveromyces marxianus* CJ3113 and perform counting statistics.



Shanghai Changing Biotechnology Co., Ltd. No: CJSW-SH-WSW-24-0923

Experimental report approved signature

Lab Leader/date	 2024.9.23
Team leader audit/date	 2024.9.23
Approved person/date	 2024.9.23

Appendix 2: Manufacturing Facility Certification

Certificate CN23/00002139

The Food Safety Management System of

Qingdao Changing Biotech Co., Ltd.

SGS

COID: CHN-1-0587-842728

Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province, P. R. China

has been assessed and certified as meeting the requirements of

Food Safety System Certification FSSC 22000

Certification scheme for food safety management systems consisting of the following elements:
ISO 22000:2018, ISO/TS22002-1:2009 and Additional FSSC 22000 requirements (version 6).

This certificate is applicable for the scope of

Production (fermenting, sterilizing and drying) of yeast (powder/ slurry) used in the food industry;
production (enzymolysing, thermal denaturing, sterilizing and drying) of yeast protein (powder) used
in the food industry; Production (blending, emulsifying and sterilizing) of margarine and cheese
products (cream cheese).

**Food Category: CIV - Processing of ambient stable products,
CI - Processing of perishable animal products**

This certificate is valid from 12 May 2024 until 24 April 2026 and remains valid subject to satisfactory surveillance audits.

Date of Certification decision 12 May 2024

Issue Date 17 May 2024

Issue 3. Certified since 25 April 2023

Authorised by

Jonathan Hall

Global Head - Certification

Services

SGS United Kingdom Ltd

Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK

t +44 (0)151 350-6666 - www.sgs.com

Authenticity of this certificate can be verified in the FSSC 22000 database of certified organizations available on www.fssc22000.com.



FSSC 22000



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This is a translation of the certificate CN23/00002140

External Number F01HACCP2300033

The management system of

SGS

Qingdao Changing Biotech Co., Ltd.

Business Registration Address: Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park, Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province, P.R. China

Business Operation Address: Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park, Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province, P.R. China

has been assessed and certified as meeting the requirements of

Certificate requirements of hazard analysis and critical control point (HACCP) system (V1.0)

For the following activities

Production (fermenting, sterilizing and drying) of yeast (powder/ slurry) used in the food industry; production (enzymolysing, thermal denaturing, sterilizing and drying) of yeast protein (powder) used in the food industry; Production (blending, emulsifying and sterilizing) of margarine and cheese products (cream cheese).

This certificate is valid from 11 October 2023 until 24 April 2026 and remains valid subject to satisfactory surveillance audits.

Issue 2. Certified since 25 April 2023

Authorised by
Jian Wang
Director -Health & Nutrition

SGS-CSTC Standards Technical Services Co., Ltd.
16F Century YuHui Mansion, No. 73 Fucheng Road, Beijing, P.R. CHINA 100142
t +86 (0)10 58251188 - www.sgsgroup.com.cn

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CNAS C033-M

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Certificate CN23/00002138
External Number F01FSMS2300053

The management system of

SGS

Qingdao Changing Biotech Co., Ltd.

Unified Social Credit Code: 91370281MA958D074G

Business Registration Address: Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province, P.R. China
Business Operation Address: Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province, P.R. China

has been assessed and certified as meeting the requirements of

ISO 22000:2018

For the following activities

Production (fermenting, sterilizing and drying) of yeast (powder/ slurry) used in the food industry; production
(enzymolysing, thermal denaturing, sterilizing and drying) of yeast protein (powder) used in the food industry;
Production (blending, emulsifying and sterilizing) of margarine and cheese products (cream cheese).

Food Category CIV - Processing of ambient stable products, CI - Processing of perishable animal products

This certificate is valid from 11 October 2023 until 24 April 2026 and remains valid subject to satisfactory surveillance audits.

Issue 2. Certified since 25 April 2023

Authorised by
Jonathan Hall
Global Head - Certification Services

SGS United Kingdom Ltd
Rossmore Business Park, Ellesmere Port, Cheshire, CH65 3EN, UK
t +44 (0)151 350-6666 - www.sgs.com

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Certificate CN23/00001781

This is to certify that

Qingdao Changing Biotech Co., Ltd.

SGS

Building No.2, Road No.7 East / Road No.6 North, Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province,
266300, P. R. China

SITE CODE 10008274

has been assessed and certified as meeting the requirements of

Global Standard Food Safety

Issue 9: August 2022

Achieved Grade: B

Audit Programme: Announced

For the following activities:

Processing (fermentation, sterilization and spray-drying) of Yeast (*Kluyveromyces marxianus*) liquid /
powder, Processing (fermentation, enzymolysis, sterilization and spray-drying) of Yeast (*Kluyveromyces*
marxianus) protein powder, all packed in plastic bags

Exclusions from scope: None

Product Categories: 7, 15

AUDITOR NUMBER: 21634

Date of Evaluation: 26 February 2024

Certificate Issue Date: 03 April 2024

Re-Evaluation Due Date 02 March 2025

Certificate Expiry Date: 13 April 2025

This certificate supersedes all other certificates bearing this certificate number with earlier certificate issue dates.

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Authorised by
Jonathan Hall
Global Head - Certification Services

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Appendix 3: COAs of Kluvy® Protein



Qingdao Changng Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.6 North
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛昌宁生物科技有限公司
山东省青岛市即墨市京沪高速工业园一二期交汇处东南角2号楼

Kluyveromyces marxianus protein Certificate of Analysis

马克斯克鲁维酵母蛋白出厂检验报告

Trade name 商品名称		Kluyveromyces marxianus protein 马克斯克鲁维酵母蛋白			
Packaging specifications 包装规格		8 kg (4 kg*2) / Box 箱	Quantity 数量		30 Box 箱
Date of manufacture/batch Number 生产日期/批号		2022 年 10 月 11 日/20221011			
Sensory examination 感官检验情况					
NO. 序号	Analysis 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Appearance 外观	Off-white to light brown powder 类白色至浅褐色粉状	/	Off-white 类白色	PASS 合格 (√) Fail 不合格 (×)
Physical and chemical 理化检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Moisture 水分/(%)	≤7.0	/	3.85	PASS 合格 (√) Fail 不合格 (×)
2	Protein 蛋白质(以干物质计)/(%)	≥55	/	59.7	PASS 合格 (√) Fail 不合格 (×)
3	Ash 灰分/(%)	<3	/	2.5	PASS 合格 (√) Fail 不合格 (×)
4	Lead 铅/ppm	≤0.1	0.010	0.011	PASS 合格 (√) Fail 不合格 (×)
5	Arsenic 砷/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)
6	Mercury 汞/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)



Qingdao Changning Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.8 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛昌宁生物科技有限公司
山东省青岛市胶州市胶东街道办事处工业园七号路北侧七号路东侧2号楼

7	Cadmium 镉/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 ()
Microorganism 微生物检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检验结果 (CFU/ g)	Single conclusion 单项结论
1	Total plate Count 菌落总数/(CFU/g)	<10000	<10	58	PASS 合格 (√) Fail 不合格 ()
2	Yeast 酵母/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
3	Mold 霉菌/(CFU/g)	<100	<10	ND	PASS 合格 (√) Fail 不合格 ()
4	<i>E. coli</i> 大肠杆菌/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
5	<i>Enterobacteriaceae</i> 肠杆菌科/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
6	<i>Salmonella</i> 沙门氏菌/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
7	<i>Staphylococcus aureus</i> 金黄色葡萄球菌 (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
Conclusion 检测结论		PASS 合格 (√)		Fail 不合格 ()	
Test 检测人		牛志华		QA 审核人	
Approved by 批准人				Date 报告 日期	2022.10.17
Note 备注					

Note: ND =not detected within the limit of detection (LOD), which is <10 CFU /gram for microbiological parameters, 0.01 ppm for arsenic, 0.01 ppm for cadmium, 0.01 ppm for lead, 0.01 ppm for mercury



Qingdao Changling Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.8 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛超域生物科技有限公司
山东省青岛市胶州市海泊河李村街道办事处三岔河二一路北侧七号路东侧2号地

Kluyveromyces marxianus protein Certificate of Analysis

马克斯克鲁维酵母蛋白出厂检验报告

Trade name 商品名称		Kluyveromyces marxianus protein 马克斯克鲁维酵母蛋白			
Packaging specifications 包装规格		8 kg (4 kg*2) / Box 箱	Quantity 数量	30 Box 箱	
Date of manufacture/batch Number 生产日期/批号		2022 年 12 月 12 日/20221212			
Sensory examination 感官检验情况					
NO. 序号	Analysis 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Appearance 外观	Off-white to light brown powder 类白色至浅褐色粉状	/	Off-white 类白色	PASS 合格 (√) Fail 不合格 (×)
Physical and chemical 理化检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Moisture 水分/(%)	≤7.0	%	4.07	PASS 合格 (√) Fail 不合格 (×)
2	Protein 蛋白质(以干物质计)/%	≥55	/	62.8	PASS 合格 (√) Fail 不合格 (×)
3	Ash 灰分/%	<3	%	2.8	PASS 合格 (√) Fail 不合格 (×)
4	Lead 铅/ppm	≤0.1	0.010	0.011	PASS 合格 (√) Fail 不合格 (×)
5	Arsenic 砷/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)
6	Mercury 汞/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)



Qingdao Changing Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛昌通生物科技有限公司
山东省青岛市胶州市院后办事处纺织工业园六号路北侧七号路西侧2号楼

7	Cadmium 镉/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 ()
Microorganism 微生物检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检验结果 (CFU/ g)	Single conclusion 单项结论
1	Total plate Count 菌落总数/(CFU/g)	<10000	<10	15	PASS 合格 (√) Fail 不合格 ()
2	Yeast 酵母/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
3	Mold 霉菌/(CFU/g)	<100	<10	ND	PASS 合格 (√) Fail 不合格 ()
4	<i>E. coli</i> 大肠杆菌/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
5	<i>Enterobacteriaceae</i> 肠杆菌科/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
6	<i>Salmonella</i> 沙门氏菌/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
7	<i>Staphylococcus aureus</i> 金黄色葡萄球菌 /(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
Conclusion 检测结论		PASS 合格 (√) Fail 不合格 ()			
Test 检测人		朱志军		QA 审核人	
Approved by 批准人				Date 报告日期	2022.12.21
Note 备注		1			

Note: ND =not detected within the limit of detection (LOD), which is <10 CFU /gram for microbiological parameters, 0.01 ppm for arsenic, 0.01 ppm for cadmium, 0.01 ppm for lead, 0.01 ppm for mercury.



Qingdao Changng Biotech Co., Ltd.
Building No 2, Road No 7 East / Road No 6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛昌宁生物科技有限公司
山东青岛莱州市莱州港办事处莱州三路四号昌宁七号路西侧2号楼

Kluyveromyces marxianus protein Certificate of Analysis

马克斯克鲁维酵母蛋白出厂检验报告

Trade name 商品名称		Kluyveromyces marxianus protein 马克斯克鲁维酵母蛋白			
Packaging specifications 包装规格		8 kg (4 kg*2) / Box 箱	Quantity 数量	30 Box 箱	
Date of manufacture/batch Number 生产日期/批号		2023 年 06 月 18 日/20230618			
Sensory examination 感官检验情况					
NO. 序号	Analysis 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Appearance 外观	Off-white to light brown powder 灰白色至浅褐色粉状	/	Off-white 灰白色	PASS 合格 (√) Fail 不合格 (×)
Physical and chemical 理化检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Moisture 水分/(%)	≤7.0	/	3.66	PASS 合格 (√) Fail 不合格 (×)
2	Protein 蛋白质(以干物质计)/%	≥55	/	60.5	PASS 合格 (√) Fail 不合格 (×)
3	Ash 灰分/%	<3	/	2.7	PASS 合格 (√) Fail 不合格 (×)
4	Lead 铅/ppm	≤0.1	0.010	0.012	PASS 合格 (√) Fail 不合格 (×)
5	Arsenic 砷/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)
6	Mercury 汞/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)



Qingdao Changing Biotech Co., Ltd.
Building No 2, Road No 7 East / Road No 6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛石湾生物科技有限公司
山东省青岛市市南区香港中路100号六号路北侧七号路东侧2号楼

7	Cadmium 镉/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 ()
Microorganism 微生物检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检验结果 (CFU/ g)	Single conclusion 单项结论
1	Total plate Count 菌落总数/(CFU/g)	<10000	<10	67	PASS 合格 (√) Fail 不合格 ()
2	Yeast 酵母/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
3	Mold 霉菌/(CFU/g)	<100	<10	ND	PASS 合格 (√) Fail 不合格 ()
4	<i>E. coli</i> 大肠杆菌/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
5	<i>Enterobacteriaceae</i> 肠杆菌科/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
6	<i>Salmonella</i> 沙门氏菌/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
7	<i>Staphylococcus aureus</i> 金黄色葡萄球菌 (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
Conclusion 检测结论		PASS 合格 (√) Fail 不合格 ()			
Test 检测人		朱志伟		QA 审核人	
Approved by 批准人				Date 报告 日期	2023.06.25
Note 备注					

Note: ND =not detected within the limit of detection (LOD), which is <10 CFU /gram for microbiological parameters, 0.01 ppm for arsenic, 0.01 ppm for cadmium, 0.01 ppm for lead, 0.01 ppm for mercury.



Qingdao Changng Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaoding Sub-district Office, Jiaochou City, Qingdao City, Shandong Province
青岛昌宁生物科技有限公司
山东省青岛市城阳街道办事处工业园内香江路7号路南侧2号楼

Kluyveromyces marxianus protein Certificate of Analysis

马克斯克鲁维酵母蛋白出厂检验报告

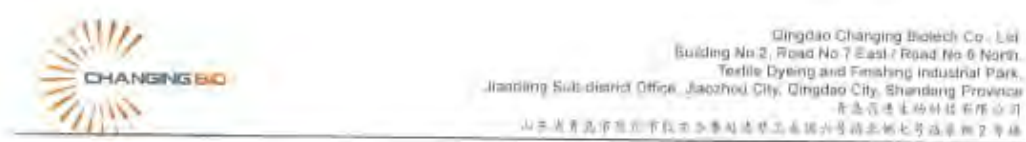
Trade name 商品名称		Kluyveromyces marxianus protein 马克斯克鲁维酵母蛋白			
Packaging specifications 包装规格		8 kg(4 kg*2) / Box 箱	Quantity 数量		30 Box 箱
Date of manufacture/batch Number 生产日期/批号		2023 年 09 月 19 日/20230919			
Sensory examination 感官检验情况					
NO. 序号	Analysis 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Appearance 外观	Off-white to light brown powder 类白色至浅褐色粉状	/	Off-white 类白色	PASS 合格 (√) Fail 不合格 (×)
Physical and chemical 理化检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Moisture 水分/(%)	≤7.0	/	4.19	PASS 合格 (√) Fail 不合格 (×)
2	Protein 蛋白质(以干物质计)/%	≥55	/	62.1	PASS 合格 (√) Fail 不合格 (×)
3	Ash 灰分/%	<3	/	2.3	PASS 合格 (√) Fail 不合格 (×)
4	Lead 铅/ppm	≤0.1	0.010	0.012	PASS 合格 (√) Fail 不合格 (×)
5	Arsenic 砷/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)
6	Mercury 汞/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)



Qingdao Changing Biotech Co., Ltd.
Building No.2, Road No.7 East / Road No.6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛智达生物科技有限公司
山东省青岛市胶州市胶东街道办事处工业园6号路东侧7号路南2号楼

7	Cadmium 镉/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 ()
Microorganism 微生物检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检验结果 (CFU/ g)	Single conclusion 单项结论
1	Total plate Count 菌落总数/ (CFU/g)	<10000	<10	37	PASS 合格 (√) Fail 不合格 ()
2	Yeast 酵母/ (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
3	Mold 霉菌/ (CFU/g)	<100	<10	ND	PASS 合格 (√) Fail 不合格 ()
4	<i>E. coli</i> 大肠杆菌/ (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
5	<i>Enterobacteriaceae</i> 肠杆菌科/ (Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
6	<i>Salmonella</i> 沙门氏菌/ (Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
7	<i>Staphylococcus aureus</i> 金黄色葡萄球菌 / (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
Conclusion 检测结论		PASS 合格 (√) Fail 不合格 ()			
Test 检测人		牛志军		QA 审核人	
Approved by 批准人				Date 报告 日期	2023.09.23
Note 备注					

Note:ND =not detected within the limit of detection (LOD), which is <10 CFU /gram for microbiological parameters, 0.01 ppm for arsenic, 0.01 ppm for cadmium, 0.01 ppm for lead, 0.01 ppm for mercury.



Kluyveromyces marxianus protein Certificate of Analysis

马克斯克鲁维酵母蛋白出厂检验报告

Trade name 商品名称		Kluyveromyces marxianus protein 马克斯克鲁维酵母蛋白			
Packaging specifications 包装规格		8 kg (4 kg*2) / Box 箱	Quantity 数量	30 Box 箱	
Date of manufacture/batch Number 生产日期/批号		2024 年 03 月 08 日/20240308			
Sensory examination 感官检验情况					
NO. 序号	Analysis 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Appearance 外观	Off-white to light brown powder 类白色至浅褐色粉状	/	Off-white 类白色	PASS 合格 (√) Fail 不合格 (×)
Physical and chemical 理化检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检测结果	Single conclusion 单项结论
1	Moisture 水分/(%)	≤7.0	/	3.79	PASS 合格 (√) Fail 不合格 (×)
2	Protein 蛋白质(以干物质计)/%	≥55	/	62.9	PASS 合格 (√) Fail 不合格 (×)
3	Ash 灰分/%	<3	/	2.5	PASS 合格 (√) Fail 不合格 (×)
4	Lead 铅/ppm	≤0.1	0.010	0.011	PASS 合格 (√) Fail 不合格 (×)
5	Arsenic 砷/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)
6	Mercury 汞/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 (×)



Qingdao Changing Biotech Co., Ltd.
Building No 2, Road No 7 East / Road No 6 North,
Textile Dyeing and Finishing Industrial Park,
Jiaodong Sub-district Office, Jiaozhou City, Qingdao City, Shandong Province
青岛昌建生物科技有限公司
山东省青岛市胶州市胶东办事处南墅工业园六号路北第七号路东侧2号楼

7	Cadmium 镉/ppm	≤0.1	0.010	ND	PASS 合格 (√) Fail 不合格 ()
Microorganism 微生物检验情况					
NO. 序号	Test item 检验项目	Specification 指标值	Limit of Detection 检出限	Result 检验结果 (CFU/ g)	Single conclusion 单项结论
1	Total plate Count 菌落总数/(CFU/g)	<10000	<10	66	PASS 合格 (√) Fail 不合格 ()
2	Yeast 酵母/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
3	Mold 霉菌/(CFU/g)	<100	<10	ND	PASS 合格 (√) Fail 不合格 ()
4	<i>E. coli</i> 大肠杆菌/(CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
5	<i>Enterobacteriaceae</i> 肠杆菌科/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
6	<i>Salmonella</i> 沙门氏菌/(Negative per 25g)	Negative	Negative	Negative	PASS 合格 (√) Fail 不合格 ()
7	<i>Staphylococcus aureus</i> 金黄色葡萄球菌 (CFU/g)	<10	<10	ND	PASS 合格 (√) Fail 不合格 ()
Conclusion 检测结论		PASS 合格 (√) Fail 不合格 ()			
Test 检测人		邢志华		QA 审核人	
Approved by 批准人				Date 报告 日期	2024.03.15
Note 备注					

Note: ND =not detected within the limit of detection (LOD), which is <10 CFU /gram for microbiological parameters, 0.01 ppm for arsenic, 0.01 ppm for cadmium, 0.01 ppm for lead, 0.01 ppm for mercury.

Appendix 4: Food Codes Used for EDI Calculation of Kluvy® Protein

Representative Food Codes for Proposed Food Uses of Kluvy® Protein in the U.S.

Baked Goods and Baking Mixes

Bagels and English Muffins, Cornbread, Corn Muffins, or Tortillas

[Kluy® Protein] = 5 g/100 g

51180010	Bagel
51180030	Bagel, with raisins
51180080	Bagel, with fruit other than raisins
51186010	Muffin, English
51186100	Muffin, English, with raisins
51186130	Muffin, English, cheese
51186160	Muffin, English, with fruit other than raisins
51187000	Melba toast
51301700	Bagel, wheat
51301750	Bagel, whole wheat
51301800	Bagel, wheat, with raisins
51301805	Bagel, whole wheat, with raisins
51301820	Bagel, wheat, with fruit and nuts
51301900	Bagel, wheat bran
51302500	Muffin, English, wheat bran
51302520	Muffin, English, wheat bran, with raisins
51303010	Muffin, English, wheat or cracked wheat
51303030	Muffin, English, whole wheat
51303050	Muffin, English, wheat or cracked wheat, with raisins
51303070	Muffin, English, whole wheat, with raisins
51303100	Muffin, English, whole grain white
51401200	Muffin, English, rye
51404500	Bagel, pumpernickel
51404550	Muffin, English, pumpernickel
51501080	Bagel, oat bran
51503000	Muffin, English, oat bran
51503040	Muffin, English, oat bran, with raisins
51630000	Bagel, multigrain
51630100	Bagel, multigrain, with raisins
51630200	Muffin, English, multigrain
52201000	Cornbread, prepared from mix
52204000	Cornbread stuffing
52206010	Cornbread muffin, stick, round
52207010	Corn flour patty or tart, fried
52208010	Corn pone, baked
52208020	Corn pone, fried
52208760	Gordita/sope shell, plain, no filling
52209010	Hush puppy
52211010	Johnnycake

52213010	Spoonbread
52215000	Tortilla, NFS
52215100	Tortilla, corn
52215200	Tortilla, flour
52215260	Tortilla, whole wheat
52215300	Taco shell, corn
52215350	Taco shell, flour
52220110	Arepa Dominicana

Crackers

[Kluy® Protein] = 5 g/100 g

54001000	Crackers, NFS
54102010	Graham crackers
54102015	Graham crackers (Teddy Grahams)
54102020	Graham crackers, chocolate covered
54102050	Crackers, oatmeal
54102060	Crackers, Cuban
54102100	Graham crackers, reduced fat
54102200	Graham crackers, sandwich, with filling
54103000	Crackers, breakfast biscuit
54200100	Crackers, butter, reduced sodium
54201010	Crackers, matzo, reduced sodium
54202020	Crackers, saltine, reduced sodium
54204020	Crackers, wheat, reduced sodium
54204030	Crackers, woven wheat, reduced sodium
54301010	Crackers, butter, plain
54301020	Crackers, butter, flavored
54301030	Crackers, butter (Ritz)
54301100	Crackers, butter, reduced fat
54304000	Crackers, cheese
54304005	Crackers, cheese (Cheez-It)
54304020	Crackers, cheese (Goldfish)
54304100	Crackers, cheese, reduced fat
54304110	Crackers, cheese, reduced sodium
54304150	Crackers, cheese, whole grain
54305010	Crackers, crispbread
54305020	Crackers, flatbread
54307000	Crackers, matzo
54308000	Crackers, milk
54313000	Crackers, oyster
54319000	Crackers, rice
54319005	Crackers, rice and nuts
54325000	Crackers, saltine
54325010	Crackers, saltine, reduced fat

54325060	Crackers, saltine, multigrain
54326000	Crackers, multigrain
54328000	Crackers, sandwich
54328100	Crackers, sandwich, peanut butter filled
54328105	Crackers, sandwich, peanut butter filled (Ritz)
54328110	Crackers, sandwich, reduced fat, peanut butter filled
54328120	Crackers, whole grain, sandwich, peanut butter filled
54328200	Crackers, sandwich, cheese filled
54328210	Crackers, sandwich, cheese filled (Ritz)
54336000	Crackers, water
54336100	Crackers, wonton
54337010	Crackers, woven wheat
54337020	Crackers, woven wheat, plain (Triscuit)
54337030	Crackers, woven wheat, flavored (Triscuit)
54337060	Crackers, woven wheat, reduced fat
54338000	Crackers, wheat
54338010	Crackers, wheat, plain (Wheat Thins)
54338020	Crackers, wheat, flavored (Wheat Thins)
54338100	Crackers, wheat, reduced fat
54339000	Crackers, corn
54340100	Crackers, gluten free, plain
54340110	Crackers, gluten free, flavored

Biscuits and Cookies

[Kluy® Protein] = 10 g/100 g

52101000	Biscuit, NFS
52101040	Crumpet
52102040	Biscuit, from refrigerated dough
52103000	Biscuit, from fast food / restaurant
52104010	Biscuit, home recipe
52104040	Biscuit, wheat
52104100	Biscuit, cheese
52104200	Biscuit with fruit
52105100	Scone
52105200	Scone, with fruit
53200100	Cookie, batter or dough, raw
53201000	Cookie, NFS
53202000	Cookie, almond
53203000	Cookie, applesauce
53203500	Cookie, biscotti
53204000	Cookie, brownie, NS as to icing
53204010	Cookie, brownie, without icing
53204100	Cookie, brownie, with icing or filling
53204840	Cookie, brownie, reduced fat, NS as to icing

53204860	Cookie, brownie, fat free, NS as to icing
53205250	Cookie, butterscotch, brownie
53205260	Cookie, bar, with chocolate
53206000	Cookie, chocolate chip
53206020	Cookie, chocolate chip, made from home recipe or purchased at a bakery
53206030	Cookie, chocolate chip, reduced fat
53206100	Cookie, chocolate chip sandwich
53206500	Cookie, chocolate, made with rice cereal
53206550	Cookie, chocolate, made with oatmeal and coconut, no bake
53207000	Cookie, chocolate or fudge
53207020	Cookie, chocolate or fudge, reduced fat
53207050	Cookie, chocolate, with chocolate filling or coating, fat free
53208000	Cookie, marshmallow, chocolate-covered
53208200	Cookie, marshmallow pie, chocolate covered
53209005	Cookie, chocolate, with icing or coating
53209010	Cookie, sugar wafer, chocolate-covered
53209015	Cookie, chocolate sandwich
53209020	Cookie, chocolate sandwich, reduced fat
53209100	Cookie, chocolate, sandwich, with extra filling
53209500	Cookie, chocolate and vanilla sandwich
53210000	Cookie, chocolate wafer
53210900	Cookie, graham cracker with chocolate and marshmallow
53211000	Cookie bar, with chocolate, nuts, and graham crackers
53215500	Cookie, coconut
53220000	Cookie, fruit-filled bar
53220010	Cookie, fruit-filled bar, fat free
53220030	Cookie, fig bar
53220040	Cookie, fig bar, fat free
53222010	Cookie, fortune
53222020	Cookie, cone shell, ice cream type, wafer or cake
53223000	Cookie, gingersnaps
53223100	Cookie, granola
53224000	Cookie, ladyfinger
53224250	Cookie, lemon bar
53225000	Cookie, macaroon
53226000	Cookie, marshmallow, with coconut
53226500	Cookie, marshmallow, with rice cereal, no bake
53226550	Cookie, marshmallow, with rice cereal and chocolate chips
53226600	Cookie, marshmallow and peanut butter, with oat cereal, no bake
53228000	Cookie, meringue
53230000	Cookie, molasses
53231000	Cookie, Lebkuchen
53231400	Cookie, multigrain, high fiber
53233000	Cookie, oatmeal
53233010	Cookie, oatmeal, with raisins
53233040	Cookie, oatmeal, reduced fat, NS as to raisins

53233050	Cookie, oatmeal sandwich, with creme filling
53233060	Cookie, oatmeal, with chocolate chips
53233080	Cookie, oatmeal sandwich, with peanut butter and jelly filling
53233100	Cookie, oatmeal, with chocolate and peanut butter, no bake
53234000	Cookie, peanut butter
53234100	Cookie, peanut butter, with chocolate
53234250	Cookie, peanut butter with rice cereal, no bake
53235000	Cookie, peanut butter sandwich
53235500	Cookie, with peanut butter filling, chocolate-coated
53235600	Cookie, Pfeffernusse
53236000	Cookie, Pizzelle
53236100	Cookie, pumpkin
53237000	Cookie, raisin
53237010	Cookie, raisin sandwich, cream-filled
53237500	Cookie, rum ball, no bake
53238000	Cookie, sandwich-type, not chocolate or vanilla
53239000	Cookie, shortbread
53239010	Cookie, shortbread, reduced fat
53239050	Cookie, shortbread, with icing or filling
53239100	Pocky
53240000	Cookie, animal
53240010	Cookie, animal, with frosting or icing
53241500	Cookie, butter or sugar
53241510	Marie biscuit
53241600	Cookie, butter or sugar, with fruit and/or nuts
53242000	Cookie, sugar wafer
53242500	Cookie, toffee bar
53243000	Cookie, vanilla sandwich
53243010	Cookie, vanilla sandwich, extra filling
53243050	Cookie, vanilla sandwich, reduced fat
53244010	Cookie, butter or sugar, with chocolate icing or filling
53244020	Cookie, butter or sugar, with icing or filling other than chocolate
53246000	Cookie, tea, Japanese
53247000	Cookie, vanilla wafer
53247050	Cookie, vanilla wafer, reduced fat
53247500	Cookie, vanilla with caramel, coconut, and chocolate coating
53251100	Cookie, rugelach
53260030	Cookie, chocolate chip, sugar free
53260200	Cookie, oatmeal, sugar free
53260300	Cookie, sandwich, sugar free
53260400	Cookie, sugar or plain, sugar free
53260500	Cookie, sugar wafer, sugar free
53260600	Cookie, peanut butter, sugar free
53261000	Cookie, gluten free
53270100	Cookies, Puerto Rican style
53344300	Dessert pizza

French toast, pancakes and waffles

[Kluyv® Protein] = 10 g/100 g

53400200	Blintz, cheese-filled
53400300	Blintz, fruit-filled
55100005	Pancakes, NFS
55100010	Pancakes, plain, from frozen
55100015	Pancakes, plain, reduced fat, from frozen
55100020	Pancakes, with fruit, from frozen
55100025	Pancakes, with chocolate, from frozen
55100030	Pancakes, whole grain, from frozen
55100035	Pancakes, whole grain, reduced fat, from frozen
55100040	Pancakes, gluten free, from frozen
55100050	Pancakes, plain, from fast food / restaurant
55100055	Pancakes, with fruit, from fast food / restaurant
55100060	Pancakes, with chocolate, from fast food / restaurant
55100065	Pancakes, whole grain, from fast food / restaurant
55100070	Pancakes, whole grain and nuts, from fast food / restaurant
55100080	Pancakes, from school, NFS
55101000	Pancakes, plain
55101015	Pancakes, plain, reduced fat
55103000	Pancakes, with fruit
55103020	Pancakes, pumpkin
55103100	Pancakes, with chocolate
55105000	Pancakes, buckwheat
55105100	Pancakes, cornmeal
55105200	Pancakes, whole grain
55105205	Pancakes, whole grain, reduced fat
55106000	Pancakes, gluten free
55200010	Waffle, NFS
55200020	Waffle, plain, from frozen
55200030	Waffle, plain, reduced fat, from frozen
55200040	Waffle, fruit, from frozen
55200050	Waffle, chocolate, from frozen
55200060	Waffle, whole grain, from frozen
55200070	Waffle, whole grain, reduced fat, from frozen
55200080	Waffle, whole grain, fruit, from frozen
55200090	Waffle, gluten free, from frozen
55200100	Waffle, plain, from fast food / restaurant
55200110	Waffle, chocolate, from fast food / restaurant
55200120	Waffle, fruit, from fast food / restaurant
55200130	Waffle, whole grain, from fast food / restaurant
55200200	Waffle, from school, NFS
55201000	Waffle, plain
55203000	Waffle, fruit

55203600	Waffle, chocolate
55203700	Waffle, cinnamon
55204000	Waffle, cornmeal
55205000	Waffle, whole grain
55208000	Waffle, gluten free
55211050	Waffle, plain, reduced fat
55212000	Waffle, whole grain, reduced fat
55300010	French toast, NFS
55300020	French toast, plain, from frozen
55300030	French toast, whole grain, from frozen
55300040	French toast, gluten free, from frozen
55300050	French toast, plain, from fast food / restaurant
55300055	French toast, whole grain, from fast food / restaurant
55300060	French toast, from school, NFS
55301000	French toast, plain
55301010	French toast, plain, reduced fat
55301015	French toast, whole grain
55301020	French toast, whole grain, reduced fat
55301025	French toast, gluten free
55301030	French toast sticks, NFS
55301031	French toast sticks, plain, from frozen
55301040	French toast sticks, plain, from fast food / restaurant
55301048	French toast sticks, from school, NFS
55301050	French toast sticks, plain
55301055	French toast sticks, whole grain
55310100	Fried bread, Puerto Rican style
55400010	Crepe, NFS
55401000	Crepe, plain
55501000	Chinese pancake
55610300	Dumpling, plain
55701000	Cake made with glutinous rice
55702000	Cake or pancake made with rice flour and/or dried beans
55702100	Dosa (Indian), plain
55703000	Cake made with glutinous rice and dried beans

Bread (high protein)

[Kluy® Protein] = 15 g/100 g

51123010	Bread, high protein
51123020	Bread, high protein, toasted

Beverages

[Kluy® Protein] = 10 g/100 g

95101000	Nutritional drink or shake, ready-to-drink (Boost)
95101010	Nutritional drink or shake, ready-to-drink (Boost Plus)
95102000	Nutritional drink or shake, ready-to-drink (Camation Instant Breakfast)
95103000	Nutritional drink or shake, ready-to-drink (Ensure)
95103010	Nutritional drink or shake, ready-to-drink (Ensure Plus)
95104000	Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)
95105000	Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)
95106000	Nutritional drink or shake, ready-to-drink (Muscle Milk)
95106010	Nutritional drink or shake, ready-to-drink, light (Muscle Milk)
95110000	Nutritional drink or shake, ready-to-drink (Slim Fast)
95110010	Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)
95110020	Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)
95120000	Nutritional drink or shake, ready-to-drink, NFS
95120010	Nutritional drink or shake, high protein, ready-to-drink, NFS
95120020	Nutritional drink or shake, high protein, light, ready-to-drink, NFS
95120050	Nutritional drink or shake, liquid, soy-based
95320200	Sports drink (Gatorade G)
95320500	Sports drink (Powerade)
95321000	Sports drink, NFS
95322200	Sports drink, low calorie (Gatorade G2)
95322500	Sports drink, low calorie (Powerade Zero)
95323000	Sports drink, low calorie

Beverages adjusted for reconstitution (factor of 7)

[Kluby® Protein] = 70 g/100 g

92900300	Sports drink, dry concentrate, not reconstituted
95201000	Nutritional powder mix (Camation Instant Breakfast)
95201010	Nutritional powder mix, sugar free (Camation Instant Breakfast)
95201200	Nutritional powder mix (EAS Whey Protein Powder)
95201300	Nutritional powder mix (EAS Soy Protein Powder)
95201500	Nutritional powder mix, high protein (Herbalife)
95201600	Nutritional powder mix (Isopure)
95201700	Nutritional powder mix (Kellogg's Special K20 Protein Water)
95202000	Nutritional powder mix (Muscle Milk)
95202010	Nutritional powder mix, light (Muscle Milk)
95210000	Nutritional powder mix (Slim Fast)
95210010	Nutritional powder mix, sugar free (Slim Fast)
95210020	Nutritional powder mix, high protein (Slim Fast)
95220000	Nutritional powder mix, NFS
95220010	Nutritional powder mix, high protein, NFS
95230000	Nutritional powder mix, whey based, NFS
95230010	Nutritional powder mix, protein, soy based, NFS
95230020	Nutritional powder mix, protein, light, NFS
95230030	Nutritional powder mix, protein, NFS

Breakfast cereals

[Kluyv® Protein] = 15 g/100 g

53710400	Cereal or granola bar (General Mills Fiber One Chewy Bar)
53710500	Cereal or granola bar (Kellogg's Nutri-Grain Cereal Bar)
53710502	Cereal or granola bar (Kellogg's Nutri-Grain Yogurt Bar)
53710504	Cereal or granola bar (Kellogg's Nutri-Grain Fruit and Nut Bar)
53710600	Milk 'n Cereal bar
53710700	Cereal or granola bar (Kellogg's Special K bar)
53710800	Cereal or granola bar (Kashi Chewy)
53710802	Cereal or granola bar (Kashi Crunchy)
53710810	Cereal or granola bar (KIND Fruit and Nut Bar)
53710900	Cereal or granola bar (General Mills Nature Valley Chewy Trail Mix)
53710902	Cereal or granola bar, with yogurt coating (General Mills Nature Valley Chewy Granola Bar)
53710904	Cereal or granola bar (General Mills Nature Valley Sweet and Salty Granola Bar)
53710906	Cereal or granola bar (General Mills Nature Valley Crunchy Granola Bar)
53711000	Cereal or granola bar (Quaker Chewy Granola Bar)
53711002	Cereal or granola bar (Quaker Chewy 90 Calorie Granola Bar)
53711004	Cereal or granola bar (Quaker Chewy 25% Less Sugar Granola Bar)
53711006	Cereal or granola bar (Quaker Chewy Dippys Granola Bar)
53711100	Cereal or granola bar (Quaker Granola Bites)
53712000	Snack bar, oatmeal
53712100	Cereal or Granola bar, NFS
53712200	Cereal or granola bar, lowfat, NFS
53712210	Cereal or granola bar, nonfat
53713000	Cereal or granola bar, reduced sugar, NFS
53713010	Cereal or granola bar, fruit and nut
53713100	Cereal or granola bar, peanuts , oats, sugar, wheat germ
53714200	Cereal or granola bar, chocolate coated, NFS
53714210	Cereal or granola bar, with coconut, chocolate coated
53714220	Cereal or granola bar with nuts, chocolate coated
53714230	Cereal or granola bar, oats, nuts, coated with non-chocolate coating
53714250	Cereal or granola bar, coated with non-chocolate coating
53714300	Cereal or granola bar, high fiber, coated with non-chocolate yogurt coating
53714400	Cereal or granola bar, with rice cereal
53714500	Breakfast bar, NFS
53714510	Breakfast bar, date, with yogurt coating
53714520	Breakfast bar, cereal crust with fruit filling, lowfat
53720100	Nutrition bar (Balance Original Bar)
53720200	Nutrition bar (Clif Bar)
53720210	Nutrition bar (Clif Kids Organic Zbar)
53720300	Nutrition bar (PowerBar)
53720400	Nutrition bar (Slim Fast Original Meal Bar)
53720500	Nutrition bar (Snickers Marathon Protein Bar)
53720600	Nutrition bar (South Beach Living Meal Bar)

53720610	Nutrition bar (South Beach Living High Protein Bar)
53720700	Nutrition bar (Tiger's Milk)
53720800	Nutrition bar (Zone Perfect Classic Crunch)
53729000	Nutrition bar or meal replacement bar, NFS
56200300	Cereal, cooked, NFS
56200990	Grits, NS as to regular, quick, or instant, NS as to fat
56201000	Grits, NS as to regular, quick, or instant, no added fat
56201040	Grits, NS as to regular, quick, or instant, fat added
56201050	Grits, regular or quick, made with water, NS as to fat
56201051	Grits, regular or quick, made with water, no added fat
56201052	Grits, regular or quick, made with water, fat added
56201055	Grits, regular or quick, made with milk, NS as to fat
56201056	Grits, regular or quick, made with milk, no added fat
56201057	Grits, regular or quick, made with milk, fat added
56201065	Grits, regular or quick, made with non-dairy milk, NS as to fat
56201066	Grits, regular or quick, made with non-dairy milk, no added fat
56201067	Grits, regular or quick, made with non-dairy milk, fat added
56201090	Grits, with cheese, NS as to fat
56201091	Grits, with cheese, no added fat
56201092	Grits, with cheese, fat added
56201210	Grits, instant, made with water, no added fat
56201220	Grits, instant, made with water, fat added
56201230	Grits, instant, made with water, NS as to fat
56201340	Grits, instant, made with milk, fat added
56201342	Grits, instant, made with milk, no added fat
56201344	Grits, instant, made with milk, NS as to fat
56201350	Grits, instant, made with non-dairy milk, NS as to fat
56201355	Grits, instant, made with non-dairy milk, no added fat
56201360	Grits, instant, made with non-dairy milk, fat added
56201515	Commeal mush, NS as to fat
56201540	Commeal, Puerto Rican Style
56201550	Commeal dumpling
56201600	Masa harina, cooked
56202100	Millet, fat added
56202900	Oatmeal, from fast food, plain
56202905	Oatmeal, from fast food, maple flavored
56202910	Oatmeal, from fast food, fruit flavored
56202920	Oatmeal, from fast food, other flavors
56202960	Oatmeal, NS as to regular, quick, or instant, NS as to fat
56203000	Oatmeal, NS as to regular, quick, or instant, no added fat
56203040	Oatmeal, NS as to regular, quick, or instant, fat added
56203055	Oatmeal, regular or quick, made with water, NS as to fat
56203056	Oatmeal, regular or quick, made with water, no added fat
56203057	Oatmeal, regular or quick, made with water, fat added
56203065	Oatmeal, regular or quick, made with milk, NS as to fat
56203066	Oatmeal, regular or quick, made with milk, no added fat

56203067	Oatmeal, regular or quick, made with milk, fat added
56203075	Oatmeal, regular or quick, made with non-dairy milk, NS as to fat
56203076	Oatmeal, regular or quick, made with non-dairy milk, no added fat
56203077	Oatmeal, regular or quick, made with non-dairy milk, fat added
56203085	Oatmeal, instant, plain, made with water, NS as to fat
56203086	Oatmeal, instant, plain, made with water, no added fat
56203087	Oatmeal, instant, plain, made with water, fat added
56203095	Oatmeal, instant, plain, made with milk, NS as to fat
56203096	Oatmeal, instant, plain, made with milk, no added fat
56203097	Oatmeal, instant, plain, made with milk, fat added
56203105	Oatmeal, instant, plain, made with non-dairy milk, NS as to fat
56203106	Oatmeal, instant, plain, made with non-dairy milk, no added fat
56203107	Oatmeal, instant, plain, made with non-dairy milk, fat added
56203125	Oatmeal, instant, maple flavored, NS as to fat
56203130	Oatmeal, instant, maple flavored, no added fat
56203135	Oatmeal, instant, maple flavored, fat added
56203150	Oatmeal, instant, fruit flavored, NS as to fat
56203155	Oatmeal, instant, fruit flavored, no added fat
56203160	Oatmeal, instant, fruit flavored, fat added
56203170	Oatmeal, instant, other flavors, NS as to fat
56203175	Oatmeal, instant, other flavors, no added fat
56203180	Oatmeal, instant, other flavors, fat added
56203500	Oatmeal, reduced sugar, plain, NS as to fat
56203510	Oatmeal, reduced sugar, plain, no added fat
56203520	Oatmeal, reduced sugar, plain, fat added
56203540	Oatmeal, made with milk and sugar, Puerto Rican style
56203550	Oatmeal, reduced sugar, flavored, NS as to fat
56203555	Oatmeal, reduced sugar, flavored, no added fat
56203560	Oatmeal, reduced sugar, flavored, fat added
56203600	Oatmeal, multigrain, NS as to fat
56203610	Oatmeal, multigrain, no added fat
56203620	Oatmeal, multigrain, fat added
56205050	Rice, cream of, cooked, no added fat
56205080	Rice, creamed, made with milk and sugar, Puerto Rican style
56205090	Rice, cream of, cooked, fat added
56205092	Rice, cream of, cooked, NS as to fat
56205094	Rice, cream of, cooked, made with milk
56205101	Congee
56206990	Cream of wheat, NS as to regular, quick, or instant, NS as to fat
56207000	Cream of wheat, NS as to regular, quick, or instant, no added fat
56207005	Cream of wheat, NS as to regular, quick, or instant, fat added
56207015	Cream of wheat, regular or quick, made with water, NS as to fat
56207016	Cream of wheat, regular or quick, made with water, no added fat
56207017	Cream of wheat, regular or quick, made with water, fat added
56207021	Cream of wheat, regular or quick, made with milk, NS as to fat
56207022	Cream of wheat, regular or quick, made with milk, no added fat

56207023	Cream of wheat, regular or quick, made with milk, fat added
56207025	Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat
56207026	Cream of wheat, regular or quick, made with non-dairy milk, no added fat
56207027	Cream of wheat, regular or quick, made with non-dairy milk, fat added
56207030	Cream of wheat, instant, made with water, no added fat
56207050	Wheat, cream of, cooked, made with milk and sugar, Puerto Rican style
56207060	Cream of wheat, instant, made with water, fat added
56207070	Cream of wheat, instant, made with water, NS as to fat
56207094	Cream of wheat, instant, made with milk, fat added
56207095	Cream of wheat, instant, made with milk, no added fat
56207096	Cream of wheat, instant, made with milk, NS as to fat
56207101	Cream of wheat, instant, made with non-dairy milk, NS as to fat
56207102	Cream of wheat, instant, made with non-dairy milk, no added fat
56207103	Cream of wheat, instant, made with non-dairy milk, fat added
56207190	Whole wheat cereal, cooked, NS as to fat
56207200	Whole wheat cereal, cooked, no added fat
56207210	Whole wheat cereal, cooked, fat added
56207370	Wheat cereal, chocolate flavored, cooked
56208500	Oat bran cereal, cooked, no added fat
56208510	Oat bran cereal, cooked, fat added
56208520	Oat bran cereal, cooked, NS as to fat
57237100	Cereal (Post Honey Bunches of Oats Honey Roasted)
57237200	Cereal (Post Honey Bunches of Oats with Vanilla Bunches)
57237300	Cereal (Post Honey Bunches of Oats with Almonds)

Condiments and relishes

[Klavy® Protein] = 15 g/100 g

27150210	Fish sauce
28520100	Oyster sauce
41420100	Miso sauce
41420250	Hoisin sauce
41420300	Soy sauce
41420350	Soy sauce, reduced sodium
41420450	Worcestershire sauce
74401010	Ketchup
74401110	Ketchup, reduced sodium
74402010	Tomato chili sauce
74402150	Salsa, red, commercially-prepared
74402210	Taco sauce
74402350	Salsa verde or salsa, green
74403000	Hot Thai sauce
75500110	Green beans, pickled
75500210	Beets, pickled
75500510	Celery, pickled

75501010	Relish, com
75502010	Cauliflower, pickled
75502500	Cabbage, fresh, pickled, Japanese style
75502510	Cabbage, red, pickled
75502520	Kimchi
75503010	Pickles, dill
75503020	Relish, pickle
75503040	Pickles, sweet
75503080	Eggplant, pickled
75503085	Ginger root, pickled
75505000	Mushrooms, pickled
75507000	Okra, pickled
75511010	Hot pepper sauce
75511020	Peppers, pickled
75511040	Pepper, hot, pickled
75512010	Radishes, pickled, Hawaiian style
75513010	Seaweed, pickled
75515100	Vegetables, pickled
75534030	Turnip, pickled
75534550	Wasabi paste
75535000	Zucchini, pickled
81302040	Sandwich spread
81302050	Tartar sauce
81302060	Horseradish sauce
81308100	Fry sauce
91361050	Duck sauce

Dairy product analogs

[Kluyv® Protein] = 8 g/100 g

11300100	Non-dairy milk, NFS
11321000	Soy milk, chocolate
11321100	Soy milk, light, chocolate
11321200	Soy milk, nonfat, chocolate
11350010	Almond milk, sweetened, chocolate
11350030	Almond milk, unsweetened, chocolate
11512030	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk
11512120	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk and whipped cream
11513310	Chocolate milk, made from dry mix with non-dairy milk
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
11513385	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)
11513395	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)
11513750	Chocolate milk, made from syrup with non-dairy milk
11513805	Chocolate milk, made from light syrup with non-dairy milk
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk

11514150	Hot chocolate / Cocoa, made with dry mix and non-dairy milk
11514360	Hot chocolate / Cocoa, made with no sugar added dry mix and non-dairy milk
11519215	Strawberry milk, non-dairy
12210520	Coffee creamer, soy, liquid
12320100	Sour cream, imitation
14502000	Imitation cheese
41420380	Yogurt, soy
42401100	Yogurt, coconut milk
42402010	Coconut cream, canned, sweetened

Frozen dairy desserts and mixes

[Kluyv® Protein] = 8 g/100 g

11459990	Frozen yogurt, NFS
11460000	Frozen yogurt, vanilla
11460100	Frozen yogurt, chocolate
11460500	Frozen yogurt, soft serve, vanilla
11460510	Frozen yogurt, soft serve, chocolate
11461200	Frozen yogurt sandwich
11461210	Frozen yogurt bar, vanilla
11461220	Frozen yogurt bar, chocolate
11461250	Frozen yogurt cone, chocolate
11461260	Frozen yogurt cone, vanilla
11461300	Frozen yogurt cone, vanilla, waffle cone
11461320	Frozen yogurt cone, chocolate, waffle cone
13110000	Ice cream, NFS
13110100	Ice cream, vanilla
13110102	Ice cream, vanilla, with additional ingredients
13110110	Ice cream, chocolate
13110112	Ice cream, chocolate, with additional ingredients
13110200	Ice cream, soft serve, vanilla
13110210	Ice cream, soft serve, chocolate
13110460	Gelato, vanilla
13110470	Gelato, chocolate
13120050	Ice cream bar, vanilla
13120100	Ice cream bar, vanilla, chocolate coated
13120110	Ice cream candy bar
13120140	Ice cream bar, chocolate
13120500	Ice cream sandwich, vanilla
13120510	Ice cream sandwich, chocolate
13120550	Ice cream cookie sandwich
13120730	Ice cream cone, scooped, vanilla
13120735	Ice cream cone, scooped, vanilla, waffle cone
13120740	Ice cream cone, NFS
13120770	Ice cream cone, scooped, chocolate

13120775	Ice cream cone, scooped, chocolate, waffle cone
13120782	Ice cream cone, soft serve, vanilla
13120784	Ice cream cone, soft serve, chocolate
13120786	Ice cream cone, soft serve, vanilla, waffle cone
13120788	Ice cream cone, soft serve, chocolate, waffle cone
13120790	Ice cream cone, vanilla, prepackaged
13120792	Ice cream cone, chocolate, prepackaged
13120800	Ice cream soda, flavors other than chocolate
13120810	Ice cream soda, chocolate
13121000	Ice cream sundae, NFS
13121100	Ice cream sundae, fruit topping
13121300	Ice cream sundae, hot fudge topping
13121400	Ice cream sundae, caramel topping
13126000	Ice cream, fried
13130100	Light ice cream, NFS
13130300	Light ice cream, vanilla
13130310	Light ice cream, chocolate
13130700	Soft serve, blended with candy or cookies, from fast food
13135000	Light ice cream sandwich, vanilla
13135010	Light ice cream sandwich, chocolate
13140000	Light ice cream bar, vanilla
13140100	Light ice cream bar, vanilla, chocolate coated
13140115	Light ice cream bar, chocolate
13140700	Creamsicle
13140710	Creamsicle, light
13140900	Fudgesicle
13142100	Light ice cream cone, vanilla, prepackaged
13142110	Light ice cream cone, chocolate, prepackaged
13150000	Sherbet, all flavors
13161600	Fudgesicle, light

Grain products and pastas

[Kluby® Protein] = 8 g/100 g

56112000	Noodles, cooked
56113000	Noodles, whole grain, cooked
56113990	Noodles, vegetable, cooked
56116000	Noodles, chow mein
56116990	Long rice noodles, made from mung beans, cooked
56117090	Rice noodles, cooked
56130000	Pasta, cooked
56132990	Pasta, whole grain, cooked
56140100	Pasta, gluten free
58122210	Gnocchi, cheese

Gravies and sauces

[Kluyv® Protein] = 10 g/100 g

13411000	White sauce or gravy
14650100	Cheese sauce
14650160	Alfredo sauce
14650165	Alfredo sauce with added vegetables
14650170	Alfredo sauce with meat
14650175	Alfredo sauce with meat and added vegetables
14650180	Alfredo sauce with poultry
14650185	Alfredo sauce with poultry and added vegetables
14650190	Alfredo sauce with seafood
14650195	Alfredo sauce with seafood and added vegetables
27150140	Seafood sauce
27150151	Spaghetti sauce with seafood
27150155	Spaghetti sauce with seafood and added vegetables
28500000	Gravy, poultry
28500040	Gravy, beef
28501010	Gravy, beef, fat free
28501110	Gravy, poultry, fat free
28520000	Gravy, made with soy sauce
28520010	Gravy, NFS
28522000	Mole sauce
41205100	Black bean sauce
41420400	Teriyaki sauce
41420410	Teriyaki sauce, reduced sodium
42204100	Gravy, vegetarian
74402250	Enchilada sauce, red
74402260	Enchilada sauce, green
74404010	Spaghetti sauce
74404020	Spaghetti sauce with added vegetables
74404050	Spaghetti sauce, reduced sodium
74404060	Spaghetti sauce, fat free
74404090	Vodka sauce with tomatoes and cream
74406010	Barbecue sauce
74406060	Buffalo sauce
74406100	Steak sauce
74406500	Cocktail sauce
81301000	Garlic sauce
81302010	Hollandaise sauce
81302070	Pesto sauce
81312100	Curry sauce
89901010	Cream sauce, for use with vegetables
89901020	Cheese sauce, for use with vegetables
89901030	Gravy, for use with vegetables

89901040	Soy based sauce, for use with vegetables
89901050	Tomato sauce, for use with vegetables

Milk products

[Kluyv® Protein] = 10 g/100 g

11120000	Milk, dry, reconstituted, NS as to fat content
11121100	Milk, dry, reconstituted, whole
11121210	Milk, dry, reconstituted, low fat (1%)
11121300	Milk, dry, reconstituted, fat free (skim)
11400000	Yogurt, NFS
11400010	Yogurt, Greek, NS as to type of milk or flavor
11410000	Yogurt, NS as to type of milk or flavor
11411010	Yogurt, NS as to type of milk, plain
11411100	Yogurt, whole milk, plain
11411200	Yogurt, low fat milk, plain
11411300	Yogurt, nonfat milk, plain
11411390	Yogurt, Greek, NS as to type of milk, plain
11411400	Yogurt, Greek, whole milk, plain
11411410	Yogurt, Greek, low fat milk, plain
11411420	Yogurt, Greek, nonfat milk, plain
11430000	Yogurt, NS as to type of milk, fruit
11431000	Yogurt, whole milk, fruit
11432000	Yogurt, low fat milk, fruit
11433000	Yogurt, nonfat milk, fruit
11433990	Yogurt, Greek, NS as to type of milk, fruit
11434000	Yogurt, Greek, whole milk, fruit
11434010	Yogurt, Greek, low fat milk, fruit
11434020	Yogurt, Greek, nonfat milk, fruit
11434090	Yogurt, NS as to type of milk, flavors other than fruit
11434100	Yogurt, whole milk, flavors other than fruit
11434200	Yogurt, low fat milk, flavors other than fruit
11434300	Yogurt, nonfat milk, flavors other than fruit
11435000	Yogurt, Greek, NS as to type of milk, flavors other than fruit
11435010	Yogurt, Greek, whole milk, flavors other than fruit
11435020	Yogurt, Greek, low fat milk, flavors other than fruit
11435030	Yogurt, Greek, nonfat milk, flavors other than fruit
11435100	Yogurt, Greek, with oats
11436000	Yogurt, liquid
11446000	Yogurt parfait, low fat, with fruit
11511000	Chocolate milk, NFS
11511100	Chocolate milk, ready to drink, whole
11511200	Chocolate milk, ready to drink, reduced fat
11511300	Chocolate milk, ready to drink, fat free
11511400	Chocolate milk, ready to drink, low fat

11511550	Chocolate milk, ready to drink, reduced sugar, NS as to milk
11511600	Chocolate milk, ready to drink, low fat (Nesquik)
11511610	Chocolate milk, ready to drink, fat free (Nesquik)
11511700	Chocolate milk, ready to drink, low fat, no sugar added (Nesquik)
11512010	Hot chocolate / Cocoa, ready to drink
11512020	Hot chocolate / Cocoa, ready to drink, made with nonfat milk
11512100	Hot chocolate / Cocoa, ready to drink, with whipped cream
11512110	Hot chocolate / Cocoa, ready to drink, made with nonfat milk and whipped cream
11513000	Chocolate milk, made from dry mix, NS as to type of milk
11513100	Chocolate milk, made from dry mix with whole milk
11513150	Chocolate milk, made from dry mix with reduced fat milk
11513200	Chocolate milk, made from dry mix with low fat milk
11513300	Chocolate milk, made from dry mix with fat free milk
11513350	Chocolate milk, made from reduced sugar mix, NS as to type of milk
11513355	Chocolate milk, made from reduced sugar mix with whole milk
11513360	Chocolate milk, made from reduced sugar mix with reduced fat milk
11513365	Chocolate milk, made from reduced sugar mix with low fat milk
11513370	Chocolate milk, made from reduced sugar mix with fat free milk
11513380	Chocolate milk, made from dry mix, NS as to type of milk (Nesquik)
11513381	Chocolate milk, made from dry mix with whole milk (Nesquik)
11513382	Chocolate milk, made from dry mix with reduced fat milk (Nesquik)
11513383	Chocolate milk, made from dry mix with low fat milk (Nesquik)
11513384	Chocolate milk, made from dry mix with fat free milk (Nesquik)
11513390	Chocolate milk, made from no sugar added dry mix, NS as to type of milk (Nesquik)
11513391	Chocolate milk, made from no sugar added dry mix with whole milk (Nesquik)
11513392	Chocolate milk, made from no sugar added dry mix with reduced fat milk (Nesquik)
11513393	Chocolate milk, made from no sugar added dry mix with low fat milk (Nesquik)
11513394	Chocolate milk, made from no sugar added dry mix with fat free milk (Nesquik)
11513400	Chocolate milk, made from syrup, NS as to type of milk
11513500	Chocolate milk, made from syrup with whole milk
11513550	Chocolate milk, made from syrup with reduced fat milk
11513600	Chocolate milk, made from syrup with low fat milk
11513700	Chocolate milk, made from syrup with fat free milk
11513800	Chocolate milk, made from light syrup, NS as to type of milk
11513801	Chocolate milk, made from light syrup with whole milk
11513802	Chocolate milk, made from light syrup with reduced fat milk
11513803	Chocolate milk, made from light syrup with low fat milk
11513804	Chocolate milk, made from light syrup with fat free milk
11513850	Chocolate milk, made from sugar free syrup, NS as to type of milk
11513851	Chocolate milk, made from sugar free syrup with whole milk
11513852	Chocolate milk, made from sugar free syrup with reduced fat milk
11513853	Chocolate milk, made from sugar free syrup with low fat milk
11513854	Chocolate milk, made from sugar free syrup with fat free milk
11514100	Hot chocolate / Cocoa, made with dry mix and water
11514110	Hot chocolate / Cocoa, made with dry mix and whole milk
11514120	Hot chocolate / Cocoa, made with dry mix and reduced fat milk

11514130	Hot chocolate / Cocoa, made with dry mix and low fat milk
11514140	Hot chocolate / Cocoa, made with dry mix and fat free milk
11514310	Hot chocolate / Cocoa, made with no sugar added dry mix and water
11514320	Hot chocolate / Cocoa, made with no sugar added dry mix and whole milk
11514330	Hot chocolate / Cocoa, made with no sugar added dry mix and reduced fat milk
11514340	Hot chocolate / Cocoa, made with no sugar added dry mix and low fat milk
11514350	Hot chocolate / Cocoa, made with no sugar added dry mix and fat free milk
11519040	Strawberry milk, NFS
11519050	Strawberry milk, whole
11519105	Strawberry milk, reduced fat
11519200	Strawberry milk, low fat
11519205	Strawberry milk, fat free
11519210	Strawberry milk, reduced sugar
11526000	Milk, malted
11541110	Milk shake, home recipe, chocolate
11541120	Milk shake, home recipe, flavors other than chocolate
11541130	Milk shake, home recipe, chocolate, light
11541135	Milk shake, home recipe, flavors other than chocolate, light
11541400	Milk shake with malt
11542100	Milk shake, fast food, chocolate
11542200	Milk shake, fast food, flavors other than chocolate
11543000	Milk shake, bottled, chocolate
11543010	Milk shake, bottled, flavors other than chocolate
11551050	Licuada or Batido
11553100	Fruit smoothie, NFS
11553110	Fruit smoothie, with whole fruit and dairy
11553120	Fruit smoothie, with whole fruit and dairy, added protein
11553130	Fruit smoothie juice drink, with dairy
11560000	Chocolate milk drink
11830160	Chocolate beverage powder, dry mix, not reconstituted
11830165	Chocolate beverage powder, light, dry mix, not reconstituted
11830400	Strawberry beverage powder, dry mix, not reconstituted
12100100	Cream, NS as to light, heavy, or half and half
12110100	Cream, light
12120100	Cream, half and half
12120106	Cream, half and half, flavored
12120110	Cream, half and half, fat free
12130100	Cream, heavy
12140000	Cream, whipped
12200100	Coffee creamer, NFS
12210200	Coffee creamer, liquid
12210210	Coffee creamer, liquid, flavored
12210260	Coffee creamer, liquid, fat free
12210270	Coffee creamer, liquid, fat free, flavored
12210280	Coffee creamer, liquid, fat free, sugar free, flavored
12210310	Coffee creamer, liquid, sugar free, flavored

12210400	Coffee creamer, powder
12210420	Coffee creamer, powder, flavored
12210430	Coffee creamer, powder, fat free
12210440	Coffee creamer, powder, fat free, flavored
12210505	Coffee creamer, powder, sugar free, flavored
12220200	Whipped topping
12220270	Whipped topping, fat free
12220280	Whipped topping, sugar free
12310100	Sour cream, regular
12310350	Sour cream, light
12310370	Sour cream, fat free

Plant protein products

[Kluyv® Protein] = 20 g/100 g

33001010	Egg substitute, omelet, scrambled, or fried, fat added
33001050	Egg substitute, omelet, scrambled, or fried, no added fat
33401000	Egg substitute, omelet, scrambled, or fried, with cheese
33401100	Egg substitute, omelet, scrambled, or fried, with meat
33401200	Egg substitute, omelet, scrambled, or fried, with vegetables
33401300	Egg substitute, omelet, scrambled, or fried, with cheese and meat
33401400	Egg substitute, omelet, scrambled, or fried, with cheese and vegetables
33401500	Egg substitute, omelet, scrambled, or fried, with meat and vegetables
33401600	Egg substitute, omelet, scrambled, or fried, with cheese, meat, and vegetables
41210000	Bean cake
59003000	Meat substitute, cereal- and vegetable protein-based, fried

Processed fruit and fruit juices

[Kluyv® Protein] = 9 g/100 g

63430150	Sorbet
64134015	Fruit smoothie, with whole fruit, no dairy
64134020	Fruit smoothie, with whole fruit, no dairy, added protein
64134025	Fruit smoothie, with whole fruit, non-dairy
64134030	Fruit smoothie juice drink, no dairy
64134100	Fruit smoothie, light
64134200	Fruit smoothie, bottled
64200100	Fruit nectar, NFS
95342000	Fruit juice, acai blend

Processed vegetable and vegetable juices

[Kluyv® Protein] = 9 g/100 g

75200700	Aloe vera juice drink
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Snack foods

[Kluyv® Protein] = 15 g/100 g

54318000	Chips, rice
54318500	Rice cake
54319020	Popcom cake
54319500	Rice paper
54401011	Corn nuts
54401021	Corn chips, plain
54401026	Corn chips, flavored
54401031	Corn chips, plain (Fritos)
54401035	Corn chips, flavored (Fritos)
54401055	Cheese flavored corn snacks
54401065	Cheese flavored corn snacks, reduced fat
54401075	Tortilla chips, plain
54401081	Cheese flavored corn snacks (Cheetos)
54401085	Tortilla chips, flavored
54401090	Corn chips, reduced sodium
54401095	Tortilla chips, popped
54401110	Tortilla chips, nacho cheese flavor (Doritos)
54401111	Tortilla chips, cool ranch flavor (Doritos)
54401112	Tortilla chips, other flavors (Doritos)
54401121	Tortilla chips, reduced fat, plain
54401122	Tortilla chips, reduced fat, flavored
54401170	Tortilla chips, low fat, unsalted
54402080	Tortilla chips, reduced sodium
54402200	Snack mix
54402610	Potato chips, restructured, multigrain
54402700	Pita chips
54403001	Popcom, NFS
54403005	Popcom, movie theater, with added butter
54403006	Popcom, movie theater, unbuttered
54403010	Popcom, air-popped, unbuttered
54403040	Popcom, air-popped, with added butter or margarine
54403051	Popcom, microwave, NFS
54403052	Popcom, microwave, plain
54403053	Popcom, microwave, plain, light
54403054	Popcom, microwave, low sodium
54403055	Popcom, microwave, unsalted
54403056	Popcom, microwave, butter flavored
54403057	Popcom, microwave, butter flavored, light
54403058	Popcom, microwave, cheese flavored
54403059	Popcom, microwave, kettle corn

54403061	Popcom, microwave, kettle corn, light
54403062	Popcom, microwave, other flavored
54403080	Popcom, ready-to-eat packaged, NFS
54403081	Popcom, ready-to-eat packaged, plain
54403082	Popcom, ready-to-eat packaged, plain, light
54403083	Popcom, ready-to-eat packaged, low sodium
54403084	Popcom, ready-to-eat packaged, unsalted
54403085	Popcom, ready-to-eat packaged, butter flavored
54403086	Popcom, ready-to-eat packaged, butter flavored, light
54403087	Popcom, ready-to-eat packaged, cheese flavored
54403088	Popcom, ready-to-eat packaged, cheese flavored, light
54403089	Popcom, ready-to-eat packaged, kettle corn
54403091	Popcom, ready-to-eat packaged, kettle corn, light
54403092	Popcom, ready-to-eat packaged, other flavored
54403110	Popcom, caramel coated
54403120	Popcom, caramel coated, with nuts
54403160	Popcom, chocolate coated
54404000	Popcorn chips, plain
54404010	Popcorn chips, other flavors
54404020	Popcorn chips, sweet flavors
54406010	Onion flavored rings
54406200	Shrimp chips
54408000	Pretzels, NFS
54408015	Pretzels, hard, NFS
54408016	Pretzels, hard, plain, salted
54408017	Pretzels, hard, plain, lightly salted
54408030	Pretzels, hard, plain, unsalted
54408035	Pretzels, hard, flavored
54408070	Pretzels, hard, multigrain
54408081	Pretzels, hard, plain, gluten free
54408082	Pretzels, hard, flavored, gluten free
54408105	Pretzel chips, hard, plain
54408110	Pretzel chips, hard, flavored
54408115	Pretzel chips, hard, gluten free
54408190	Pretzels, hard, coated, NFS
54408200	Pretzels, hard, chocolate coated
54408210	Pretzels, hard, white chocolate coated
54408250	Pretzels, hard, yogurt coated
54408260	Pretzels, hard, coated, gluten free
54408290	Pretzels, hard, filled, NFS
54408300	Pretzels, hard, cheese filled
54408310	Pretzels, hard, peanut butter filled
54420210	Multigrain chips (Sun Chips)
54420220	Snack mix, plain (Chex Mix)
54440010	Bagel chips
54440020	Cracker chips

71200010	Potato chips, NFS
71200100	Potato chips, plain
71200110	Potato chips, barbecue flavored
71200120	Potato chips, sour cream and onion flavored
71200130	Potato chips, cheese flavored
71200140	Potato chips, other flavored
71200200	Potato chips, ruffled, plain
71200210	Potato chips, ruffled, barbecue flavored
71200220	Potato chips, ruffled, sour cream and onion flavored
71200230	Potato chips, ruffled, cheese flavored
71200240	Potato chips, ruffled, other flavored
71200300	Potato chips, restructured, plain
71200310	Potato chips, restructured, flavored
71200400	Potato chips, baked, plain
71200410	Potato chips, baked, flavored
71201050	Potato chips, reduced fat
71201060	Potato chips, fat free
71201200	Potato chips, restructured, reduced fat, lightly salted
71201210	Potato chips, restructured, fat free
71202000	Potato chips, unsalted
71202100	Potato chips, reduced fat, unsalted
71202500	Potato chips, lightly salted
71202510	Potato chips, restructured, lightly salted
71203010	Potato chips, popped, plain
71203020	Potato chips, popped, flavored
71203030	Potato chips, popped, NFS
71205020	Potato sticks, plain
71205030	Potato sticks, flavored
71205040	Potato sticks, fry shaped
71220000	Vegetable chips
71980200	Taro chips
73410210	Sweet potato chips

Soft candy

[Kluyv® Protein] = 5 g/100 g

91700010	Candy, NFS
91700500	M&M's Almond Chocolate Candies
91701010	Almonds, chocolate covered
91701020	Almonds, sugar-coated
91701030	Almonds, yogurt-covered
91702010	Butterscotch morsels
91703010	Caramel, chocolate-flavored roll
91703020	Caramel, flavor other than chocolate
91703030	Caramel, with nuts

91703040	Caramel candy, chocolate covered
91703050	Caramel with nuts and cereal, chocolate covered
91703060	Caramel with nuts, chocolate covered
91703080	Caramel, all flavors, sugar free
91703150	Toblerone, milk chocolate with honey and almond nougat
91703200	TWIX Caramel Cookie Bars
91703250	TWIX Chocolate Fudge Cookie Bars
91703300	TWIX Peanut Butter Cookie Bars
91703600	Espresso coffee beans, chocolate-covered
91705010	Milk chocolate candy, plain
91705020	Milk chocolate candy, with cereal
91705040	Chocolate, milk, with nuts, not almond or peanuts
91705050	Milk chocolate candy, with fruit and nuts
91705060	Milk chocolate candy, with almonds
91705070	Chocolate, milk, with peanuts
91705090	Chocolate candy with fondant and caramel
91705200	Chocolate, semi-sweet morsel
91705300	Chocolate, sweet or dark
91705310	Chocolate, sweet or dark, with almonds
91705400	Chocolate, white
91705410	Chocolate, white, with almonds
91705420	Chocolate, white, with cereal
91705500	Mexican chocolate, tablet
91706000	Coconut candy, chocolate covered
91706100	Coconut candy, no chocolate covering
91706400	Coconut candy, Puerto Rican style
91707000	Fondant
91707010	Fondant, chocolate covered
91708000	Fruit peel, candied
91708010	Date candy
91708020	Soft fruit confections
91708030	Fruit leather and fruit snacks candy
91708040	Fun Fruits Creme Supremes
91708070	Tamarind candy
91708100	Fruit snacks candy, with high vitamin C
91708150	Yogurt covered fruit snacks candy, with added vitamin C
91708160	Yogurt covered fruit snacks candy rolls, with high vitamin C
91709000	Gumdrops, chocolate covered
91713010	Fudge, chocolate, chocolate-coated
91713020	Fudge, chocolate, chocolate-coated, with nuts
91713030	Fudge, chocolate
91713040	Fudge, chocolate, with nuts
91713050	Fudge, peanut butter
91713060	Fudge, peanut butter, with nuts
91713070	Fudge, vanilla
91713080	Fudge, vanilla, with nuts

91713090	Fudge, divinity
91713100	Fudge, brown sugar, penuche
91715000	Fudge, caramel and nut, chocolate-coated candy
91715100	SNICKERS Bar
91716110	Halvah, chocolate covered
91718100	Butterfinger
91718110	Butterfinger Crisp
91718200	Chocolate-flavored sprinkles
91718300	Ladoo, round ball, Asian-Indian dessert
91726000	Nougat, plain
91726110	Nougat, with caramel, chocolate covered
91726130	MILKY WAY Bar
91726140	MILKY WAY MIDNIGHT Bar
91726150	MARS Almond Bar
91726410	Nougat, chocolate covered
91726420	3 MUSKETEERS Bar
91726425	3 Musketeers Truffle Crisp Bar
91731010	M&M's Peanut Chocolate Candies
91731060	M&M's Peanut Butter Chocolate Candies
91733200	Peanut Bar, chocolate covered candy
91736000	Pineapple candy, Puerto Rican style
91746010	Sugar-coated chocolate discs
91746100	M&M's Milk Chocolate Candies
91746150	Easter egg, candy coated chocolate
91746200	M&M's Pretzel Chocolate Candies
91770000	Dietetic or low calorie candy, NFS
91770030	Dietetic or low calorie candy, chocolate covered
91770050	Dietetic or low calorie mints

Soups and soup mixes

[Kluy® Protein] = 8 g/100 g

14710100	Cheddar cheese soup, home recipe, canned or ready-to-serve
14710200	Beer cheese soup, made with milk
28310150	Oxtail soup
28310320	Beef noodle soup, Puerto Rican style
28310420	Beef and rice soup, Puerto Rican style
28311010	Pepperpot soup
28311020	Menudo soup, home recipe
28311030	Menudo soup, canned, prepared with water or ready-to-serve
28315050	Beef vegetable soup with potato, pasta, or rice, chunky style, canned, or ready-to-serve
28315140	Beef vegetable soup, home recipe, Mexican style
28315150	Meat and corn hominy soup, home recipe, Mexican style
28315160	Italian Wedding Soup
28317010	Beef stroganoff soup, chunky style, home recipe, canned or ready-to-serve

28320140	Ham, noodle, and vegetable soup, Puerto Rican style
28320160	Pork vegetable soup with potato, pasta, or rice, stew type, chunky style
28321130	Bacon soup, cream of, prepared with water
28331110	Lamb, pasta, and vegetable soup, Puerto Rican style
28340150	Mexican style chicken broth soup stock
28340210	Chicken rice soup, Puerto Rican style
28340220	Chicken soup with noodles and potatoes, Puerto Rican style
28340310	Chicken or turkey gumbo soup, home recipe, canned or ready-to-serve
28340510	Chicken or turkey noodle soup, chunky style, canned or ready-to-serve
28340550	Sweet and sour soup
28340580	Chicken or turkey soup with vegetables, broccoli, carrots, celery, potatoes and onions, Asian style
28340590	Chicken or turkey corn soup with noodles, home recipe
28340600	Chicken or turkey vegetable soup, canned, prepared with water or ready-to-serve
28340610	Chicken or turkey vegetable soup, stew type
28340630	Chicken or turkey vegetable soup with rice, stew type, chunky style
28340640	Chicken or turkey vegetable soup with noodles, stew type, chunky style, canned or ready-to-serve
28340660	Chicken or turkey vegetable soup, home recipe
28340670	Chicken or turkey vegetable soup with rice, home recipe, Mexican style
28340680	Chicken or turkey and corn hominy soup, home recipe, Mexican style
28340690	Chicken or turkey vegetable soup with potato and cheese, chunky style, canned or ready-to-serve
28340700	Bird's nest soup
28340750	Hot and sour soup
28340800	Chicken or turkey soup with vegetables and fruit, Asian Style
28345010	Chicken or turkey soup, cream of, canned, reduced sodium, NS as to made with milk or water
28345020	Chicken or turkey soup, cream of, canned, reduced sodium, made with milk
28345030	Chicken or turkey soup, cream of, canned, reduced sodium, made with water
28345110	Chicken or turkey soup, cream of, NS as to prepared with milk or water
28345120	Chicken or turkey soup, cream of, prepared with milk
28345130	Chicken or turkey soup, cream of, prepared with water
28345160	Chicken or turkey mushroom soup, cream of, prepared with milk
28350110	Crab soup, NS as to tomato-base or cream style
28350120	Crab soup, tomato-base
28350310	Turtle and vegetable soup
28351110	Fish and vegetable soup, no potatoes, Mexican style
28351120	Fish soup with potatoes, Mexican style
28351160	Codfish, rice, and vegetable soup, Puerto Rican style
28351170	Codfish soup with noodles, Puerto Rican style
28355110	Clam chowder, New England, NS as to prepared with water or milk
28355120	Clam chowder, New England, prepared with milk
28355130	Clam chowder, New England, prepared with water
28355140	Clam chowder, New England, reduced sodium, canned or ready-to-serve
28355210	Crab soup, cream of, prepared with milk
28355350	Salmon soup, cream style
28355410	Shrimp soup, cream of, NS as to prepared with milk or water

28355420	Shrimp soup, cream of, prepared with milk
28355430	Shrimp soup, cream of, prepared with water
28355450	Seafood soup with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy
28355460	Seafood soup with potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy
28355470	Seafood soup with vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes
28355480	Seafood soup with vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes
28360210	Spanish vegetable soup, Puerto Rican style
32300100	Egg drop soup
32301100	Garlic egg soup, Puerto Rican style
41601020	Bean with bacon or ham soup, canned or ready-to-serve
41601030	Black bean soup, home recipe, canned or ready-to-serve
41601040	Lima bean soup, home recipe, canned or ready-to-serve
41601070	Soybean soup, miso broth
41601080	Pinto bean soup, home recipe, canned or ready-to-serve
41601090	Bean soup, with macaroni, home recipe, canned, or ready-to-serve
41601110	Bean and ham soup, chunky style, canned or ready-to-serve
41601130	Bean soup, mixed beans, home recipe, canned or ready-to-serve
41601140	Bean soup, home recipe
41601160	Bean and ham soup, canned, reduced sodium, prepared with water or ready-to-serve
41601180	Bean and ham soup, home recipe
41602010	Pea and ham soup, chunky style, canned or ready-to-serve
41602020	Garbanzo bean or chickpea soup, home recipe, canned or ready-to-serve
41602030	Split pea and ham soup
41602050	Split pea soup
41602070	Split pea soup, canned, reduced sodium, prepared with water or ready-to-serve
41602090	Split pea and ham soup, canned, reduced sodium, prepared with water or ready-to-serve
41603010	Lentil soup, home recipe, canned, or ready-to-serve
58400000	Soup, NFS
58400100	Noodle soup, NFS
58400200	Rice soup, NFS
58401010	Barley soup, home recipe, canned, or ready-to-serve
58401200	Barley soup, sweet, with or without nuts, Asian Style
58402010	Beef noodle soup, canned or ready-to-serve
58402020	Beef dumpling soup, home recipe, canned or ready-to-serve
58402030	Beef rice soup, home recipe, canned or ready-to-serve
58403010	Chicken or turkey noodle soup, canned or ready-to-serve
58403050	Chicken or turkey noodle soup, cream of, home recipe, canned, or ready-to-serve
58403060	Chicken or turkey noodle soup, reduced sodium, canned or ready-to-serve
58403100	Noodle and potato soup, Puerto Rican style
58404010	Chicken or turkey rice soup, canned, or ready-to-serve
58404040	Chicken or turkey rice soup, reduced sodium, canned, prepared with water or ready-to-serve
58404050	Chicken or turkey rice soup, reduced sodium, canned, prepared with milk
58404100	Rice and potato soup, Puerto Rican style
58404500	Matzo ball soup
58404510	Chicken or turkey soup with dumplings and potatoes, home recipe, canned, or ready-to-serve
58404520	Chicken or turkey soup with dumplings, home recipe, canned or ready-to-serve

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Shanghai Changing Biotechnology Co., Ltd.

58407010	Instant soup, noodle
58407030	Soup, mostly noodles
58407035	Soup, mostly noodles, reduced sodium
58407050	Instant soup, noodle with egg, shrimp or chicken
58408010	Wonton soup
58408500	Noodle soup with vegetables, Asian style
58409000	Noodle soup, with fish ball, shrimp, and dark green leafy vegetable
58421000	Sopa seca, Mexican style, NFS
71801000	Potato soup, NS as to made with milk or water
71801010	Potato soup, cream of, prepared with milk
71801020	Potato soup, prepared with water
71801100	Potato and cheese soup
71803010	Potato chowder
71851010	Plantain soup, Puerto Rican style
73501000	Carrot soup, cream of, prepared with milk, home recipe, canned or ready-to-serve
73501010	Carrot with rice soup, cream of, prepared with milk, home recipe, canned or ready-to-serve
74601000	Tomato soup, NFS
74601010	Tomato soup, cream of, prepared with milk
74602010	Tomato soup, prepared with water, or ready-to-serve
74602050	Tomato soup, instant type, prepared with water
74602200	Tomato soup, canned, reduced sodium, prepared with water, or ready-to-serve
74602300	Tomato soup, canned, reduced sodium, prepared with milk
74603010	Tomato beef soup, prepared with water
74604010	Tomato beef noodle soup, prepared with water
74604100	Tomato beef rice soup, prepared with water
74604500	Tomato noodle soup, canned, prepared with water or ready-to-serve
74604600	Tomato noodle soup, canned, prepared with milk
74605010	Tomato rice soup, prepared with water
74606010	Tomato vegetable soup, prepared with water
74606020	Tomato vegetable soup with noodles, prepared with water
75600150	Soup, cream of, NFS
75601000	Asparagus soup, cream of, NS as to made with milk or water
75601010	Asparagus soup, cream of, prepared with milk
75601020	Asparagus soup, cream of, prepared with water
75601100	Borscht
75601200	Cabbage soup, home recipe, canned or ready-to-serve
75601210	Cabbage with meat soup, home recipe, canned or ready-to-serve
75603010	Celery soup, cream of, prepared with milk, home recipe, canned or ready-to-serve
75603020	Celery soup, cream of, prepared with water, home recipe, canned or ready-to-serve
75604010	Corn soup, cream of, prepared with milk
75604020	Corn soup, cream of, prepared with water
75604600	Gazpacho
75605010	Leek soup, cream of, prepared with milk
75607000	Mushroom soup, NFS
75607010	Mushroom soup, cream of, prepared with milk
75607020	Mushroom soup, cream of, prepared with water

75607040	Mushroom soup, with meat broth, prepared with water
75607050	Mushroom soup, cream of, low sodium, prepared with water
75607060	Mushroom soup, cream of, NS as to made with milk or water
75607080	Mushroom with chicken soup, cream of, prepared with milk
75607090	Mushroom soup, cream of, canned, reduced sodium, NS as to made with milk or water
75607100	Mushroom soup, cream of, canned, reduced sodium, prepared with milk
75607140	Mushroom soup, cream of, canned, reduced sodium, prepared with water
75608010	Onion soup, cream of, prepared with milk
75608100	Onion soup, French
75608200	Onion soup, made from dry mix
75609010	Pea soup, prepared with milk
75611010	Vegetable soup, cream of, prepared with milk
75612010	Zucchini soup, cream of, prepared with milk
75646010	Shav soup
75647000	Seaweed soup
75649010	Vegetable soup, canned, prepared with water or ready-to-serve
75649040	Vegetable soup, reduced sodium, canned, ready to serve
75649050	Vegetable soup, made from dry mix
75649110	Vegetable soup, home recipe
75649150	Vegetable noodle soup, home recipe
75650990	Minestrone soup, reduced sodium, canned or ready-to-serve
75651000	Minestrone soup, home recipe
75651010	Minestrone soup, canned, prepared with water, or ready-to-serve
75651020	Vegetable beef soup, canned, prepared with water, or ready-to-serve
75651030	Vegetable beef noodle soup, prepared with water
75651040	Vegetable noodle soup, canned, prepared with water, or ready-to-serve
75651070	Vegetable rice soup, canned, prepared with water or ready-to-serve
75651080	Vegetable beef soup with rice, canned, prepared with water or ready-to-serve
75651110	Vegetable chicken rice soup, canned, prepared with water or ready-to-serve
75651140	Vegetable soup with chicken broth, home recipe, Mexican style
75651150	Vegetable noodle soup, reduced sodium, canned, prepared with water or ready-to-serve
75652010	Vegetable beef soup, home recipe
75652030	Vegetable beef soup, canned, prepared with milk
75652040	Vegetable beef soup with noodles or pasta, home recipe
75652050	Vegetable beef soup with rice, home recipe
75654010	Vegetarian vegetable soup, prepared with water
75656010	Vegetable soup, Spanish style, stew type
75656020	Vegetable soup, chunky style
75656040	Vegetable soup, with pasta, chunky style
75656060	Vegetable beef soup, chunky style
75657000	Vegetable broth, bouillon

Other

Desserts

[Kluyv® Protein] = 10 g/100 g

13252200	Milk dessert or milk candy, Puerto Rican style
13252500	Barfi or Burfi, Indian dessert
13252590	Trifle
13252600	Tiramisu

Sweet bread and pastry

[Kluyv® Protein] = 15 g/100 g

58123110	Sweet bread dough, filled with meat, steamed
58123120	Sweet bread dough, filled with bean paste, meatless, steamed
58124210	Pastry, cheese-filled
58124220	Pastry, egg and cheese filled
58124230	Pastry, meat / poultry-filled

Appendix 5: Allergenicity Evaluation of Kluvy® Protein

RE Goodman Consulting
Kluyveromyces marxianus

Changjing Bio

REG-Changing
Page 1 of 9

Report Title

Review of literature, bioinformatics comparisons and stability data for potential food allergy for *Kluyveromyces marxianus*

Authors

Richard E. Goodman, Ph.D.

Completed On

20 September 2024

Performed by

**RE Goodman Consulting, LLC
8110 Dougan Circle
Lincoln, NE 68516**

Client

**Anchor Center for Certification
No. 1295 Chuangqiao Road, Building#2, Suite 302
Pudong, Shanghai, China 201206**

Laboratory Project ID

REG-ACC 20 Sept 2024

Summary

The Shanghai Changing Biotechnology Co., Ltd. (referred to as “Changing Bio”) has developed a native yeast organism grown in culture to produce yeast protein as a food ingredient source. They are working with the Anchor Center for Certification (ACC) to evaluate the safety of their product through a GRAS notification by the United States Food and Drug Administration. The yeast organism taxonomy is *Kluyveromyces marxianus* and it is closely related to other similar species of the larger taxonomic group called *Saccharomyces*. It is a budding yeast, closely related to *Kluyveromyces lactis* that has been used in the dairy industry to produce yogurt and other fermented products (Lane, MM, et al., 2010). It has been used in food production over multiple decades.

The nutritional properties of the product were characterized in the review by ACC and results appear reasonable as a protein source by mc. A literature search for this species for allergy or allergenicity was found to have only two references and they were not showing risks, only benefits of probiotics (Tullio, 2024; Ameli and Ciprandi, 2019). There are no peer-reviewed publications reporting food allergy from this yeast species.

The protein amino acid sequences provided by the company were compared to the major allergenic proteins in the eight priority allergen sources including peanut, milk, eggs, tree nuts, wheat, soybeans, fish and shellfish and did not identify any significant matches in www.AllergenOnline.org or other databases. Together with a lack of published findings of allergy from the literature, that suggests the possibility of food allergy from these ingredients does not exist.

RE Goodman Consulting	Changjing Bio	REG-Changing
<i>Kluyveromyces marxianus</i>		Page 3 of 9

Study Number: REG- Changing Yeast Protein

Title: Review of literature, bioinformatics comparisons and stability data for potential food allergy for *Kluyveromyces marxianus*

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Study Start Date: 1 September 2024

Study Completion Date: 20 September 2024

Records Retention: All study specific raw data and a copy of the final report will be retained by Richard E. Goodman.

	20 September 2024
Principle Investigator: Richard E. Goodman	Date

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Abbreviations and Definitions

AA	Amino acid
AC	Accession number from NCBI or UniProt
AOL v22	http://www.AllergenOnline.com/ database version 21
BLASTP	Algorithm used to find local high scoring alignments between a pair of protein sequences (using databases on Entrez)
Entrez NCBI	A public genetic database maintained by the National Center for Biotechnology Information (NCBI) at the National Institutes of Health, Bethesda, MD. Protein entries in the Entrez search and retrieval system are maintained by the NCBI of the National Institutes of Health (U.S.A.)
FARRP	Food Allergy Research and Resource Program, University of Nebraska
FASTA3	Algorithm used to find local high scoring alignments between a pair or protein sequences (using the AllergenOnline database)
GI	A unique identification number assigned by NCBI to each sequence in the database
PubMed	A public information database of scientific journal articles and abstracts maintained by the National Library of Medicine, National Institutes of Health (U.S.A.)
80mer	Sliding window of 80 amino acids of query protein are compared to AOL v 22 by FASTA3

1.0 Introduction.

The Shanghai Changing Biotechnology Company, Ltd has developed a non-transgenic, or native yeast organism culture as a protein-rich food protein source for foods. The company and the Anchor Center for Certification have considered food characteristics including nutritional properties, potential toxicity and potential food allergy for the product. Based on the literature search for risks for possible toxicity and allergenicity, they have concluded that the food products should be generally recognized as safe. I have reviewed the product description and their evaluation focusing on food allergy risks based on my knowledge of food allergy from GMO evaluations with CODEX guidelines, and based on allergen identification in managing the www.AllergenOnline.org database and have concluded the products should be considered Generally Recognized as Safe (GRAS).

A report from Changing Biotechnology or ACC describes the species compositional analysis, protein identification based on liquid chromatography-tandem mass spectrometry (LC-MSMS) and comparison of protein sequences to the major allergenic proteins in the eight-priority allergenic sources listed by the US FDA. I have reviewed their report and performed independent investigation of public data to consider possible risks of food allergy from the material that would be used as a human food protein source. My analysis is reported here.

Comparisons in www.AllergenOnline.org used a sliding 80 amino acid window looking for minimum identity matches of 35% or more. Additional searches were made with the NCBI Protein database using BLASTP and looking for long alignments of > 35% identity.

2. Purpose. The purpose of this study is to perform an evaluation of potential food allergy risks that might arise for consumers with existing allergies. The risks are best judged by whether the source organism is a common food allergy source, or if the protein amino acid sequence matches the identity of known allergens.

3. Methods.

3.1 Scientific literature search strategies. The PubMed database (<https://www.ncbi.nlm.nih.gov/pubmed>) maintained by the U.S. National Library of Medicine was used as the primary data source for scientific literature on allergy. Allergy to the yeast species, *Kluyveromyces marxianus* was performed by a general search of the species name AND "allergen" AND "allergy". In addition, Wikipedia was also searched with the species name. Information from those two searches were evaluated for evidence of allergy. The NCBI Taxonomy database was searched with the species name to investigate the apparent relationships with other fungi.

3.2 Amino acid sequence queries for allergenicity. The report from the Changing Biotechnology Company / ACC lists the top 100 proteins identified by LC-MSMS based on proportion relative to the total protein mass of the product. They compared the detected

sequences to the major allergens of the eight priority allergenic foods and did not find significant sequence identity matches. The list of 100 was reviewed based on my knowledge of allergens, and reviews in www.AllergenOnline.org and the WHO/IUIS Allergen Nomenclature database (www.Allergen.org).

4. Results.

4.1 Literature searches. No evidence of any allergy was found for this species. Not in PubMed, nor in Wikipedia. There were no publications identified in PubMed with keywords “*Kluyveromyces marxianus*” AND “allergen”. Two papers were identified with “*Kluyveromyces marxianus*” AND “allergy” but they showed a lack of allergy with exposure. One was by Amelia and Ciprandi in 2019, suggesting recurrent respiratory infections could be avoided by supplementary feeding with this species plus bovine colostrum. The other by Tullio in 2024 described probiotic use of this species or a few other yeasts that might reduce allergy.

Other publications were reviewed to describe the AllergenOnline database (Goodman et al., 2016), and to consider possible allergy and cross-reactivity for those who are already allergic (Aalberse, 2000). Proteins predicted from genomes are sometimes used to compare to allergen databases (Machida et al, 2005 and Abdelmoteleb et al., 2020). Most allergy protocols were to consider possible risks from transgenic or genetically modified organisms (Goodman, 2006), but are often over-predictive if based on the CODEX guidelines (2003 (Goodman et al. 2008), and would lead to unnecessary human serum IgE studies and possibly to basophil histamine release assays (Knol, 2006).

4.2 Bioinformatics using proteins identified by LC-MS/MS detected peptides. The report by Shanghai Changing Biotechnology Company describes and shows the results of LC-MS/MS identification of proteins. The more abundant ones are of the same overall protein types that have sometimes been called allergen for instance in WHO/IUIS Allergen Nomenclature database but the proof of reactivity for those proteins are not listed in peer-reviewed publications. Furthermore, there are no publications demonstrating allergy to proteins from this species, *Kluyveromyces marxianus*. The report from the company did not find any significant sequence match to clear allergens from the eight major allergenic foods. There are many proteins listed in both the AllergenOnline.org database and the WHO/IUIS allergen nomenclature database that do not have clear IgE binding or clinical allergy associated with them. It is therefore reasonable to assume based on scientific evidence, that proteins from this species are unlikely to cause allergy.

5. CONCLUSIONS

My overall conclusion from reviewing the data and information provided by the developer and based on literature searches is that the protein material generated from this product should be assumed to be Generally Recognized as Safe (GRAS).

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7.0 Appendix: Allergenonline.org database, version 22, 25 May 2023

Appendix 6: Report of Proteins Identification of Kluvy® Protein and Sequence Comparison with Major Allergens Using AllergenOnline

Study on the Allergenicity of *Kluyveromyces marxianus* Protein

Shanghai Changing Biotechnology Co., Ltd.

Project NO.: CJSW-SH-WSW-24-0920



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1. Test purpose

An evaluation of the potential for a new food protein to cause allergic reactions in consumers is a key component of a safety assessment. *Kluyveromyces marxianus* protein powder has been assessed for potential allergenicity risks by: 1) composition analysis including proteins, peptides and amino acids; 2) protein identification and comparison of protein sequence information.

2. Method

2.1 In-solution digestion

The samples were mixed with 200 µl of UA buffer (8 M Urea, 150 mM Tris HCl pH8.0) loaded into the filter devices and centrifuged at 14,000 g for 15 min. The concentrates were diluted in the device with 200 µl of UA buffer and centrifuged again. Then the concentrates were mixed with 100 µl of 50mM IAAs (3-Indoleacetic acid) in UA buffer and incubated in darkness at room temperature for 30 min followed by centrifugation for 15 min. Then, the concentrate was diluted with 100ul of UA buffer and concentrated again. This step was repeated twice. The concentrate was diluted with 100 µl of 25 mM NH₄HCO₃ and concentrated again. This step was repeated twice. The resulting concentrate was diluted to 40 µL of 25 mM NH₄HCO₃ containing 2 µg of Trypsin. After overnight incubation at 37°C, peptides were collected by centrifugation of the filter units for 20 min.

2.2 LC-MS/MS

Experiments were performed on a Q Exactive mass spectrometer that was coupled to Easy nLC (Thermo Fisher Scientific). The peptide mixture was loaded onto a the C18-reversed phase column (15 cm long, 75 µm inner diameter) packed in-house with RP-C18 5 µm resin in buffer A (0.1% Formic acid in HPLC-grade water) and separated with a linear gradient of buffer B (0.1% Formic acid in 84% acetonitrile) at a flow rate of 250 nl/min controlled by IntelliFlow technology over 60 min. MS data was acquired using a data-dependent top10 method dynamically choosing the most abundant precursor ions from the survey scan (300–1800 m/z) for HCD fragmentation. Determination of the target value is based on predictive Automatic Gain Control (pAGC). Dynamic exclusion duration was 20 s. Survey scans were acquired at a resolution of 70,000 at m/z 200 and resolution for HCD spectra was set to 17,500 at m/z 200. Normalized collision energy was 27 eV and the underfill ratio, which specifies the minimum percentage of the target value likely to be reached at maximum fill time, was defined as 0.1%. The instrument was run with peptide recognition mode enabled.

2.3 Data Analysis

MS/MS spectra were searched using MASCOT engine (Matrix Science, London, UK; version



2.2) against *Vibrio cholerae* sequence database (downloaded on November 4th, 2020; 114251 sequences). For protein identification, the following options were used. Peptide mass tolerance=20 ppm, MS/MS tolerance=0.1 Da, Enzyme=Trypsin, Missed cleavage=2, Fixed modification: Carbamidomethyl (C), Variable modification: Oxidation(M), ion score> 20 at peptide and protein level.

3. Detailed Results and Discussion

3.1 Analysis of protein content, peptide content and amino acid composition of *Kluyveromyces marxianus* protein powder

The water content, protein content, peptide content, and amino acid composition of *Kluyveromyces marxianus* protein powder were analyzed and the results were shown in Table 1. The protein content of *Kluyveromyces marxianus* protein powder (N×6,25) was 40.86 ± 0.24 g/100g and peptide content was 2.89 ± 0.17 g/100g.



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Table 1 Protein content, peptide content and amino acid composition of *Kluyveromyces marxianus* protein powder

Items	Methods	Results
Water content, g/100g	AOAC 930.04	4.22±0.12
Protein content (N × 6.25), g/100g	AOAC 979.09	40.86±0.24
Peptide content, g/100g	/	2.89±0.17
Asp, g/100g	ISO 13903:2005	5.23
Glu, g/100g	ISO 13903:2005	5.29
Ser, g/100g	ISO 13903:2005	1.97
His, g/100g	ISO 13903:2005	0.98
Gly, g/100g	ISO 13903:2005	1.94
Thr, g/100g	ISO 13903:2005	2.09
Arg, g/100g	ISO 13903:2005	2.13
Ala, g/100g	ISO 13903:2005	2.52
Tyr, g/100g	ISO 13903:2005	1.24
Cys, g/100g	ISO 13903:2005	0.27
Val, g/100g	ISO 13903:2005	2.86
Met, g/100g	ISO 13903:2005	0.46
Phe, g/100g	ISO 13903:2005	2.16
Ile, g/100g	ISO 13903:2005	2.47
Leu, g/100g	ISO 13903:2005	3.45
Lys, g/100g	ISO 13903:2005	3.47
Pro, g/100g	ISO 13903:2005	1.77
Trp, g/100g	ISO 13903:2005	0.45

Amino acid composition was studied for *Kluyveromyces marxianus* protein, along with digestible indispensable amino acid score (DIAAS) analysis, since DIAAS is the latest method for assessing the nutritional value of proteins. As shown in Table 2, *Kluyveromyces marxianus* protein contained approximately 25.82% of aspartic acid and glutamic acid, followed by leucine, lysine and valine. Compared to the FAO/WHO recommended amino acid pattern for children or adults, sulphur-containing amino acids were the limiting amino acids for *Kluyveromyces marxianus* protein. This



resulted in a relatively lower DIAAS score for *Kluyveromyces marxianus* protein.

Table 2 Amino acid composition (g/100g) and DIAAS of *Kluyveromyces marxianus* protein

Amino acids	<i>Kluyveromyces marxianus</i> protein	FAO/WHO recommended pattern	
		Children	Adult
Asp	12.84		
Glu	12.98		
Ser	4.83		
His*	2.40	2.0	1.6
Gly	4.76		
Thr*	5.13	3.1	2.5
Arg	5.23		
Ala	6.19		
Tyr	3.04		
Val*	7.02	4.3	4.0
Phe*	5.30		
Ile*	6.06	3.2	3.0
Leu*	8.47	6.6	6.1
Lys*	8.52	5.7	4.8
Pro	4.34		
Sulphur-containing amino acids (SAA)	1.79	2.7	2.3
Aromatic amino acids (AAA)	8.80	5.2	4.1
in vitro DIAAS (SAA, %)	49.07 (children)		
	58.00 (adults)		

"*" indicated essential amino acids; FAO/WHO recommended pattern referred to FAO/WHO Amino Acid Scoring Standard; the grey column indicates no specified value.

3.2 Identification of *Kluyveromyces marxianus* protein composition using mass spectrometry and comparison of protein sequences to known allergen using database

Kluyveromyces marxianus protein samples were pre-treated, and enzymatically hydrolyzed. The resulting peptides were quantified and analyzed by mass spectrometry and the protein identification results were shown in Table 3. According to the proportion of identified proteins, the top 100 yeast



proteins with an abundance more than 0.24% were screened out, and their specific protein names, gene names, molecular weights, and their proportions were shown in Table 3, respectively. The proteins listed in Table 3 comprised 79.21% of the total yeast proteins, while the remaining proteins were low-abundance, each constituting less than 0.24% of the total.

All 100 proteins identified by mass spectrometry were compared with those of allergenic proteins including the major food allergens (peanut, tree nuts, egg, milk, crustacean shellfish, fish, soybean, wheat). The results indicated that *Kluyveromyces marxianus* protein did not show strong or consistent matches with the common and potent allergenic proteins mentioned above.

There were no matches to the major peanut protein allergens such as Ara h 1, Ara h 2 and Ara h 3, no matches to soybean protein allergens such as Gly m 1a, Gly m 1b, Gly m 2, Gly m 3, no matches with soybean whey proteins, 2S globulin, 11S globulin (A1a B1b; A2B1a), the 7S globulin (α' -subunit, α -subunit and β -subunit from soybean β -conglycinin; Gly m Bd 28K; Gly m Bd 30K), Kunitz-type trypsin inhibitor, soybean lectin and lipoxygenase. There were no matches with vicilins or tropomyosins, which were major allergens of peanuts, tree nuts, shrimp, etc.

Table 4 shows the protein information that matches the eight major allergen sources based on the identified peptide sequences. There were no significant sequence matches to clinically important proteins from the major allergenic sources. Some minor proteins such as glyceraldehyde-3-phosphate dehydrogenase and alcohol dehydrogenase are listed as allergens in other databases (WHO/IUIS Allergen Nomenclature). However, they have not been demonstrated to cause clinical allergy. Those proteins are also conserved in sequence across many species and that lack of broad taxonomic species shared allergy demonstrates doubt on their role in clinical allergy. Thus, there does not appear to be a risk of allergy from those matched proteins.

4. Conclusion

Based on the study outcomes, *Kluyveromyces marxianus* protein is concluded to be unlikely to present allergenicity risks.



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Table 3 Protein mass spectrometry analysis of *Kluyveromyces marxianus* protein

No	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
1	Glyceraldehyde-3-phosphate dehydrogenase	GAP1 KLMA_40218	35.225	7.68
2	Phosphopyruvate hydratase	ENO KLMA_10462	46.802	6.13
3	Phosphoglycerate mutase	GPM1 KLMA_20098	28.854	3.12
4	Alcohol dehydrogenase 1	ADH1	37.158	2.73
5	Phosphoglycerate kinase	PGK KLMA_10540	44.444	2.66
6	Elongation factor 1-alpha		41.206	2.12
7	Glyceraldehyde-3-phosphate dehydrogenase	GAP3 KLMA_80059	35.523	2.11
8	ATP synthase subunit beta	ATP2 KLMA_60435	54.15	1.88
9	40S ribosomal protein S12	RPS12 KLMA_50008	14.463	1.84
10	Histone H4	HHF1 KLMA_80118	11.409	1.78
11	Heat shock protein SSA3	SSA3 KLMA_80367	70.078	1.77
12	Fructose-bisphosphate aldolase	FBA1 KLMA_40388	39.496	1.32
13	5-methyltetrahydropteroyltriglutamate-homocysteine S-methyltransferase	MET6 KLMA_80348	86.395	1.25
14	Sphingolipid long chain base-responsive protein LSP1	LSP1 KLMA_70451	35.665	1.18
15	40S ribosomal protein S15	RPS15 KLMA_20356	15.999	1.17
16	Peptidyl-prolyl cis-trans isomerase	CPR1 KLMA_50501	17.361	1.14
17	Uncharacterised protein	OM45 KLMA_40373	62.697	1.13
18	Phosphotransferase	RAG5 KLMA_60412	53.767	1.04



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No	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
19	Mitochondrial outer membrane protein porin 1	POR1 KLMA_80163	29.262	1.01
20	Heat shock protein SSB	SSB KLMA_50592	66.115	0.99
21	Citrate synthase	CIT1 KLMA_20105	53.266	0.95
22	60S ribosomal protein L2	RPL2 KLMA_50473	27.379	0.94
23	60S ribosomal protein L13)	rpl13a klma_10701	22.652	0.9
24	Ribosomal_L22	RPL17B KLMA_10376	20.655	0.86
25	ATP synthase subunit alpha	ATP1 KLMA_70140	59.117	0.85
26	60S ribosomal protein L19	RPL19B KLMA_70358	22.254	0.82
27	60S ribosomal protein L4-B	RPL4B KLMA_50245	39.268	0.76
28	Meiotic sister chromatid recombination protein 1	MSC1 KLMA_20659	56.285	0.75
29	60S ribosomal protein L35	rpl35b klma_60504	14.046	0.71
30	40S ribosomal protein S4	RPS4B KLMA_40482	29.537	0.69
31	40S ribosomal protein S8	RPS8B KLMA_80365	23.8	0.66
32	Glucose-6-phosphate isomerase	RAG2 KLMA_10763	61.591	0.66
33	Cytochrome b-c1 complex subunit 2	QCR2 KLMA_40479	39.023	0.66
34	Sphingolipid long chain base-responsive protein PIL1	PIL1 KLMA_20343	34.891	0.65
35	40S ribosomal protein S25	RPS25B KLMA_50210	11.957	0.62
36	Histone H2B	HTB1 KLMA_20169	14.291	0.6
37	Aconitate hydratase, mitochondrial	ACO2b	85.061	0.59



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No	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
		KLMA_30312		
38	60S ribosomal protein L24	RPL24 KLMA_20054	17.529	0.59
39	Cytochrome b-c ₁ complex subunit 1	COR1 KLMA_70095	48.406	0.58
40	60S ribosomal protein L3	RPL3 KLMA_30264	43.703	0.57
41	Potassium-activated aldehyde dehydrogenase	ALD4 KLMA_50012	56.58	0.56
42	ATP synthase subunit 5	atpH KLMA_30431	21.912	0.54
43	60S ribosomal protein L27-A	RPL27A KLMA_20558	15.565	0.54
44	Isocitrate dehydrogenase	IDP2 KLMA_30149	46.883	0.51
45	40S ribosomal protein S24)	RPS24B KLMA_40405	15.171	0.48
46	60S ribosomal protein L32	RPL32 KLMA_70149	14.721	0.48
47	Enoate reductase 1	KYE1 KLMA_10458	45.931	0.47
48	Pyruvate decarboxylase	PDC1 KLMA_60075	61.9	0.47
49	Alpha-1,4 glucan phosphorylase	GPH1 KLMA_20256	103.44	0.47
50	60S ribosomal protein L8	RPL8B KLMA_80413	28.086	0.45
51	Elongation factor 2	EFT1 KLMA_20582	93.351	0.45
52	6-Phosphogluconate dehydrogenase	GND1 KLMA_10468	53.429	0.45
53	Heat shock protein 60	HSP60 KLMA_20282	60.996	0.44
54	Heat shock protein SSC1	SSC1 KLMA_10712	69.437	0.43
55	Cell wall mannoprotein HSP150	PIR1 KLMA_50253	32.29	0.43
56	ATP synthase subunit d	ATP7 KLMA_60151	19.76	0.43
57	40S ribosomal protein S7	RPS7A	21.485	0.4



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No	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
		KLMA_80222		
58	Guanine nucleotide-binding protein subunit beta-like	ASC1 KLMA_30115	35.765	0.39
59	60S ribosomal protein L20	rpl20b klma_20318	20.453	0.38
60	60S ribosomal protein L31-A	RPL31 KLMA_40452	13.139	0.37
61	60S ribosomal protein L14-B	rpl14b klma_30066	15.104	0.36
62	NAD	XYL1 KLMA_10683	37.443	0.36
63	40S ribosomal protein S20	KLMA_60313	13.335	0.35
64	Ribosomal protein L23	RPL25 KLMA_40393	15.759	0.35
65	Transaldolase	TAL1 KLMA_30634	36.474	0.34
66	Transketolase	TKL1 KLMA_80302	73.84	0.33
67	Pyruvate kinase	PYK1 KLMA_60219	54.804	0.33
68	Acetyl-CoA carboxylase	ACC1 KLMA_60469	250.05	0.33
69	Alcohol dehydrogenase	ADH3	39.772	0.32
70	Hyaluronic acid-binding protein 4	KLMA_50529	27.35	0.32
71	Ribosomal_L18e super family	RPL28 KLMA_20403	16.516	0.32
72	Single-stranded nucleic acid-binding protein	SBP1 KLMA_80414	27.499	0.31
73	60S ribosomal protein L12	RPL12B KLMA_30664	17.838	0.31
74	60S ribosomal protein L26-B	RPL26B KLMA_50194	17.716	0.3
75	SDO1-like protein YHR087W	RTC3 KLMA_40247	12.513	0.3
76	Ketol-acid reductoisomerase	ILV5 KLMA_30631	44.714	0.3
77	Malate dehydrogenase	MDH1 KLMA_60167	36.089	0.3



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No.	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
78	Dihydrolipoyl dehydrogenase	LPD1 KLMA_60420	52.895	0.3
79	Cell wall protein ECM33	ECM33 KLMA_30554	42.756	0.29
80	Protoplast secreted protein 2	PST2 KLMA_60549	20.908	0.29
81	40S ribosomal protein S6	RPS6	26.702	0.29
82	40S ribosomal protein S26	RPS26A KLMA_20492	13.464	0.29
83	Actin	ACT KLMA_70051	42.04	0.28
84	Cell division control protein 48	CDC48 KLMA_60484	91.387	0.28
85	Fatty acid synthase subunit alpha	FAS2 KLMA_70219	207.52	0.28
86	Protein BMH2	BMH2 KLMA_20132	28.328	0.28
87	Elongation factor 1-beta	EFB1 KLMA_40028	23.065	0.27
88	alcohol dehydrogenase	ADH2 KLMA_40220	36.947	0.26
89	60S ribosomal protein L6-B	RPL6B KLMA_40443	23.32	0.25
90	Catalase	CTT1 KLMA_50419	62.475	0.25
91	40S ribosomal protein S16	RPS16 KLMA_10700	15.86	0.25
92	40S ribosomal protein S18	rps13 KLMA_40577	17.094	0.25
93	ATP-dependent RNA helicase	TIF1 KLMA_10348	44.596	0.25
94	Superoxide dismutase	sod1	15.917	0.25
95	Plasma membrane ATPase	PMA1 KLMA_10457	97.976	0.25
96	40S ribosomal protein S14	RPS14 KLMA_50264	14.178	0.25
97	ATP-dependent molecular chaperone	HSP82	81.407	0.25



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No.	Protein names	Gene names	Molecular Weight (kDa)	Proportion (%)
	HSC82	KLMA_50344		
98	60S ribosomal protein L5	RPL5 KLMA_20800	33.918	0.24
99	40S ribosomal protein S17-B	RPS17A KLMA_40581	15.756	0.24
100	40S ribosomal protein S0	RPS0 KLMA_60069	28.207	0.24



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Table 4 Protein information matched to the eight major allergen sources based on the identified peptide sequences

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
1	Keratin 5	62.623	Bos taurus (Bovine)	no	LRSEIDNVKK, NKYEDEINKR, SLDLDSIIAEVK, YEELQQTAGR	7.1	39.44	0.3275
2	Histone H2A	13.833	Bos taurus (Bovine)	no	AGLTFPVGR, HLQLAIR, HLQLAIRNDDELNK, KGNYAQR	23.1	26.01	0.3272
3	Triosephosphate isomerase	26.995	Triticum aestivum (Wheat)	no	RTYFHETDEFIADK, TYFHETDEFIADK, TYFHETDEFIADKTK, VILCIGETLEEK, VQDWTNVVVVAYEPVWAIGTGL AATSDDAQDIHHSIR, AQVSVA AQNAYLK, ASGAFTGENSVEQIK. DVGAEWVILGHSE, ASGAFTGENSVEQIKDVGAEWVI	46.4	142.56	0.3112



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No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
					LGHSEK, GITLEVVQR			
4	Heat shock protein	96.295	Bos taurus (Bovine)	no	LINEPTAAAIAYGLDK, LINEPTAAAIAYGLDKK	2	24.90	0.2612
5	60S acidic ribosomal protein P2-B	10.935	Bos taurus (Bovine)	no	AVLESVGIEAEDAK, SINELIAEGQTK, YLAAYLLIAQGGVEAPSATEIK	44	71.56	0.2293
6	Glyceraldehyde 3-phosphate dehydrogenase NAD(P) binding domain-containing protein	18.618	Bos taurus (Bovine)	no	IVSDASCTTNCLAPLAK	10.1	12.34	0.1700
7	60S	19.797	Bos taurus	no	LVLNISVGESGDR	21.3	25.07	0.1630



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No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	ribosomal protein L11		(Bovine)		QRYDAEVLDK, VLEQLSGQTPVHSK			
8	40S ribosomal protein S9	16.814	Bos taurus (Bovine)	no	IEDFLER, LFEGNALLR	11.4	12.06	0.1494
9	Photosystem II 5 kDa protein	11.256	Bos taurus (Bovine)	no	MASITMTTSIGGGMVLR	15.9	6.56	0.1464
10	Keratin 10	54.849	Arachis hypogaea (Peanut)	no	QSLEASLAETEGR QSVEADINGLRR DAEAWFNEK LENEIQTYR	8.2	36.20	0.1218
11	40S ribosomal protein S27	8.8692	Bos taurus (Bovine)	no	LSEGTSFR VLVQDLLHPTAASEAR	29.3	13.23	0.1062
12	Ubiquitin	104.29	Bos taurus	no	GLGGLGGLAGR	1.1	6.08	0.0678



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No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	carboxyl-terminal hydrolase		(Bovine)					
13	Late embryogenes is abundant protein LEA-2 subgroup domain-containing protein	26.351	Triticum aestivum (Wheat)	no	ATAVSATLSR	4.2	6.02	0.0459
14	fumarate hydratase	52.343	Glycine max (Soybean)	no	AIELLGGELGSK	2.4	6.20	0.0418
15	60S ribosomal	12.935	Glycine max	no	KSELEYYYAMLSK	10.2	7.04	0.0347



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No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	protein L30		(Soybean) (Glycine hispida)					
16	ADP/ATP translocase (ADP, ATP carrier protein)	36.997	Glycine max (Soybean)	no	LLLQNQDEMIK	3.3	8.95	0.0324
17	Ring finger protein 213	582.16	Glycine max (Soybean)	no	QLLDGQK	0.1	6.39	0.0320
18	Glutathione peroxidase	18.152	Glycine max (Soybean)	no	YSSLTKPSSLEATVEK	9.9	53.06	0.0319
19	Protein transport	9.3728	Juglans regia	no	QAGFGGSSSSILK	14.8	6.65	0.0241



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	protein Sec61 subunit beta		(English walnut)					
20	Probable trans-2-enoyl-CoA reductase	41.783	Juglans regia (English walnut)	no	SLIYSSHDQDVSK	3.4	6.89	0.0228
21	Non-histone chromosomal protein 6	10.634	Juglans regia (English walnut)	no	ALSEDEKAPYEAK	14	6.66	0.0207
22	Uncharacterised protein	7.0934	Glycine max (Soybean)	no	QIDGKMCYDQLVR	21.7	5.96	0.0184
23	Ras-related protein SEC4	23.622	Triticum aestivum (Wheat)	no	TVSYEQGEALAK	5.6	6.01	0.0181



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
24	L1 cell adhesion molecule	139.77	Triticum aestivum (Wheat)	no	DKEDTQVDSEAR	1	5.95	0.0179
25	NADH: flavin oxidoreductase/NADH oxidase N-terminal domain-containing protein	25.093	Triticum aestivum (Wheat)	no	TDEYGGSLNR	4.9	9.18	0.0167
26	NAB domain-containing protein	202.22	Triticum aestivum (Wheat)	no	KDLAELNEEKESLAVLYHQCLEK	1.3	5.84	0.0164



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
27	Protein FUN14	18.693	Triticum aestivum (Wheat)	no	GDLQQELGLPTR	7	10.50	0.0158
28	40S ribosomal protein S29	8.77	Triticum aestivum (Wheat)	no	ISGSHSGLIR	13.5	9.33	0.0153
29	Histone H2A	14.1	Triticum aestivum (Wheat)	no	AGLQFPVGR, GDDELDLIR, HLQLAIR	19.7	12.23	0.0147
30	Succinate dehydrogenase [ubiquinone] cytochrome b subunit	20.097	Bos taurus (Bovine)	no	TTTAEYELLVAQR	7.7	21.19	0.0140
31	Ribosomal	18.177	Bos taurus	no	VLNSYWVNEDSTYK	9	6.08	0.0137



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CISW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	protein L15		(Bovine)					
32	26S proteasome regulatory subunit RPN13	33.51	Penaeus vannamei (Whiteleg shrimp)	no	TAGSSVDPIEFILIPGETK	6.4	12.90	0.0131
33	phenylalanine - tRNA ligase	67.37	Penaeus vannamei (Whiteleg shrimp)	no	TASFELIQGLLGK	2.2	13.84	0.0130
34	Cytochrome c oxidase subunit I	58.673	Arachis hypogaea (Peanut)	no	GPDFVESNQIFAINK	2.8	16.66	0.0119
35	Histone H2B	14.414	Arachis hypogaea (Peanut)	no	HAVSEGTR, HAVSEGTRAVTK, KETYSSYIYK, QTHPDTGISQK, KTASSIDPSK	32.6	7.25	0.0118



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
36	ADP/ATP translocase (ADP,ATP carrier protein)	42.338	Arachis hypogaea (Peanut)	no	TAAAPIER, SDGLAGLYR	4.4	6.30	0.0115
37	transmembrane emp24 domain containing protein p24delta9-like	24.279	Arachis hypogaea (Peanut)	no	KGQIEVMELELKK	6	6.53	0.0115
38	Cofilin (Actin-depolymerising)	16.084	Arachis hypogaea (Peanut)	no	ETSADQDYDAFLEK	9.8	11.99	0.0114



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	factor 1)							
39	Uncharacterised protein	56.02	Arachis hypogaea (Peanut)	no	PDATIDTQR	1.7	6.09	0.0114
40	Aspartic proteinase 3	63.097	Arachis hypogaea (Peanut)	no	SGSILFGAVDHNK	2.2	5.92	0.0111
41	Uncharacterised protein	8.9183	Arachis hypogaea (Peanut)	no	SSATIAINHAAQTSK	19.3	11.13	0.0108
42	D-arabinitol 2-dehydrogenase	33.846	Glycine max (Soybean)	no	QVISQSPEGAALQK	4.5	37.14	0.0106
43	Succinate dehydrogenase	21.734	Danio rerio (Zebrafish)	no	GILETDLLLSR	5.8	8.93	0.0105



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	e assembly factor 2, mitochondria 1 (SDH assembly factor 2)							
44	Histone H2B	14.74	Gallus gallus (Chicken)	no	PAAEKAPAEK	7.5	6.99	0.0103
45	Keratin 24	55,128	Danio rerio (Zebrafish)	no	SDLEMQIESLTEELAYLKK	3.6	25.33	0.0099
46	Dihydrolipoamide acetyltransferase	56.387	Gallus gallus (Chicken)	no	ISINDLVIK	1.7	8.67	0.0093



Shanghai Changing Biotechnology Co., Ltd. Project NO.: CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
	component of pyruvate dehydrogenase complex							
47	Actin-related protein 9	66.139	Gallus gallus (Chicken)	no	NMYSAVLVLPETFDNREIK	3.2	5.92	0.0090
48	MICOS complex subunit MIC10	8.8101	Danio rerio (Zebrafish)	no	GYAEGDAIFR	12.2	5.92	0.0089
49	Actin	37.202	Bos taurus (Bovine)	no	SYELPDGQVITIGDER	10.4	9.98	0.0082
50	Patatin	46.488	Glycine max	no	LELFGGELIK	2.3	6.18	0.0081



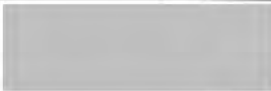
Shanghai Changing Biotechnology Co., Ltd. Project NO. CJSW-SH-WSW-24-0920

No.	Protein name	Molecular weight (kDa)	Source Organism	Whether reported as allergens	Peptide sequences identified	Sequence coverage %	score	Percentage %
			(Soybean)					
51	E3 ubiquitin ligase complex SCF subunit	20.918	Triticum aestivum (Wheat)	no	ENQ			



Shanghai Changing Biotechnology Co., Ltd. Project NO. : CJSW-SH-WSW-24-0920

Approval signature for experimental report

Experiment leader/date	 2024.09.20
Team leader review/date	 2024.9.20
Approved by/Date	 2024.9.20

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001248	DATE OF RECEIPT Oct 31, 2024
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____
	☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Yu Xu		Position or Title Director of Science Center	
	Organization (<i>if applicable</i>) Shanghai Changing Biotechnology Co., Ltd.			
	Mailing Address (<i>number and street</i>) 3rd Floor, Building 4, No.789/799/855, Zi Ping Road, Pudong District, Shanghai, China 201318			
City Shanghai		State or Province Shanghai	Zip Code/Postal Code 201318	Country China
Telephone Number + 86 185 1630 6886		Fax Number	E-Mail Address xuy@chgbio.com	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person Stella Si		Position or Title Managing Director	
	Organization (<i>if applicable</i>) Anchor Center for Certification			
	Mailing Address (<i>number and street</i>) No. 1295 Chuan Qiao Road, Building 2, Suite 302, Pudong, Shanghai, China.			
City Shanghai		State or Province Shanghai	Zip Code/Postal Code 201206	Country China
Telephone Number +86 186 2167 9927		Fax Number	E-Mail Address stella.si@acclab.cn	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Kluby® Protein

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)*

☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))*

☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Kluby® Protein is a source of protein in food ingredient and intended to serve as a substitute for other animal or vegetable proteins, in various food categories, excluding infant formulas, meat and poultry products. The addition levels of Kluby® Protein range from 1% to 20%, depending on food category.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☒ Yes ☐ No

Did you include this other information in the list of attachments?

☒ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Shanghai Changing Biotechnology Co., Ltd.

(name of notifier)

has concluded that the intended use(s) of Kluyv[®] Protein

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Shanghai Changing Biotechnology Co., Ltd. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

No. 1295 Chuan Qiao Road, Building 2, Suite 302, Pudong, Shanghai, China 201206

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Stella Si

Digitally signed by Stella Si
Date: 2024.10.31 15:14:10 +08'00'

Printed Name and Title

Stella Si, Managing Director of ACC Lab

Date (mm/dd/yyyy)

10/30/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	FDAGRASKluby®ProteinGRASDossier.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.